

Gödel's and Scott's variants of the ontological argument in Isabelle/HOL

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Presentation is based on ...

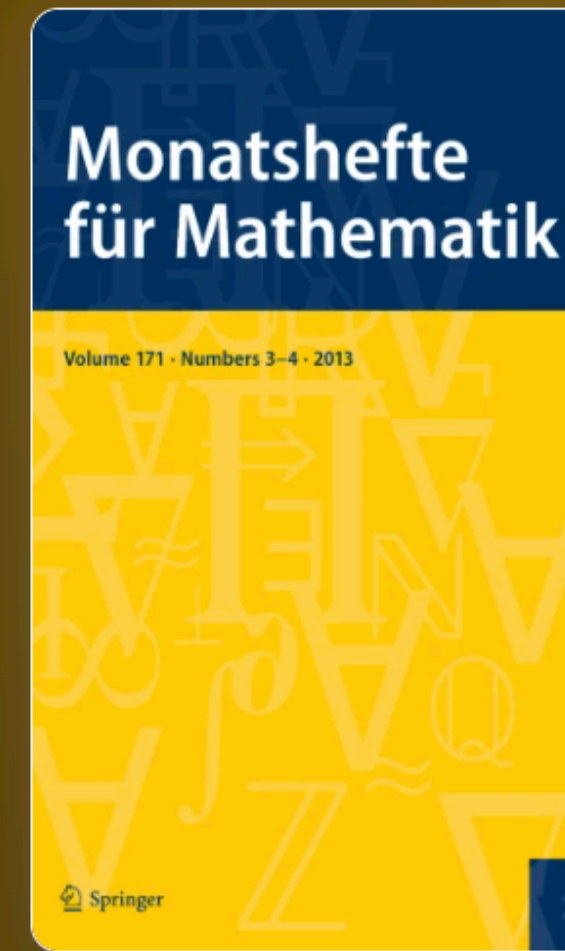
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Notes on Gödel's and Scott's variants of the ontological argument

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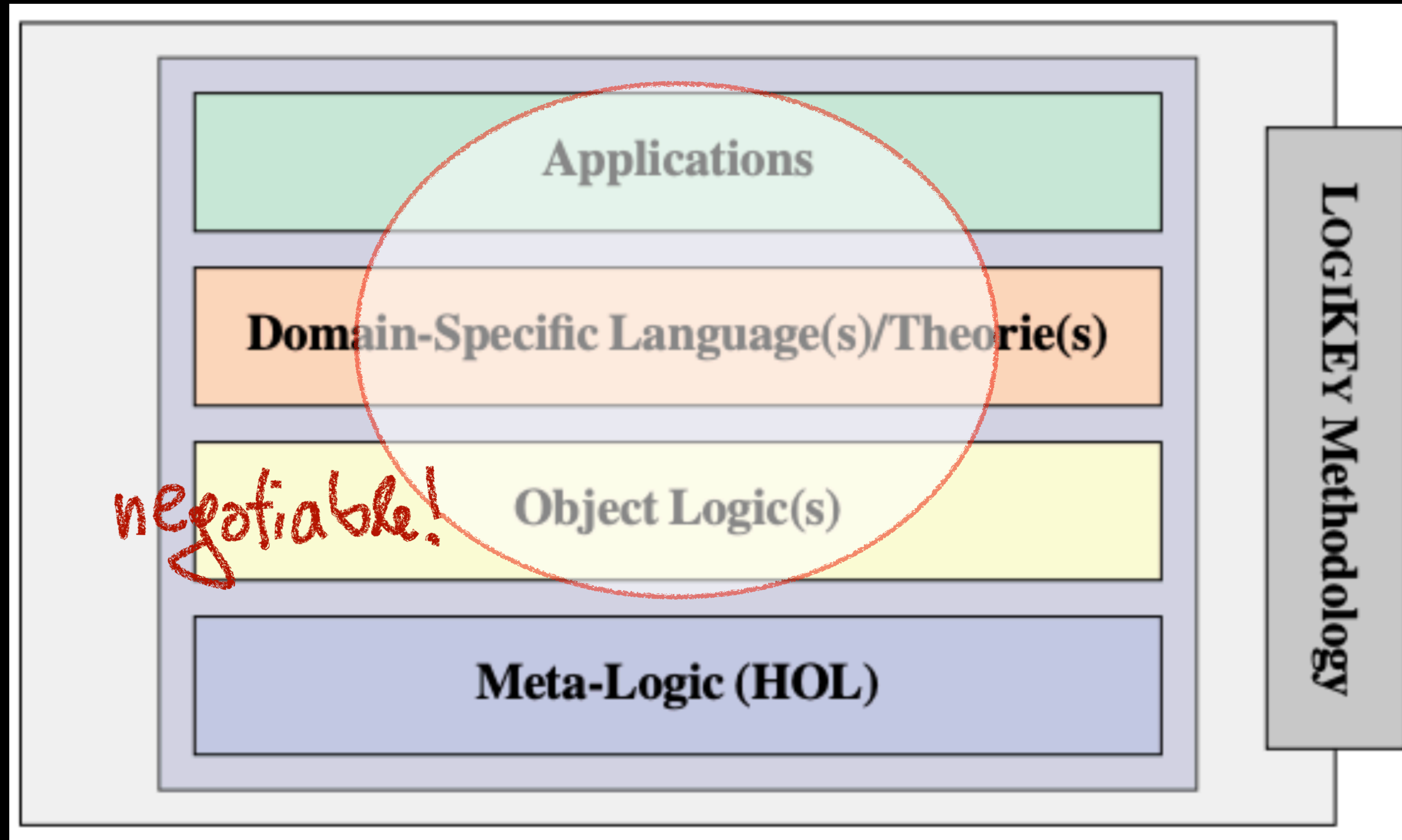
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... but some novel findings are added.

LogiKEy: Logic-pluralistic KR&R Methodology

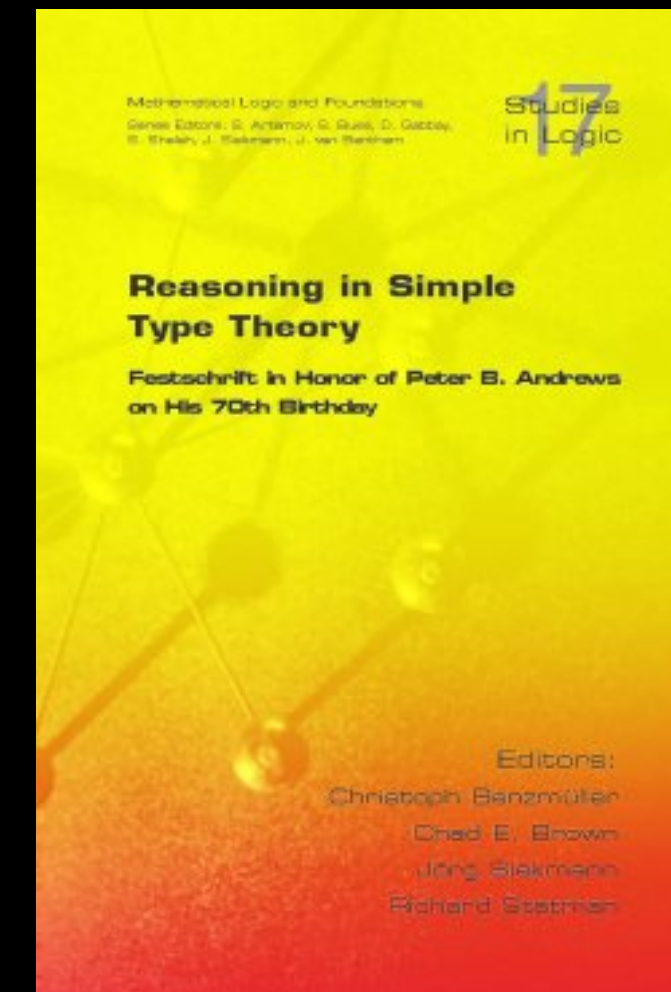


Origins: 2007-2008

LogiKEy: Logic-pluralistic KR&R Methodology

Selection of logic embeddings:

- Multimodal Logic [Andrews Festschrift, 2008]
- Intuitionistic Logic [LJ IGPL, 2010]
- Quantified Multimodal Logic [Log Univers, 2012]
- Quantified Conditional Logic [J Phil Log, 2016]
- Multivalued Logic [LLP, 2016]
- Free Logic [JAR, 2020]
- Dyadic Deontic Logic [AIJ, 2020] [JANCL, 2024]
- Hyper-Intensional HO Modal Logic [RSL, 2020]
- Public Announcement Logic [JLC, 2022]
- Logic of the Right to Know [ECAI, 2025]



Term
LogiKEy
introduced
in 2020

HOL ITP & ATP



Isabelle

**UNIVERSITY OF CAMBRIDGE**
Computer Laboratory

**TECHNISCHE UNIVERSITÄT MÜNCHEN**

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What is Isabelle?

Isabelle is a generic proof assistant. It allows mathematical formulas to be expressed in a formal language and provides tools for proving those formulas in a logical calculus. Isabelle was originally developed at the [University of Cambridge](#) and [Technische Universität München](#), but now includes numerous contributions from institutions and individuals worldwide. See the [Isabelle overview](#) for a brief introduction.

Now available: Isabelle2025 (March 2025)



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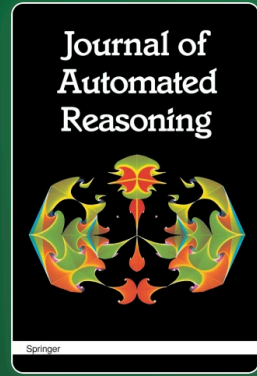


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Christoph Benz Müller , Nik Sultana, Lawrence C. Paulson & Frank Thei ß

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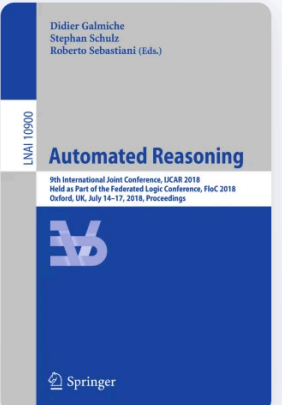
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Automated Reasoning
(IJCAR 2018)

Alexander Steen  & Christoph Benz Müller

- Other provers could be used instead:
- ITPs: Lean, HOL4, Coq, Agda, ...
 - ATPs: Vampire, E, Zipperpin, CVC5, ...

Isabelle/HOL: My favorite tool for logic education (and choice for this paper)

- Proof automation with **Sledgehammer**
- Model- and countermodel-finding with **Nitpick**
- Powerful **User Interface**

Ontological Beweis

Feb 10, 1970

$P(\varphi)$ φ is positive ($\varphi \in P$)

At 1 $P(\varphi) \cdot P(\psi) \supset P(\varphi \cdot \psi)$ At 2 $P(\varphi) \vee P(\neg \varphi)$

P1 $G(x) \equiv (\varphi) [P(\varphi) \supset \varphi(x)]$ (God)

P2 $\varphi \text{ Ess } x \equiv (\psi) [\psi(x) \supset N(y) [\varphi(y) \supset \psi(y)]]$ (Essence of x)

$p \supset_N q = N(p \supset q)$ Necessity

At 2 $P(\varphi) \supset NP(\varphi)$
 $\neg P(\varphi) \supset N \neg P(\varphi)$ } because it follows from the nature of the property

Th. $G(x) \supset G \text{ Ess } x$

Df. $E(x) \equiv (\varphi) [\varphi \text{ Ess } x \supset N \exists x \varphi(x)]$ necessary Existence

Ax 3 $P(E)$

Th. $G(x) \supset N(\exists y) G(y)$

hence $(\exists x) G(x) \supset N(\exists y) G(y)$

" $M(\exists x) G(x) \supset MN(\exists y) G(y)$

" $\supset N(\exists y) G(y)$

M = possibility

any two essences of x are nec. equivalent

exclusive or and for any number of humanoids

$M(\exists x) G(x)$ means ^{the system of} all pos. prop. is compatible
 This is true because of:

At 4: $P(\varphi) \cdot \varphi \supset_N \psi \supset P(\psi)$ which impl

~~hence~~ $\begin{cases} x=x & \text{is positive} \\ x \neq x & \text{is negative} \end{cases}$

But if a system S of pos. prop. were incomp.
 It would mean that the same prop. is (which is positive) would be $x \neq x$

Positive means positive in the moral aesthetic sense (independently of the accidental structure of the world). Only then the ax. true. It also means "attribution" as opposed to "privation" (or containing privation). This interprets simpler proof

If $\neg \varphi$ pos. then $(x) N \neg \varphi(x)$. Otherwise $\varphi(x) \supset_N x \neq x$

hence $x \neq x$ positive so $x=x$ neg. contradictory At

or the exist. of pos. prop.

x i.e. the normal form in terms of elem. prop. contains a member without negation.

Overview

- Preliminaries: HOML in HOL
 - Gödel's 1970 scriptum variant of the modal ontological argument
 - with notion of essence from scriptum — inconsistency
 - with adapted notion of essence
 - with adapted notion of property inclusion
 - no symmetries, no Evil
 - Scott's variant
 - this is the work you (most likely) already know about
 - Conclusion
- new formal studies
- new variant(?)

Preliminaries:

HOML in HOL

HOML in HOL

Unimportant Isabelle/HOL settings

```
1 theory HOMLinHOL imports Main
2 begin
3 —<Global parameters setting for the model finder nitpick and the parser; unimport for the reader>
4 nitpick_params[user_axioms,expect=genuine,show_all,format=2,max_genuine=3]
5 declare[[syntax_ambiguity_warning=false]]
```

Fig. 3 Shallow embedding of higher-order modal logic (HOML) in the classical higher-order logic (HOL) of Isabelle/HOL utilizing the LogiKEy methodology

HOML in HOL

Type **i** for possible worlds

Type **e** for entities

Type $\sigma := (i \Rightarrow \text{bool})$ for modal formulas

Types $\tau := (e \Rightarrow \text{bool})$ for modal predicates

Constant symbol **r** for accessibility between worlds

r is reflexive, transitive and symmetric (**S5** logic)

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10 type_synonym  $\tau$  = "e $\Rightarrow$  $\sigma$ " —<Modal properties>
11 consts R::"i $\Rightarrow$ i $\Rightarrow$ bool" ("_r_") —<Accessibility relation between worlds>
12 axiomatization where
13 Rrefl: " $\forall x. xrx$ " and Rsymm: " $\forall x y. xry \longrightarrow yrx$ " and Rtrans: " $\forall x y z. xry \wedge yrz \longrightarrow xrz$ "
```

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HOML in HOL

Modal logic connectives:

- abbreviations for λ -terms in HOL

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18 abbreviation Mand::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl " $\wedge$ " 50) where " $\varphi \wedge \psi \equiv \lambda w. \varphi w \wedge \psi w$ "
19 abbreviation Mor::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl " $\vee$ " 49) where " $\varphi \vee \psi \equiv \lambda w. \varphi w \vee \psi w$ "
20 abbreviation Mimp::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixr " $\supset$ " 48) where " $\varphi \supset \psi \equiv \lambda w. \varphi w \longrightarrow \psi w$ "
21 abbreviation Mequiv::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl " $\leftrightarrow$ " 47) where " $\varphi \leftrightarrow \psi \equiv \lambda w. \varphi w \longleftrightarrow \psi w$ "
```

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HOML in HOL

Constant symbol r used to define Box and Diamond

```
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```

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HOML in HOL

Polymorphic primitive equality (rigid)

```

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HOML in HOL

Negation ($\sim$) and conjunction ($\cdot$) of modal properties and exclusive-or ($\vee^e$)

```

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27 abbreviation Mconpred::" $\tau \Rightarrow \tau \Rightarrow \tau$ " (infixl "." 50) where " $\Phi . \Psi \equiv \lambda x. \lambda w. \Phi x w \wedge \Psi x w$ "
28 abbreviation Mexclor::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl "∨e" 49) where " $\varphi \vee^e \psi \equiv (\varphi \vee \psi) \wedge \neg(\varphi \wedge \psi)$ "

```

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HOML in HOL

Possibilist quantifiers (polymorphic, for all types)

```
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29 —<Possibilist quantifiers (polymorphic)>
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31 abbreviation Mallpossb (binder "∀" [8]9) where " $\forall x. \varphi(x) \equiv \forall \varphi$ "
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HOML in HOL

Actualist quantifiers (for entity type **e** only) where constant symbol **x@w** denotes “existence of **x** in world **w**”

```

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17 abbreviation Mneg::" $\sigma \Rightarrow \sigma$ " ("¬") [52]53) where " $\neg \varphi \equiv \lambda w. \neg(\varphi w)$ "
18 abbreviation Mand::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl "∧" 50) where " $\varphi \wedge \psi \equiv \lambda w. \varphi w \wedge \psi w$ "
19 abbreviation Mor::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl "∨" 49) where " $\varphi \vee \psi \equiv \lambda w. \varphi w \vee \psi w$ "
20 abbreviation Mimp::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixr "⊃" 48) where " $\varphi \supset \psi \equiv \lambda w. \varphi w \longrightarrow \psi w$ "
21 abbreviation Mequiv::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl "↔" 47) where " $\varphi \leftrightarrow \psi \equiv \lambda w. \varphi w \longleftrightarrow \psi w$ "
22 abbreviation Mbox::" $\sigma \Rightarrow \sigma$ " ("□") [54]55) where " $\Box \varphi \equiv \lambda w. \forall v. w r v \longrightarrow \varphi v$ "
23 abbreviation Mdia::" $\sigma \Rightarrow \sigma$ " ("◇") [54]55) where " $\Diamond \varphi \equiv \lambda w. \exists v. w r v \wedge \varphi v$ "
24 abbreviation Mprimeq::" $'a \Rightarrow 'a \Rightarrow \sigma$ " ("=_") where " $x = y \equiv \lambda w. x = y$ "
25 abbreviation Mprimneg::" $'a \Rightarrow 'a \Rightarrow \sigma$ " ("≠") where " $x \neq y \equiv \lambda w. x \neq y$ "
26 abbreviation Mnegpred::" $\tau \Rightarrow \tau$ " ("¬~") where " $\sim \Phi \equiv \lambda x. \lambda w. \neg \Phi x w$ "
27 abbreviation Mconpred::" $\tau \Rightarrow \tau \Rightarrow \tau$ " (infixl "." 50) where " $\Phi . \Psi \equiv \lambda x. \lambda w. \Phi x w \wedge \Psi x w$ "
28 abbreviation Mexclor::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl "∨e" 49) where " $\varphi \vee^e \psi \equiv (\varphi \vee \psi) \wedge \neg(\varphi \wedge \psi)$ "
29 —<Possibilist quantifiers (polymorphic)>
30 abbreviation Mallposs::" $('a \Rightarrow \sigma) \Rightarrow \sigma$ " ("∀") where " $\forall \Phi \equiv \lambda w. \forall x. \Phi x w$ "
31 abbreviation Mallpossb (binder "∀" [8]9) where " $\forall x. \varphi(x) \equiv \forall \varphi$ "
32 abbreviation Mexiposs::" $('a \Rightarrow \sigma) \Rightarrow \sigma$ " ("∃") where " $\exists \Phi \equiv \lambda w. \exists x. \Phi x w$ "
33 abbreviation Mexipossb (binder "∃" [8]9) where " $\exists x. \varphi(x) \equiv \exists \varphi$ "
34 —<Actualist quantifiers (for individuals/entities)>
35 consts existsAt::" $e \Rightarrow \sigma$ " ("_@_")
36 abbreviation Mallact::" $(e \Rightarrow \sigma) \Rightarrow \sigma$ " ("∀E") where " $\forall^E \Phi \equiv \lambda w. \forall x. x@w \longrightarrow \Phi x w$ "
37 abbreviation Mallactb (binder "∀E" [8]9) where " $\forall^E x. \varphi(x) \equiv \forall^E \varphi$ "
38 abbreviation Mexiact::" $(e \Rightarrow \sigma) \Rightarrow \sigma$ " ("∃E") where " $\exists^E \Phi \equiv \lambda w. \exists x. x@w \wedge \Phi x w$ "
39 abbreviation Mexiactb (binder "∃E" [8]9) where " $\exists^E x. \varphi(x) \equiv \exists^E \varphi$ "

```

Fig. 3 Shallow embedding of higher-order modal logic (HOML) in the classical higher-order logic (HOL) of Isabelle/HOL utilizing the LogiKEy methodology

HOML in HOL

Polymorphic Leibniz equality

```

1 theory HOMLinHOL imports Main
2 begin
3 —<Global parameters setting for the model finder nitpick and the parser; unimport for the reader>
4 nitpick_params[user_axioms,expect=genuine,show_all,format=2,max_genuine=3]
5 declare[[syntax_ambiguity_warning=false]]
6 —<Type i is associated with possible worlds and type e with entities:>
7 typedecl i —<Possible worlds>
8 typedecl e —<Individuals/entities>
9 type_synonym  $\sigma$  = "i $\Rightarrow$ bool" —<World-lifted propositions>
10 type_synonym  $\tau$  = "e $\Rightarrow$  $\sigma$ " —<Modal properties>
11 consts R::"i $\Rightarrow$ i $\Rightarrow$ bool" ("_r_") —<Accessibility relation between worlds>
12 axiomatization where
13   Rrefl: " $\forall x. xrx$ " and Rsymm: " $\forall x y. xry \longrightarrow yrx$ " and Rtrans: " $\forall x y z. xry \wedge yrz \longrightarrow xrz$ "
14 —<Logical connectives (operating on truth-sets):>
15 abbreviation Mbot:: $\sigma$  ("⊥") where " $\bot \equiv \lambda w. \text{False}$ "
16 abbreviation Mtop:: $\sigma$  ("⊤") where " $\top \equiv \lambda w. \text{True}$ "
17 abbreviation Mneg::" $\sigma \Rightarrow \sigma$ " ("¬") [52]53) where " $\neg \varphi \equiv \lambda w. \neg(\varphi w)$ "
18 abbreviation Mand::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl "∧" 50) where " $\varphi \wedge \psi \equiv \lambda w. \varphi w \wedge \psi w$ "
19 abbreviation Mor::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl "∨" 49) where " $\varphi \vee \psi \equiv \lambda w. \varphi w \vee \psi w$ "
20 abbreviation Mimp::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixr "⊃" 48) where " $\varphi \supset \psi \equiv \lambda w. \varphi w \longrightarrow \psi w$ "
21 abbreviation Mequiv::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl "↔" 47) where " $\varphi \leftrightarrow \psi \equiv \lambda w. \varphi w \longleftrightarrow \psi w$ "
22 abbreviation Mbox::" $\sigma \Rightarrow \sigma$ " ("□") [54]55) where " $\Box \varphi \equiv \lambda w. \forall v. w r v \longrightarrow \varphi v$ "
23 abbreviation Mdia::" $\sigma \Rightarrow \sigma$ " ("◇") [54]55) where " $\Diamond \varphi \equiv \lambda w. \exists v. w r v \wedge \varphi v$ "
24 abbreviation Mprimeq::" $'a \Rightarrow 'a \Rightarrow \sigma$ " ("=_") where " $x = y \equiv \lambda w. x = y$ "
25 abbreviation Mprimneg::" $'a \Rightarrow 'a \Rightarrow \sigma$ " ("≠") where " $x \neq y \equiv \lambda w. x \neq y$ "
26 abbreviation Mnegpred::" $\tau \Rightarrow \tau$ " ("¬~") where " $\sim \Phi \equiv \lambda x. \lambda w. \neg \Phi x w$ "
27 abbreviation Mconpred::" $\tau \Rightarrow \tau \Rightarrow \tau$ " (infixl "." 50) where " $\Phi . \Psi \equiv \lambda x. \lambda w. \Phi x w \wedge \Psi x w$ "
28 abbreviation Mexclor::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl "∨e" 49) where " $\varphi \vee^e \psi \equiv (\varphi \vee \psi) \wedge \neg(\varphi \wedge \psi)$ "
29 —<Possibilist quantifiers (polymorphic)>
30 abbreviation Mallposs::" $('a \Rightarrow \sigma) \Rightarrow \sigma$ " ("∀") where " $\forall \Phi \equiv \lambda w. \forall x. \Phi x w$ "
31 abbreviation Mallpossb (binder "∀" [8]9) where " $\forall x. \varphi(x) \equiv \forall \varphi$ "
32 abbreviation Mexiposs::" $('a \Rightarrow \sigma) \Rightarrow \sigma$ " ("∃") where " $\exists \Phi \equiv \lambda w. \exists x. \Phi x w$ "
33 abbreviation Mexipossb (binder "∃" [8]9) where " $\exists x. \varphi(x) \equiv \exists \varphi$ "
34 —<Actualist quantifiers (for individuals/entities)>
35 consts existsAt::" $e \Rightarrow \sigma$ " ("_@_")
36 abbreviation Mallact::" $(e \Rightarrow \sigma) \Rightarrow \sigma$ " ("∀E") where " $\forall^E \Phi \equiv \lambda w. \forall x. x @ w \longrightarrow \Phi x w$ "
37 abbreviation Mallactb (binder "∀E" [8]9) where " $\forall^E x. \varphi(x) \equiv \forall^E \varphi$ "
38 abbreviation Mexiact::" $(e \Rightarrow \sigma) \Rightarrow \sigma$ " ("∃E") where " $\exists^E \Phi \equiv \lambda w. \exists x. x @ w \wedge \Phi x w$ "
39 abbreviation Mexiactb (binder "∃E" [8]9) where " $\exists^E x. \varphi(x) \equiv \exists^E \varphi$ "
40 —<Leibniz equality (polymorphic):>
41 abbreviation Mleibeq::" $'a \Rightarrow 'a \Rightarrow \sigma$ " ("≡") where " $x \equiv y \equiv \forall P. P x \supset P y$ "

```

Fig. 3 Shallow embedding of higher-order modal logic (HOML) in the classical higher-order logic (HOL) of Isabelle/HOL utilizing the LogiKey methodology

HOML in HOL

Notion of validity for modal formulas

```

1 theory HOMLinHOL imports Main
2 begin
3   <Global parameters setting for the model finder nitpick and the parser; unimport for the reader>
4   nitpick_params[user_axioms,expect=genuine,show_all,format=2,max_genuine=3]
5   declare[[syntax_ambiguity_warning=false]]
6   <Type i is associated with possible worlds and type e with entities:>
7   typedecl i   <Possible worlds>
8   typedecl e   <Individuals/entities>
9   type_synonym  $\sigma$  = "i $\Rightarrow$ bool"   <World-lifted propositions>
10  type_synonym  $\tau$  = "e $\Rightarrow\sigma$ "      <Modal properties>
11  consts R::"i $\Rightarrow$ i $\Rightarrow$ bool" ("_r_")  <Accessibility relation between worlds>
12  axiomatization where
13    Rrefl: " $\forall x. xrx$ " and Rsymm: " $\forall x y. xry \longrightarrow yrx$ " and Rtrans: " $\forall x y z. xry \wedge yrz \longrightarrow xrz$ "
14  <Logical connectives (operating on truth-sets):>
15  abbreviation Mbot:: $\sigma$  ("⊥") where " $\bot \equiv \lambda w. \text{False}$ "
16  abbreviation Mtop:: $\sigma$  ("⊤") where " $\top \equiv \lambda w. \text{True}$ "
17  abbreviation Mneg::" $\sigma \Rightarrow \sigma$ " ("¬") [52]53) where " $\neg \varphi \equiv \lambda w. \neg(\varphi w)$ "
18  abbreviation Mand::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl "∧" 50) where " $\varphi \wedge \psi \equiv \lambda w. \varphi w \wedge \psi w$ "
19  abbreviation Mor::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl "∨" 49) where " $\varphi \vee \psi \equiv \lambda w. \varphi w \vee \psi w$ "
20  abbreviation Mimp::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixr "⊃" 48) where " $\varphi \supset \psi \equiv \lambda w. \varphi w \longrightarrow \psi w$ "
21  abbreviation Mequiv::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl "↔" 47) where " $\varphi \leftrightarrow \psi \equiv \lambda w. \varphi w \longleftrightarrow \psi w$ "
22  abbreviation Mbox::" $\sigma \Rightarrow \sigma$ " ("□") [54]55) where " $\Box \varphi \equiv \lambda w. \forall v. w r v \longrightarrow \varphi v$ "
23  abbreviation Mdia::" $\sigma \Rightarrow \sigma$ " ("◇") [54]55) where " $\Diamond \varphi \equiv \lambda w. \exists v. w r v \wedge \varphi v$ "
24  abbreviation Mprimeq::" $'a \Rightarrow 'a \Rightarrow \sigma$ " ("=") where " $x = y \equiv \lambda w. x = y$ "
25  abbreviation Mprimneg::" $'a \Rightarrow 'a \Rightarrow \sigma$ " ("≠") where " $x \neq y \equiv \lambda w. x \neq y$ "
26  abbreviation Mnegpred::" $\tau \Rightarrow \tau$ " ("¬") where " $\sim \Phi \equiv \lambda x. \lambda w. \neg \Phi x w$ "
27  abbreviation Mconpred::" $\tau \Rightarrow \tau \Rightarrow \tau$ " (infixl "." 50) where " $\Phi . \Psi \equiv \lambda x. \lambda w. \Phi x w \wedge \Psi x w$ "
28  abbreviation Mexclor::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl "⊕" 49) where " $\varphi \oplus \psi \equiv (\varphi \vee \psi) \wedge \neg(\varphi \wedge \psi)$ "
29  <Possibilist quantifiers (polymorphic)>
30  abbreviation Mallposs::" $('a \Rightarrow \sigma) \Rightarrow \sigma$ " ("∀") where " $\forall \Phi \equiv \lambda w. \forall x. \Phi x w$ "
31  abbreviation Mallpossb (binder "∀" [8]9) where " $\forall x. \varphi(x) \equiv \forall \varphi$ "
32  abbreviation Mexiposs::" $('a \Rightarrow \sigma) \Rightarrow \sigma$ " ("∃") where " $\exists \Phi \equiv \lambda w. \exists x. \Phi x w$ "
33  abbreviation Mexipossb (binder "∃" [8]9) where " $\exists x. \varphi(x) \equiv \exists \varphi$ "
34  <Actualist quantifiers (for individuals/entities)>
35  consts existsAt::" $e \Rightarrow \sigma$ " ("@")
36  abbreviation Mallact::" $(e \Rightarrow \sigma) \Rightarrow \sigma$ " ("∀E") where " $\forall^E \Phi \equiv \lambda w. \forall x. x @ w \longrightarrow \Phi x w$ "
37  abbreviation Mallactb (binder "∀E" [8]9) where " $\forall^E x. \varphi(x) \equiv \forall^E \varphi$ "
38  abbreviation Mexiact::" $(e \Rightarrow \sigma) \Rightarrow \sigma$ " ("∃E") where " $\exists^E \Phi \equiv \lambda w. \exists x. x @ w \wedge \Phi x w$ "
39  abbreviation Mexiactb (binder "∃E" [8]9) where " $\exists^E x. \varphi(x) \equiv \exists^E \varphi$ "
40  <Leibniz equality (polymorphic):>
41  abbreviation Mleibeq::" $'a \Rightarrow 'a \Rightarrow \sigma$ " ("≡") where " $x \equiv y \equiv \forall P. P x \supset P y$ "
42  <Meta-logical predicate for global validity:>
43  abbreviation Mvalid::" $\sigma \Rightarrow \text{bool}$ " ("⊢") where " $\vdash \psi \equiv \forall w. \psi w$ "
44  end

```

Fig. 3 Shallow embedding of higher-order modal logic (HOML) in the classical higher-order logic (HOL) of Isabelle/HOL utilizing the LogiKEy methodology

HOML in HOL

A mathematician would only look at this

```

1 theory HOMLinHOL imports Main
2 begin
3 —<Global parameters setting for the model finder nitpick and the parser; unimport for the reader>
4 nitpick_params[user_axioms,expect=genuine,show_all,format=2,max_genuine=3]
5 declare[[syntax_ambiguity_warning=false]]
6 —<Type i is associated with possible worlds and type e with entities:>
7 typedecl i —<Possible worlds>
8 typedecl e —<Individuals/entities>
9 type_synonym  $\sigma$  = "i $\Rightarrow$ bool" —<World-lifted propositions>
10 type_synonym  $\tau$  = "e $\Rightarrow$  $\sigma$ " —<Modal properties>
11 consts R::"i $\Rightarrow$ i $\Rightarrow$ bool" ("_r_") —<Accessibility relation between worlds>
12 axiomatization where
13   Rrefl: " $\forall x. xrx$ " and Rsymm: " $\forall x y. xry \longrightarrow yrx$ " and Rtrans: " $\forall x y z. xry \wedge yrz \longrightarrow xrz$ "
14 —<Logical connectives (operating on truth-sets):>
15 abbreviation Mbot:: $\sigma$  ("⊥") where
16 abbreviation Mtop:: $\sigma$  ("⊤") where
17 abbreviation Mneg::" $\sigma \Rightarrow \sigma$ " ("¬_") [52]53) where
18 abbreviation Mand::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl "∧" 50) where
19 abbreviation Mor::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl "∨" 49) where
20 abbreviation Mimp::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixr "⊃" 48) where
21 abbreviation Mequiv::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl "↔" 47) where
22 abbreviation Mbox::" $\sigma \Rightarrow \sigma$ " ("□_" [54]55) where
23 abbreviation Mdia::" $\sigma \Rightarrow \sigma$ " ("◇_" [54]55) where
24 abbreviation Mprimeq::" $'a \Rightarrow 'a \Rightarrow \sigma$ " ("=_") where
25 abbreviation Mprimneg::" $'a \Rightarrow 'a \Rightarrow \sigma$ " ("≠_") where
26 abbreviation Mnegpred::" $\tau \Rightarrow \tau$ " ("¬~_") where
27 abbreviation Mconpred::" $\tau \Rightarrow \tau \Rightarrow \tau$ " (infixl "." 50) where
28 abbreviation Mexclor::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixl "∨e" 49) where
29 —<Possibilist quantifiers (polymorphic)>
30 abbreviation Mallposs::" $'a \Rightarrow \sigma \Rightarrow \sigma$ " ("∀") where
31 abbreviation Mallpossb (binder "∀" [8]9) where
32 abbreviation Mexiposs::" $'a \Rightarrow \sigma \Rightarrow \sigma$ " ("∃") where
33 abbreviation Mexipossb (binder "∃" [8]9) where
34 —<Actualist quantifiers (for individuals/entities)>
35 consts existsAt::"e $\Rightarrow$  $\sigma$ " ("_@_")
36 abbreviation Mallact::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("∀E") where
37 abbreviation Mallactb (binder "∀E" [8]9) where
38 abbreviation Mexiact::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("∃E") where
39 abbreviation Mexiactb (binder "∃E" [8]9) where
40 —<Leibniz equality (polymorphic):>
41 abbreviation Mleibeq::" $'a \Rightarrow 'a \Rightarrow \sigma$ " ("≡_") where
42 —<Meta-logical predicate for global validity:>
43 abbreviation Mvalid::" $\sigma \Rightarrow$ bool" ("⊢_") where
44 end

```

```

"⊥ ≡ λw. False"
"⊤ ≡ λw. True"
"¬φ ≡ λw. ¬(φ w)"
"φ ∧ ψ ≡ λw. φ w ∧ ψ w"
"φ ∨ ψ ≡ λw. φ w ∨ ψ w"
"φ ⊃ ψ ≡ λw. φ w ⟶ ψ w"
"φ ↔ ψ ≡ λw. φ w ⟷ ψ w"
"□φ ≡ λw. ∀v. w r v ⟶ φ v"
"◇φ ≡ λw. ∃v. w r v ∧ φ v"
"x = y ≡ λw. x = y"
"x ≠ y ≡ λw. x ≠ y"
"¬Φ ≡ λx. λw. ¬Φ x w"
"Φ.Ψ ≡ λx. λw. Φ x w ∧ Ψ x w"
"φ ∨e ψ ≡ (φ ∨ ψ) ∧ ¬(φ ∧ ψ)"

"∀Φ ≡ λw. ∀x. Φ x w"
"∀x. φ(x) ≡ ∀φ"
"∃Φ ≡ λw. ∃x. Φ x w"
"∃x. φ(x) ≡ ∃φ"

"∀EΦ ≡ λw. ∀x. x@w ⟶ Φ x w"
"∀Ex. φ(x) ≡ ∀Eφ"
"∃EΦ ≡ λw. ∃x. x@w ∧ Φ x w"
"∃Ex. φ(x) ≡ ∃Eφ"

"x ≡ y ≡ ∀P. P x ⊃ P y"

"⊢ψ ≡ ∀w. ψ w"

```

Fig. 3 Shallow embedding of higher-order modal logic (HOML) in the classical higher-order logic (HOL) of Isabelle/HOL utilizing the LogiKEy methodology

HOML in HOL: tests

The embedding is verified by proving the following

- S5 axioms: valid

```
1 theory TestsHOML imports HOMLinHOL
2 begin
3 -<Test for S5 modal logic>
4 lemma axM: "[ $\Box \varphi \supset \varphi$ ]" using Rrefl by blast
5 lemma axD: "[ $\Box \varphi \supset \Diamond \varphi$ ]" using Rrefl by blast
6 lemma axB: "[ $\varphi \supset \Box \Diamond \varphi$ ]" using Rsymm by blast
7 lemma ax4: "[ $\Box \varphi \supset \Box \Box \varphi$ ]" using Rtrans by blast
8 lemma ax5: "[ $\Diamond \varphi \supset \Box \Diamond \varphi$ ]" using Rsymm Rtrans by blast
```

Fig. 4 Tests and verifications of properties for the embedding of HOML in HOL from Fig. 3

HOML in HOL: tests

The embedding is verified by proving the following

- S5 axioms: valid
- Barcan formulas: not valid for actualist quantifiers
- Barcan formulas: valid for possibilist quantifiers

```
1 theory TestsHOML imports HOLLinHOL
2 begin
3 -<Test for S5 modal logic>
4 lemma axM: "[ $\Box \varphi \supset \varphi$ ]" using Rrefl by blast
5 lemma axD: "[ $\Box \varphi \supset \Diamond \varphi$ ]" using Rrefl by blast
6 lemma axB: "[ $\varphi \supset \Box \Diamond \varphi$ ]" using Rsymm by blast
7 lemma ax4: "[ $\Box \varphi \supset \Box \Box \varphi$ ]" using Rtrans by blast
8 lemma ax5: "[ $\Diamond \varphi \supset \Box \Diamond \varphi$ ]" using Rsymm Rtrans by blast
9 -<Test for Barcan and converse Barcan formula:>
10 lemma BarcanAct: "[ $(\forall^E x. \Box(\varphi x)) \supset \Box(\forall^E x. (\varphi x))$ ]" nitpick oops -<Countermodel found>
11 lemma ConvBarcanAct: "[ $\Box(\forall^E x. (\varphi x)) \supset (\forall^E x. \Box(\varphi x))$ ]" nitpick oops -<Countermodel found>
12 lemma BarcanPoss: "[ $(\forall x. \Box(\varphi x)) \supset \Box(\forall x. \varphi x)$ ]" by blast
13 lemma ConvBarcanPoss: "[ $\Box(\forall x. (\varphi x)) \supset (\forall x. \Box(\varphi x))$ ]" by blast
```

Fig. 4 Tests and verifications of properties for the embedding of HOML in HOL from Fig. 3

HOML in HOL: tests

The embedding is verified by proving the following

- S5 axioms: valid
- Barcan formulas: not valid for actualist quantifiers
- Barcan formulas: valid for possibilist quantifiers
- Hilbert axioms and rules: valid as expected

```
1 theory TestsHOML imports HOMLinHOL
2 begin
3 -<Test for S5 modal logic>
4 lemma axM: "[ $\Box \varphi \supset \varphi$ ]" using Rrefl by blast
5 lemma axD: "[ $\Box \varphi \supset \Diamond \varphi$ ]" using Rrefl by blast
6 lemma axB: "[ $\varphi \supset \Box \Diamond \varphi$ ]" using Rsymm by blast
7 lemma ax4: "[ $\Box \varphi \supset \Box \Box \varphi$ ]" using Rtrans by blast
8 lemma ax5: "[ $\Diamond \varphi \supset \Box \Diamond \varphi$ ]" using Rsymm Rtrans by blast
9 -<Test for Barcan and converse Barcan formula:>
10 lemma BarcanAct: "[ $(\forall x. \Box(\varphi x)) \supset \Box(\forall x. (\varphi x))$ ]" nitpick oops -<Countermodel found>
11 lemma ConvBarcanAct: "[ $\Box(\forall x. (\varphi x)) \supset (\forall x. \Box(\varphi x))$ ]" nitpick oops -<Countermodel found>
12 lemma BarcanPoss: "[ $(\forall x. \Box(\varphi x)) \supset \Box(\forall x. \varphi x)$ ]" by blast
13 lemma ConvBarcanPoss: "[ $\Box(\forall x. (\varphi x)) \supset (\forall x. \Box(\varphi x))$ ]" by blast
14 -<A simple Hilbert system for classical propositional logic is derived>
15 lemma Hilbert_A1: "[ $A \supset (B \supset A)$ ]" by blast
16 lemma Hilbert_A2: "[ $(A \supset (B \supset C)) \supset ((A \supset B) \supset (A \supset C))$ ]" by blast
17 lemma Hilbert_MP: assumes "[A]" and "[A  $\supset$  B]" shows "[B]" using assms by blast
```

Fig. 4 Tests and verifications of properties for the embedding of HOML in HOL from Fig. 3

HOML in HOL: tests

The embedding is verified by proving the following

- S5 axioms: valid
- Barcan formulas: not valid for actualist quantifiers
- Barcan formulas: valid for possibilist quantifiers
- Hilbert axioms and rules: valid as expected
- existential import: valid for possibilist quantifiers

```
1 theory TestsHOML imports HOLLinHOL
2 begin
3 <Test for S5 modal logic>
4 lemma axM: "[ $\Box \varphi \supset \varphi$ ]" using Rrefl by blast
5 lemma axD: "[ $\Box \varphi \supset \Diamond \varphi$ ]" using Rrefl by blast
6 lemma axB: "[ $\varphi \supset \Box \Diamond \varphi$ ]" using Rsymm by blast
7 lemma ax4: "[ $\Box \varphi \supset \Box \Box \varphi$ ]" using Rtrans by blast
8 lemma ax5: "[ $\Diamond \varphi \supset \Box \Diamond \varphi$ ]" using Rsymm Rtrans by blast
9 <Test for Barcan and converse Barcan formula:>
10 lemma BarcanAct: "[ $(\forall x. \Box(\varphi x)) \supset \Box(\forall x. (\varphi x))$ ]" nitpick oops <Countermodel found>
11 lemma ConvBarcanAct: "[ $\Box(\forall x. (\varphi x)) \supset (\forall x. \Box(\varphi x))$ ]" nitpick oops <Countermodel found>
12 lemma BarcanPoss: "[ $(\forall x. \Box(\varphi x)) \supset \Box(\forall x. \varphi x)$ ]" by blast
13 lemma ConvBarcanPoss: "[ $\Box(\forall x. (\varphi x)) \supset (\forall x. \Box(\varphi x))$ ]" by blast
14 <A simple Hilbert system for classical propositional logic is derived>
15 lemma Hilbert_A1: "[ $A \supset (B \supset A)$ ]" by blast
16 lemma Hilbert_A2: "[ $(A \supset (B \supset C)) \supset ((A \supset B) \supset (A \supset C))$ ]" by blast
17 lemma Hilbert_MP: assumes "[A]" and "[A  $\supset$  B]" shows "[B]" using assms by blast
18 <We have a polymorphic possibilist quantifier for which existential import holds>
19 lemma Quant_1: assumes "[A]" shows "[ $\forall x::'a. A$ ]" using assms by auto
20 <Existential import holds for possibilist quantifiers>
21 lemma ExImPossibilist1: "[ $\exists x::e. x = x$ ]" by blast
22 lemma ExImPossibilist2: "[ $\exists x::e. x \equiv x$ ]" by blast
23 lemma ExImPossibilist3: "[ $\exists x::e. x = t$ ]" by blast
24 lemma ExImPossibilist4: "[ $\exists x::'a. x \equiv t::'a$ ]" by blast
25 lemma ExImPossibilist: "[ $\exists x::'a. T$ ]" by blast
```

Fig. 4 Tests and verifications of properties for the embedding of HOML in HOL from Fig. 3

HOML in HOL: tests

The embedding is verified by proving the following

- S5 axioms: valid
- Barcan formulas: not valid for actualist quantifiers
- Barcan formulas: valid for possibilist quantifiers
- Hilbert axioms and rules: valid as expected
- existential import: valid for possibilist quantifiers
- existential import: not valid for actualist quantifiers

```
1 theory TestsHOML imports HOLLinHOL
2 begin
3 <Test for S5 modal logic>
4 lemma axM: "[ $\Box \varphi \supset \varphi$ ]" using Rrefl by blast
5 lemma axD: "[ $\Box \varphi \supset \Diamond \varphi$ ]" using Rrefl by blast
6 lemma axB: "[ $\varphi \supset \Box \Diamond \varphi$ ]" using Rsymm by blast
7 lemma ax4: "[ $\Box \varphi \supset \Box \Box \varphi$ ]" using Rtrans by blast
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53 lemma EmptyProperty3: "[ $\exists x. \varphi x \longrightarrow \varphi \neq (\lambda x::e. \bot)$ ]" by blast
54 lemma EmptyProperty4: "[ $\varphi \neq (\lambda x::e. \bot) \longrightarrow \exists x. \varphi x$ ]" nitpick oops <Countermodel found>
55 lemma EmptyProperty5: "[ $\varphi \neq (\lambda x::e. \bot) \longrightarrow \exists x. \varphi x$ ]" nitpick oops <Countermodel found>
56 end
```

Fig. 4 Tests and verifications of properties for the embedding of HOML in HOL from Fig. 3

HOML in HOL: modal (ultra-)filter

(Ultra-)Filter on sets of modal properties

- elementhood
- empty set
- universal set
- subset relation (using poss. quantifier)
- subset relation (using act. quantifier)
- set intersection
- set inverse

```
1 theory ModalFilter imports HOMLinHOL
2 begin
3 type_synonym  $\tau = "e \Rightarrow \sigma"$ 
4 abbreviation Element:: $\tau \Rightarrow (\tau \Rightarrow \sigma) \Rightarrow \sigma$  (infix  $\in$  90) where  $\varphi \in S \equiv S \varphi$ 
5 abbreviation EmptySet:: $\tau$  (" $\emptyset$ ") where  $\emptyset \equiv \lambda x. \bot$ 
6 abbreviation UniversalSet:: $\tau$  (" $U$ ") where  $U \equiv \lambda x. \top$ 
7 abbreviation Subset:: $\tau \Rightarrow \tau \Rightarrow \sigma$  (infix  $\subseteq$  80) where  $\varphi \subseteq \psi \equiv \forall x. ((\varphi x) \supset (\psi x))$ 
8 abbreviation SubsetE:: $\tau \Rightarrow \tau \Rightarrow \sigma$  (infix  $\subseteq^E$  80) where  $\varphi \subseteq^E \psi \equiv \forall^E x. ((\varphi x) \supset (\psi x))$ 
9 abbreviation Intersection:: $\tau \Rightarrow \tau \Rightarrow \tau$  (infix  $\sqcap$  91) where  $\varphi \sqcap \psi \equiv \lambda x. ((\varphi x) \wedge (\psi x))$ 
10 abbreviation Inverse:: $\tau \Rightarrow \tau$  (" $^{-1}$ ") where  $^{-1}\psi \equiv \lambda x. \neg(\psi x)$ 
```

Fig. 5 Set filter and ultrafilter formalized for our modal logic setting

HOML in HOL: modal (ultra-)filter

(Ultra-)Filter on sets of modal properties

- elementhood
- empty set
- universal set
- subset relation (using poss. quantifier)
- subset relation (using act. quantifier)
- set intersection
- set inverse
- Filter and Ultrafilter (**act. quantifier**)
 - contains universal set
 - not contains empty set
 - closed under **supersets**
 - closed under intersection
 - maximal

```
1 theory ModalFilter imports HOLinHOL
2 begin
3 type_synonym  $\tau = "e \Rightarrow \sigma"$ 
4 abbreviation Element:: $\tau \Rightarrow (\tau \Rightarrow \sigma) \Rightarrow \sigma$  (infix  $\in$  90) where  $\varphi \in S \equiv S \varphi$ 
5 abbreviation EmptySet:: $\tau$  ("∅") where  $\emptyset \equiv \lambda x. \bot$ 
6 abbreviation UniversalSet:: $\tau$  ("U") where  $U \equiv \lambda x. \top$ 
7 abbreviation Subset:: $\tau \Rightarrow \tau \Rightarrow \sigma$  (infix  $\subseteq$  80) where  $\varphi \subseteq \psi \equiv \forall x. ((\varphi x) \supset (\psi x))$ 
8 abbreviation SubsetE:: $\tau \Rightarrow \tau \Rightarrow \sigma$  (infix  $\subseteq^E$  80) where  $\varphi \subseteq^E \psi \equiv \forall^E x. ((\varphi x) \supset (\psi x))$ 
9 abbreviation Intersection:: $\tau \Rightarrow \tau \Rightarrow \tau$  (infix  $\sqcap$  91) where  $\varphi \sqcap \psi \equiv \lambda x. ((\varphi x) \wedge (\psi x))$ 
10 abbreviation Inverse:: $\tau \Rightarrow \tau$  (" $\neg$ ") where  $\neg \psi \equiv \lambda x. \neg(\psi x)$ 
11 abbreviation "Filter  $\Phi$   $\equiv U \in \Phi \wedge \neg(\emptyset \in \Phi) \wedge (\forall \varphi \psi. \varphi \in \Phi \wedge \varphi \subseteq^E \psi \supset \psi \in \Phi) \wedge (\forall \varphi \psi. \varphi \in \Phi \wedge \psi \in \Phi \supset \varphi \sqcap \psi \in \Phi)$ "
12 abbreviation "UFilter  $\Phi$   $\equiv$  Filter  $\Phi \wedge (\forall \varphi. \varphi \in \Phi \vee (\neg \varphi) \in \Phi)$ "
```

Fig. 5 Set filter and ultrafilter formalized for our modal logic setting

HOML in HOL: modal (ultra-)filter

(Ultra-)Filter on sets of modal properties

- elementhood
- empty set
- universal set
- subset relation (using poss. quantifier)
- subset relation (using act. quantifier)
- set intersection
- set inverse
- Filter and Ultrafilter (**act. quantifier**)
 - contains universal set
 - not contains empty set
 - closed under **supersets**
 - closed under intersection
 - maximal
- Filter and Ultrafilter (**poss. quantifier**)
 - contains universal set
 - not contains empty set
 - closed under **supersets**
 - closed under intersection
 - maximal

```
1 theory ModalFilter imports HOLinHOL
2 begin
3 type_synonym  $\tau = "e \Rightarrow \sigma"$ 
4 abbreviation Element:: $\tau \Rightarrow (\tau \Rightarrow \sigma) \Rightarrow \sigma$  (infix  $\in$  90) where  $\varphi \in S \equiv S \varphi$ 
5 abbreviation EmptySet:: $\tau$  (" $\emptyset$ ") where  $\emptyset \equiv \lambda x. \bot$ 
6 abbreviation UniversalSet:: $\tau$  (" $U$ ") where  $U \equiv \lambda x. \top$ 
7 abbreviation Subset:: $\tau \Rightarrow \tau \Rightarrow \sigma$  (infix  $\subseteq$  80) where  $\varphi \subseteq \psi \equiv \forall x. ((\varphi x) \supset (\psi x))$ 
8 abbreviation SubsetE:: $\tau \Rightarrow \tau \Rightarrow \sigma$  (infix  $\subseteq^E$  80) where  $\varphi \subseteq^E \psi \equiv \forall^E x. ((\varphi x) \supset (\psi x))$ 
9 abbreviation Intersection:: $\tau \Rightarrow \tau \Rightarrow \tau$  (infix  $\sqcap$  91) where  $\varphi \sqcap \psi \equiv \lambda x. ((\varphi x) \wedge (\psi x))$ 
10 abbreviation Inverse:: $\tau \Rightarrow \tau$  (" $^{-1}$ ") where  $^{-1}\psi \equiv \lambda x. \neg(\psi x)$ 
11 abbreviation "Filter  $\Phi$ "  $\equiv U \in \Phi \wedge \neg(\emptyset \in \Phi) \wedge (\forall \varphi \psi. \varphi \in \Phi \wedge \varphi \subseteq^E \psi \supset \psi \in \Phi) \wedge (\forall \varphi \psi. \varphi \in \Phi \wedge \psi \in \Phi \supset \varphi \sqcap \psi \in \Phi)$ 
12 abbreviation "UFilter  $\Phi$ "  $\equiv \text{Filter } \Phi \wedge (\forall \varphi. \varphi \in \Phi \vee (^{-1}\varphi) \in \Phi)$ 
13 abbreviation "FilterP  $\Phi$ "  $\equiv U \in \Phi \wedge \neg(\emptyset \in \Phi) \wedge (\forall \varphi \psi. \varphi \in \Phi \wedge \varphi \subseteq \psi \supset \psi \in \Phi) \wedge (\forall \varphi \psi. \varphi \in \Phi \wedge \psi \in \Phi \supset \varphi \sqcap \psi \in \Phi)$ 
14 abbreviation "UFilterP  $\Phi$ "  $\equiv \text{FilterP } \Phi \wedge (\forall \varphi. \varphi \in \Phi \vee (^{-1}\varphi) \in \Phi)$ 
15 end
```

Fig. 5 Set filter and ultrafilter formalized for our modal logic setting

Formalisation of
Gödel's 1970 manuscript
of the modal ontological argument
(original notion of essence: Gödel1)

— inconsistency —

Gödel 1

Positive properties: P

Axiom Ax1: $P\varphi \wedge P\psi \supset P(\varphi . \psi)$

Axiom Ax2a: $P\varphi \vee^e P\sim\varphi$

```
1 theory GoedelVariantHOML1 imports HOMLinHOL
2 begin
3 consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4 axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi . \psi$ )]"
5 axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
```

Fig. 6 Gödel's axioms and definitions, as presented in the 1970 manuscript, are inconsistent

Gödel 1

Definition of God-like: $G x \equiv \forall \varphi. P \varphi \supset \varphi x$

```
1 theory GoedelVariantHOML1 imports HOMLinHOL
2 begin
3   consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4   axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5   axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6   definition God ("G") where "G x  $\equiv$   $\forall \varphi. P \varphi \supset \varphi x$ "
```

Fig. 6 Gödel's axioms and definitions, as presented in the 1970 manuscript, are inconsistent

Gödel 1

Definition of Essence: $\varphi \text{ Ess. } x \equiv \forall \psi. \psi x \supset (\varphi \supset_N \psi)$

```
1 theory GoedelVariantHOML1 imports HOMLinHOL
2 begin
3   consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4   axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5   axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6   definition God ("G") where "G x  $\equiv$   $\forall\varphi. P \varphi \supset \varphi x$ "
7   abbreviation PropertyInclusion ("_ $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\forall^E y. \varphi y \supset \psi y)$ "
8   definition Essence ("_Ess._") where " $\varphi \text{ Ess. } x \equiv \forall\psi. \psi x \supset (\varphi \supset_N \psi)$ "
```

Fig. 6 Gödel's axioms and definitions, as presented in the 1970 manuscript, are inconsistent

Gödel 1

Axiom Ax2b: $P \varphi \supset \Box P \varphi$ and $\neg P \varphi \supset \Box(\neg P \varphi)$

```
1 theory GoedelVariantHOML1 imports HOMLinHOL
2 begin
3   consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4   axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5   axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6   definition God ("G") where "G x  $\equiv$   $\forall\varphi$ . P  $\varphi$   $\supset$   $\varphi$  x"
7   abbreviation PropertyInclusion ("_ $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\forall^E y. \varphi y \supset \psi y)$ "
8   definition Essence ("_Ess._") where " $\varphi$  Ess. x  $\equiv \forall\psi$ .  $\psi$  x  $\supset$  ( $\varphi \supset_N \psi$ )"
9   axiomatization where Ax2b: "[P  $\varphi$   $\supset$   $\Box$  P  $\varphi$ ]"
```

Fig. 6 Gödel's axioms and definitions, as presented in the 1970 manuscript, are inconsistent

Gödel 1

Axiom Ax2b: $P \varphi \supset \Box P \varphi$ and $\neg P \varphi \supset \Box(\neg P \varphi)$

```
1 theory GoedelVariantHOML1 imports HOMLinHOL
2 begin
3 consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4 axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5 axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6 definition God ("G") where "G x  $\equiv$   $\forall\varphi$ . P  $\varphi$   $\supset$   $\varphi$  x"
7 abbreviation PropertyInclusion ("_ $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\forall^E y. \varphi y \supset \psi y)$ "
8 definition Essence ("_Ess._") where " $\varphi$  Ess. x  $\equiv$   $\forall\psi$ .  $\psi$  x  $\supset$  ( $\varphi \supset_N \psi$ )"
9 axiomatization where Ax2b: "[P  $\varphi$   $\supset$   $\Box$  P  $\varphi$ ]"
10 lemma Ax2b': "[ $\neg$ P  $\varphi$   $\supset$   $\Box(\neg$ P  $\varphi$ )]" using Ax2a Ax2b by blast
```

Fig. 6 Gödel's axioms and definitions, as presented in the 1970 manuscript, are inconsistent

Gödel 1

Theorem Th1: $G x \supset G \text{Ess. } x$

```
1 theory GoedelVariantHOML1 imports HOMLinHOL
2 begin
3 consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4 axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi \supset$  P ( $\varphi . \psi$ )]"
5 axiomatization where Ax2a: "[P  $\varphi \vee^e$  P  $\sim\varphi$ ]"
6 definition God ("G") where "G x  $\equiv \forall\varphi. P \varphi \supset \varphi x$ "
7 abbreviation PropertyInclusion ("_ $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\forall^e y. \varphi y \supset \psi y)$ "
8 definition Essence ("_Ess._") where " $\varphi \text{Ess. } x \equiv \forall\psi. \psi x \supset (\varphi \supset_N \psi)$ "
9 axiomatization where Ax2b: "[P  $\varphi \supset \Box P \varphi$ ]"
10 lemma Ax2b': "[ $\neg P \varphi \supset \Box(\neg P \varphi)$ ]" using Ax2a Ax2b by blast
11 theorem Th1: "[G x  $\supset$  G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
```

Fig. 6 Gödel's axioms and definitions, as presented in the 1970 manuscript, are inconsistent

Gödel 1

Definition of Necessary Existence: $E x \equiv \forall \varphi. \varphi \text{ Ess. } x \supset \Box(\exists^E x. \varphi x)$

Axiom Ax3: $P E$

```
1 theory GoedelVariantHOML1 imports HOMLinHOL
2 begin
3   consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4   axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5   axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6   definition God ("G") where "G x  $\equiv$   $\forall\varphi. P \varphi \supset \varphi x$ "
7   abbreviation PropertyInclusion ("_ $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\forall^E y. \varphi y \supset \psi y)$ "
8   definition Essence ("_Ess._") where " $\varphi \text{ Ess. } x \equiv \forall\psi. \psi x \supset (\varphi \supset_N \psi)$ "
9   axiomatization where Ax2b: "[P  $\varphi \supset \Box P \varphi$ ]"
10  lemma Ax2b': "[ $\neg P \varphi \supset \Box(\neg P \varphi)$ ]" using Ax2a Ax2b by blast
11  theorem Th1: "[G x  $\supset$  G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12  definition NecExist ("E") where "E x  $\equiv \forall\varphi. \varphi \text{ Ess. } x \supset \Box(\exists^E x. \varphi x)$ "
13  axiomatization where Ax3: "[P E]"
```

Fig. 6 Gödel's axioms and definitions, as presented in the 1970 manuscript, are inconsistent

Gödel 1

Theorem Th2: $G x \supset \Box(\exists^E y. G y)$

```
1 theory GoedelVariantHOML1 imports HOMLinHOL
2 begin
3   consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4   axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi \supset$  P ( $\varphi . \psi$ )]"
5   axiomatization where Ax2a: "[P  $\varphi \vee^e$  P  $\sim \varphi$ ]"
6   definition God ("G") where "G x  $\equiv \forall \varphi. P \varphi \supset \varphi x$ "
7   abbreviation PropertyInclusion ("_ $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\forall^E y. \varphi y \supset \psi y)$ "
8   definition Essence ("_Ess._") where " $\varphi$  Ess. x  $\equiv \forall \psi. \psi x \supset (\varphi \supset_N \psi)$ "
9   axiomatization where Ax2b: "[P  $\varphi \supset \Box P \varphi$ ]"
10  lemma Ax2b': "[ $\neg P \varphi \supset \Box(\neg P \varphi)$ ]" using Ax2a Ax2b by blast
11  theorem Th1: "[G x  $\supset$  G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12  definition NecExist ("E") where "E x  $\equiv \forall \varphi. \varphi$  Ess. x  $\supset \Box(\exists^E x. \varphi x)$ "
13  axiomatization where Ax3: "[P E]"
14  theorem Th2: "[G x  $\supset \Box(\exists^E y. G y)$ ]" using Ax3 Th1 God_def NecExist_def by smt
```

Fig. 6 Gödel's axioms and definitions, as presented in the 1970 manuscript, are inconsistent

Gödel 1

Theorem Th3: $\Diamond(\exists^E x. G x) \supset \Box(\exists^E y. G y)$

Logic KB (Rsymm) needed here

```
1 theory GoedelVariantHOML1 imports HOMLinHOL
2 begin
3   consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4   axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5   axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6   definition God ("G") where "G x  $\equiv$   $\forall\varphi. P \varphi \supset \varphi x$ "
7   abbreviation PropertyInclusion ("_ $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\forall^E y. \varphi y \supset \psi y)$ "
8   definition Essence ("_Ess._") where " $\varphi$  Ess. x  $\equiv$   $\forall\psi. \psi x \supset (\varphi \supset_N \psi)$ "
9   axiomatization where Ax2b: "[P  $\varphi \supset \Box P \varphi$ ]"
10  lemma Ax2b': "[ $\neg P \varphi \supset \Box(\neg P \varphi)$ ]" using Ax2a Ax2b by blast
11  theorem Th1: "[G x  $\supset$  G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12  definition NecExist ("E") where "E x  $\equiv$   $\forall\varphi. \varphi$  Ess. x  $\supset \Box(\exists^E x. \varphi x)$ "
13  axiomatization where Ax3: "[P E]"
14  theorem Th2: "[G x  $\supset \Box(\exists^E y. G y)$ ]" using Ax3 Th1 God_def NecExist_def by smt
15  theorem Th3: "[ $\Diamond(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" sledgehammer(Th2 Rsymm) —<Proof found>
16  proof -
17    have 1: "[ $(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" using Th2 by blast
18    have 2: "[ $\Diamond(\exists^E x. G x) \supset \Diamond\Box(\exists^E y. G y)$ ]" using 1 by blast
19    have 3: "[ $\Diamond(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" using 2 Rsymm by blast
20    thus ?thesis by blast
21  qed
```

Fig. 6 Gödel's axioms and definitions, as presented in the 1970 manuscript, are inconsistent

Gödel 1

Axiom Ax4: $P \varphi \wedge (\varphi \supset_N \psi) \supset P \psi$

```
1 theory GoedelVariantHOML1 imports HOMLinHOL
2 begin
3   consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4   axiomatization where Ax1: "[P  $\varphi \wedge P \psi \supset P (\varphi \cdot \psi)$ ]"
5   axiomatization where Ax2a: "[P  $\varphi \vee^e P \sim \varphi$ ]"
6   definition God ("G") where "G x  $\equiv \forall \varphi. P \varphi \supset \varphi x$ "
7   abbreviation PropertyInclusion ("_ $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\forall^E y. \varphi y \supset \psi y)$ "
8   definition Essence ("_Ess._") where " $\varphi$  Ess. x  $\equiv \forall \psi. \psi x \supset (\varphi \supset_N \psi)$ "
9   axiomatization where Ax2b: "[P  $\varphi \supset \Box P \varphi$ ]"
10  lemma Ax2b': "[ $\neg P \varphi \supset \Box(\neg P \varphi)$ ]" using Ax2a Ax2b by blast
11  theorem Th1: "[G x  $\supset$  G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12  definition NecExist ("E") where "E x  $\equiv \forall \varphi. \varphi$  Ess. x  $\supset \Box(\exists^E x. \varphi x)$ "
13  axiomatization where Ax3: "[P E]"
14  theorem Th2: "[G x  $\supset \Box(\exists^E y. G y)$ ]" using Ax3 Th1 God_def NecExist_def by smt
15  theorem Th3: "[ $\Diamond(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" sledgehammer(Th2 Rsymm) —<Proof found>
16  proof -
17    have 1: "[ $(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" using Th2 by blast
18    have 2: "[ $\Diamond(\exists^E x. G x) \supset \Diamond \Box(\exists^E y. G y)$ ]" using 1 by blast
19    have 3: "[ $\Diamond(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" using 2 Rsymm by blast
20    thus ?thesis by blast
21  qed
22  axiomatization where Ax4: "[P  $\varphi \wedge (\varphi \supset_N \psi) \supset P \psi$ ]"
```

Fig. 6 Gödel's axioms and definitions, as presented in the 1970 manuscript, are inconsistent

Gödel 1

No satisfying model found by nitpick!!!

```
1 theory GoedelVariantHOML1 imports HOMLinHOL
2 begin
3   consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4   axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5   axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6   definition God ("G") where "G x  $\equiv$   $\forall\varphi$ . P  $\varphi$   $\supset$   $\varphi$  x"
7   abbreviation PropertyInclusion ("_ $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\forall^e y. \varphi y \supset \psi y)$ "
8   definition Essence ("_Ess._") where " $\varphi$  Ess. x  $\equiv$   $\forall\psi$ .  $\psi$  x  $\supset$  ( $\varphi \supset_N \psi$ )"
9   axiomatization where Ax2b: "[P  $\varphi$   $\supset$   $\Box$  P  $\varphi$ ]"
10  lemma Ax2b': "[ $\neg$ P  $\varphi$   $\supset$   $\Box(\neg$ P  $\varphi$ )]" using Ax2a Ax2b by blast
11  theorem Th1: "[G x  $\supset$  G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12  definition NecExist ("E") where "E x  $\equiv$   $\forall\varphi$ .  $\varphi$  Ess. x  $\supset$   $\Box(\exists^e x. \varphi x)$ "
13  axiomatization where Ax3: "[P E]"
14  theorem Th2: "[G x  $\supset$   $\Box(\exists^e y. G y)$ ]" using Ax3 Th1 God_def NecExist_def by smt
15  theorem Th3: "[ $\Diamond(\exists^e x. G x) \supset \Box(\exists^e y. G y)$ ]" sledgehammer(Th2 Rsymm) —<Proof found>
16  proof -
17    have 1: "[ $(\exists^e x. G x) \supset \Box(\exists^e y. G y)$ ]" using Th2 by blast
18    have 2: "[ $\Diamond(\exists^e x. G x) \supset \Diamond\Box(\exists^e y. G y)$ ]" using 1 by blast
19    have 3: "[ $\Diamond(\exists^e x. G x) \supset \Box(\exists^e y. G y)$ ]" using 2 Rsymm by blast
20    thus ?thesis by blast
21  qed
22  axiomatization where Ax4: "[P  $\varphi$   $\wedge$  ( $\varphi \supset_N \psi$ )  $\supset$  P  $\psi$ ]"
23  lemma True nitpick[satisfy] oops —<No model found>
```

Fig. 6 Gödel's axioms and definitions, as presented in the 1970 manuscript, are inconsistent

Gödel 1

Inconsistency detected
by theorem provers

The inconsistency follows from what
is called in own prior work the “empty
essence lemma”:

$$(\lambda x. \perp) \text{Ess. } x.$$

This lemma states that the absurd,
empty property (or, alternatively, the
extensionally equivalent property of
self-difference), which does not apply
to any x , is nevertheless an essential
property of every x .

```
1 theory GoedelVariantHOML1 imports HOMLinHOL
2 begin
3 consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4 axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5 axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6 definition God ("G") where "G x  $\equiv$   $\forall\varphi$ . P  $\varphi$   $\supset$   $\varphi$  x"
7 abbreviation PropertyInclusion (" $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\forall^e y. \varphi y \supset \psi y)$ "
8 definition Essence ("_Ess._") where " $\varphi$  Ess. x  $\equiv$   $\forall\psi$ .  $\psi$  x  $\supset$  ( $\varphi \supset_N \psi$ )"
9 axiomatization where Ax2b: "[P  $\varphi$   $\supset$   $\Box$  P  $\varphi$ ]"
10 lemma Ax2b': "[ $\neg$ P  $\varphi$   $\supset$   $\Box(\neg$ P  $\varphi$ )]" using Ax2a Ax2b by blast
11 theorem Th1: "[G x  $\supset$  G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12 definition NecExist ("E") where "E x  $\equiv$   $\forall\varphi$ .  $\varphi$  Ess. x  $\supset$   $\Box(\exists^e x. \varphi x)$ "
13 axiomatization where Ax3: "[P E]"
14 theorem Th2: "[G x  $\supset$   $\Box(\exists^e y. G y)$ ]" using Ax3 Th1 God_def NecExist_def by smt
15 theorem Th3: "[ $\Diamond(\exists^e x. G x) \supset \Box(\exists^e y. G y)$ ]" sledgehammer(Th2 Rsymm) —<Proof found>
16 proof -
17   have 1: "[ $(\exists^e x. G x) \supset \Box(\exists^e y. G y)$ ]" using Th2 by blast
18   have 2: "[ $\Diamond(\exists^e x. G x) \supset \Diamond\Box(\exists^e y. G y)$ ]" using 1 by blast
19   have 3: "[ $\Diamond(\exists^e x. G x) \supset \Box(\exists^e y. G y)$ ]" using 2 Rsymm by blast
20   thus ?thesis by blast
21 qed
22 axiomatization where Ax4: "[P  $\varphi$   $\wedge$  ( $\varphi \supset_N \psi$ )  $\supset$  P  $\psi$ ]"
23 lemma True nitpick[satisfy] oops —<No model found>
24 lemma EmptyEssL: "[ $(\lambda y. \perp)$  Ess. x]" using Essence_def by auto
25 theorem Inconsistency: False sledgehammer(Ax2a Ax3 Ax4 EmptyEssL NecExist_def) —<Proof found>
26 proof -
27   have 1: "[ $\neg$ (P ( $\lambda x. \perp$ ))]" using Ax2a Ax4 by blast
28   have 2: "[P ( $\lambda x. (\lambda y. \perp)$  Ess. x)  $\supset$   $\Box(\exists^e z. (\lambda y. \perp) z)$ ]" using Ax3 Ax4 NecExist_def by smt
29   have 3: "[P ( $\lambda x. \Box(\exists^e z. (\lambda x. \perp) z)$ )]" using 2 EmptyEssL by simp
30   have 4: "[P ( $\lambda x. \Box\perp$ )]" using 3 by auto
31   have 5: "[P ( $\lambda x. \perp$ )]" using 4 Ax2a Ax4 by smt
32   have 6: "[ $\perp$ ]" using 1 5 by blast
33   thus ?thesis by blast
34 qed
35 end
```

Fig. 6 Gödel’s axioms and definitions, as presented in the 1970 manuscript, are inconsistent

Formalisation of
Gödel's 1970 manuscript
of the modal ontological argument
(adapted notion of essence: Gödel2)

— proof in logic KB —

Gödel 2

Same as before in Gödel 1

```
1 theory GoedelVariantHOML2 imports HOMLinHOL ModalFilter
2 begin
3 consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4 axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi \supset$  P ( $\varphi . \psi$ )]"
5 axiomatization where Ax2a: "[P  $\varphi \vee^e$  P  $\sim\varphi$ ]"
6 definition God ("G") where "G x  $\equiv \forall\varphi. P \varphi \supset \varphi x$ "
7 abbreviation PropertyInclusion ("_ $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\forall^ey. \varphi y \supset \psi y)$ "
```

Fig. 7 After an appropriate modification of the definition of essence in Gödel's 1970 ontological proof, the inconsistency revealed in Fig. 6 is avoided, and the argument can be successfully verified in modal logic S5 (indeed, as shown, only symmetry of the accessibility relation is actually needed)

Gödel 2

Modified notion of essence:

$$\varphi \text{ Ess. } x \equiv \varphi x \wedge \forall \psi. \psi x \supset (\varphi \supset_N \psi))$$

```
1 theory GoedelVariantHOML2 imports HOMLinHOL ModalFilter
2 begin
3 consts PositiveProperty::"(e⇒σ)⇒σ" ("P")
4 axiomatization where Ax1: "[P φ ∧ P ψ ⊃ P (φ . ψ)]"
5 axiomatization where Ax2a: "[P φ Ve P ~φ]"
6 definition God ("G") where "G x ≡ ∀φ. P φ ⊃ φ x"
7 abbreviation PropertyInclusion ("_⊃_N_") where "φ ⊃_N ψ ≡ □(∀ey. φ y ⊃ ψ y)"
8 (**) definition Essence ("_Ess._") where "φ Ess. x ≡ φ x ∧ (∀ψ. ψ x ⊃ (φ ⊃_N ψ))"
```

Fig. 7 After an appropriate modification of the definition of essence in Gödel’s 1970 ontological proof, the inconsistency revealed in Fig. 6 is avoided, and the argument can be successfully verified in modal logic S5 (indeed, as shown, only symmetry of the accessibility relation is actually needed)

Gödel 2

Same as before in Gödel 1

```
1 theory GoedelVariantHOML2 imports HOMLinHOL ModalFilter
2 begin
3 consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4 axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5 axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6 definition God ("G") where "G x  $\equiv$   $\forall\varphi$ . P  $\varphi$   $\supset$   $\varphi$  x"
7 abbreviation PropertyInclusion ("_ $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\forall^ey. \varphi y \supset \psi y)$ "
8 (**) definition Essence ("_Ess._") where " $\varphi$  Ess. x  $\equiv$   $\varphi$  x  $\wedge$  ( $\forall\psi$ .  $\psi$  x  $\supset$  ( $\varphi \supset_N \psi$ ))"
9 axiomatization where Ax2b: "[P  $\varphi$   $\supset$   $\Box$  P  $\varphi$ ]"
10 lemma Ax2b': "[ $\neg$ P  $\varphi$   $\supset$   $\Box(\neg$ P  $\varphi$ )]" using Ax2a Ax2b by blast
11 theorem Th1: "[G x  $\supset$  G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12 definition NecExist ("E") where "E x  $\equiv$   $\forall\varphi$ .  $\varphi$  Ess. x  $\supset$   $\Box(\exists^ex. \varphi x)$ "
13 axiomatization where Ax3: "[P E]"
14 theorem Th2: "[G x  $\supset$   $\Box(\exists^ey. G y)$ ]" using Ax3 Th1 God_def NecExist_def by smt
15 theorem Th3: "[ $\Diamond(\exists^ex. G x) \supset \Box(\exists^ey. G y)$ ]" sledgehammer(Th2 Rsymm) —<Proof found>
16 proof -
17   have 1: "[ $(\exists^ex. G x) \supset \Box(\exists^ey. G y)$ ]" using Th2 by blast
18   have 2: "[ $\Diamond(\exists^ex. G x) \supset \Diamond\Box(\exists^ey. G y)$ ]" using 1 by blast
19   have 3: "[ $\Diamond(\exists^ex. G x) \supset \Box(\exists^ey. G y)$ ]" using 2 Rsymm by blast
20   thus ?thesis by blast
21 qed
22 axiomatization where Ax4: "[P  $\varphi \wedge (\varphi \supset_N \psi) \supset P \psi$ ]"
```

Fig. 7 After an appropriate modification of the definition of essence in Gödel's 1970 ontological proof, the inconsistency revealed in Fig. 6 is avoided, and the argument can be successfully verified in modal logic S5 (indeed, as shown, only symmetry of the accessibility relation is actually needed)

Gödel 2

But now nitpick finds a model!

```
1 theory GoedelVariantHOML2 imports HOMLinHOL ModalFilter
2 begin
3 consts PositiveProperty::"(e⇒σ)⇒σ" ("P")
4 axiomatization where Ax1: "[P φ ∧ P ψ ⊃ P (φ . ψ)]"
5 axiomatization where Ax2a: "[P φ ∨e P ~φ]"
6 definition God ("G") where "G x ≡ ∀φ. P φ ⊃ φ x"
7 abbreviation PropertyInclusion ("_⊃N_") where "φ ⊃N ψ ≡ □(∀ey. φ y ⊃ ψ y)"
8 (**) definition Essence ("_Ess._") where "φ Ess. x ≡ φ x ∧ (∀ψ. ψ x ⊃ (φ ⊃N ψ))"
9 axiomatization where Ax2b: "[P φ ⊃ □ P φ]"
10 lemma Ax2b': "[¬P φ ⊃ □(¬P φ)]" using Ax2a Ax2b by blast
11 theorem Th1: "[G x ⊃ G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12 definition NecExist ("E") where "E x ≡ ∀φ. φ Ess. x ⊃ □(∃ex. φ x)"
13 axiomatization where Ax3: "[P E]"
14 theorem Th2: "[G x ⊃ □(∃ey. G y)]" using Ax3 Th1 God_def NecExist_def by smt
15 theorem Th3: "[◇(∃ex. G x) ⊃ □(∃ey. G y)]" sledgehammer(Th2 Rsymm) —<Proof found>
16 proof -
17   have 1: "[(∃ex. G x) ⊃ □(∃ey. G y)]" using Th2 by blast
18   have 2: "[◇(∃ex. G x) ⊃ ◇□(∃ey. G y)]" using 1 by blast
19   have 3: "[◇(∃ex. G x) ⊃ □(∃ey. G y)]" using 2 Rsymm by blast
20   thus ?thesis by blast
21 qed
22 axiomatization where Ax4: "[P φ ∧ (φ ⊃N ψ) ⊃ P ψ]"
23 lemma True nitpick[satisfy,card=1,eval="[P (λx. ⊥)]"] oops —<One model found of cardinality one>
```

Nitpicking formula...
Nitpick found a model for card e = 1 and card i = 1:
Evaluated term:
[P (λx. ⊥)] = False
Constants:
P = (λx. _)((λx. _)((e₁, i₁) := True), i₁) := True, ((λx. _)((e₁, i₁) := False), i₁) := False)
R = (λx. _)((i₁, i₁) := True)
existsAt = (λx. _)((e₁, i₁) := True)

Fig. 7 After an appropriate modification of the definition of essence in Gödel’s 1970 ontological proof, the inconsistency revealed in Fig. 6 is avoided, and the argument can be successfully verified in modal logic S5 (indeed, as shown, only symmetry of the accessibility relation is actually needed)

Gödel 2

But now nitpick finds a model!

Unfortunately, however, we do not succeed in proving

Lemma L: $P \rightarrow G$

or

Theorem Th4: $\Diamond(\exists^E x. G x)$

```
1 theory GoedelVariantHOML2 imports HOMLinHOL ModalFilter
2 begin
3 consts PositiveProperty::"(e⇒σ)⇒σ" ("P")
4 axiomatization where Ax1: "[P φ ∧ P ψ ⊃ P (φ . ψ)]"
5 axiomatization where Ax2a: "[P φ ∨e P ~φ]"
6 definition God ("G") where "G x ≡ ∀φ. P φ ⊃ φ x"
7 abbreviation PropertyInclusion ("_⊃N_") where "φ ⊃N ψ ≡ □(∀Ey. φ y ⊃ ψ y)"
8 (**) definition Essence ("_Ess._") where "φ Ess. x ≡ φ x ∧ (∀ψ. ψ x ⊃ (φ ⊃N ψ))"
9 axiomatization where Ax2b: "[P φ ⊃ □ P φ]"
10 lemma Ax2b': "[¬P φ ⊃ □(¬P φ)]" using Ax2a Ax2b by blast
11 theorem Th1: "[G x ⊃ G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12 definition NecExist ("E") where "E x ≡ ∀φ. φ Ess. x ⊃ □(∃Ex. φ x)"
13 axiomatization where Ax3: "[P E]"
14 theorem Th2: "[G x ⊃ □(∃Ey. G y)]" using Ax3 Th1 God_def NecExist_def by smt
15 theorem Th3: "[◇(∃Ex. G x) ⊃ □(∃Ey. G y)]" sledgehammer(Th2 Rsymm) —<Proof found>
16 proof -
17   have 1: "[ (∃Ex. G x) ⊃ □(∃Ey. G y)]" using Th2 by blast
18   have 2: "[◇(∃Ex. G x) ⊃ ◇□(∃Ey. G y)]" using 1 by blast
19   have 3: "[◇(∃Ex. G x) ⊃ □(∃Ey. G y)]" using 2 Rsymm by blast
20   thus ?thesis by blast
21 qed
22 axiomatization where Ax4: "[P φ ∧ (φ ⊃N ψ) ⊃ P ψ]"
23 lemma True nitpick[satisfy,card=1,eval="[P (λx.⊥)]"] oops —<One model found of cardinality one>
```

Fig. 7 After an appropriate modification of the definition of essence in Gödel’s 1970 ontological proof, the inconsistency revealed in Fig. 6 is avoided, and the argument can be successfully verified in modal logic S5 (indeed, as shown, only symmetry of the accessibility relation is actually needed)

Gödel 2

Generalization of Ax1 needed; remember Gödel's footnote:

• and for any number of minimality

Axiom Ax1Gen: $(PosProps \Phi \wedge ConjOfPropsFrom \varphi \Phi) \supset P \varphi$

```
1 theory GoedelVariantHOML2 imports HOMLinHOL ModalFilter
2 begin
3 consts PositiveProperty::"(e⇒σ)⇒σ" ("P")
4 axiomatization where Ax1: "[P φ ∧ P ψ ⊃ P (φ . ψ)]"
5 axiomatization where Ax2a: "[P φ ∨e P ~φ]"
6 definition God ("G") where "G x ≡ ∀φ. P φ ⊃ φ x"
7 abbreviation PropertyInclusion ("_⊃N_") where "φ ⊃N ψ ≡ □(∀ey. φ y ⊃ ψ y)"
8 (**) definition Essence ("_Ess._") where "φ Ess. x ≡ φ x ∧ (∀ψ. ψ x ⊃ (φ ⊃N ψ))"
9 axiomatization where Ax2b: "[P φ ⊃ □ P φ]"
10 lemma Ax2b': "[¬P φ ⊃ □(¬P φ)]" using Ax2a Ax2b by blast
11 theorem Th1: "[G x ⊃ G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12 definition NecExist ("E") where "E x ≡ ∀φ. φ Ess. x ⊃ □(∃ex. φ x)"
13 axiomatization where Ax3: "[P E]"
14 theorem Th2: "[G x ⊃ □(∃ey. G y)]" using Ax3 Th1 God_def NecExist_def by smt
15 theorem Th3: "[◇(∃ex. G x) ⊃ □(∃ey. G y)]" sledgehammer(Th2 Rsymm) —<Proof found>
16 proof -
17   have 1: "[ (∃ex. G x) ⊃ □(∃ey. G y)]" using Th2 by blast
18   have 2: "[◇(∃ex. G x) ⊃ ◇□(∃ey. G y)]" using 1 by blast
19   have 3: "[◇(∃ex. G x) ⊃ □(∃ey. G y)]" using 2 Rsymm by blast
20   thus ?thesis by blast
21 qed
22 axiomatization where Ax4: "[P φ ∧ (φ ⊃N ψ) ⊃ P ψ]"
23 lemma True nitpick[satisfy,card=1,eval="[P (λx.⊥)]"] oops —<One model found of cardinality one>
24 abbreviation "PosProps Φ ≡ ∀φ. Φ φ ⊃ P φ"
25 abbreviation "ConjOfPropsFrom φ Φ ≡ □(∀ez. φ z ↔ (∀ψ. Φ ψ ⊃ ψ z))"
26 axiomatization where Ax1Gen: "[ (PosProps Φ ∧ ConjOfPropsFrom φ Φ) ⊃ P φ]"
```

Fig. 7 After an appropriate modification of the definition of essence in Gödel's 1970 ontological proof, the inconsistency revealed in Fig. 6 is avoided, and the argument can be successfully verified in modal logic S5 (indeed, as shown, only symmetry of the accessibility relation is actually needed)

See [Anderson & Gettings 1996, p. 170]; a related proposal can be found in Gödel's Nachlass; see [Kanckos & Lethen 2021, p. 1019]

Gödel 2

Generalization of Ax1 needed; remember Gödel's footnote:

• and for any number of minimality

Axiom Ax1Gen: $(PosProps \Phi \wedge ConjOfPropsFrom \varphi \Phi) \supset P \varphi$

... then we can prove:

Lemma L: $P G$ and **Theorem Th4:** $\Diamond(\exists^E x. G x)$

```
1 theory GoedelVariantHOML2 imports HOMLinHOL ModalFilter
2 begin
3 consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4 axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5 axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6 definition God ("G") where "G x  $\equiv$   $\forall\varphi. P \varphi \supset \varphi x$ "
7 abbreviation PropertyInclusion ("_ $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\forall y. \varphi y \supset \psi y)$ "
8 (**) definition Essence ("_Ess._") where " $\varphi$  Ess. x  $\equiv$   $\varphi x$   $\wedge (\forall\psi. \psi x \supset (\varphi \supset_N \psi))$ "
9 axiomatization where Ax2b: "[P  $\varphi \supset \Box P \varphi$ ]"
10 lemma Ax2b': "[ $\neg P \varphi \supset \Box(\neg P \varphi)$ ]" using Ax2a Ax2b by blast
11 theorem Th1: "[G x  $\supset$  G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12 definition NecExist ("E") where "E x  $\equiv$   $\forall\varphi. \varphi$  Ess. x  $\supset \Box(\exists^E x. \varphi x)$ "
13 axiomatization where Ax3: "[P E]"
14 theorem Th2: "[G x  $\supset \Box(\exists^E y. G y)$ ]" using Ax3 Th1 God_def NecExist_def by smt
15 theorem Th3: "[ $\Diamond(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" sledgehammer(Th2 Rsymm) —<Proof found>
16 proof -
17   have 1: "[ $(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" using Th2 by blast
18   have 2: "[ $\Diamond(\exists^E x. G x) \supset \Diamond\Box(\exists^E y. G y)$ ]" using 1 by blast
19   have 3: "[ $\Diamond(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" using 2 Rsymm by blast
20   thus ?thesis by blast
21 qed
22 axiomatization where Ax4: "[P  $\varphi \wedge (\varphi \supset_N \psi) \supset P \psi$ ]"
23 lemma True nitpick[satisfy,card=1,eval="[P ( $\lambda x. \bot$ )]" ] oops —<One model found of cardinality one>
24 abbreviation "PosProps  $\Phi \equiv \forall\varphi. \Phi \varphi \supset P \varphi$ "
25 abbreviation "ConjOfPropsFrom  $\varphi \Phi \equiv \Box(\forall z. \varphi z \leftrightarrow (\forall\psi. \Phi \psi \supset \psi z))$ "
26 axiomatization where Ax1Gen: "[ $(PosProps \Phi \wedge ConjOfPropsFrom \varphi \Phi) \supset P \varphi$ ]"
27 lemma L: "[P G]" using Ax1Gen God_def by smt
28 theorem Th4: "[ $\Diamond(\exists^E x. G x)$ ]" using Ax2a Ax4 L by blast
```

Fig. 7 After an appropriate modification of the definition of essence in Gödel's 1970 ontological proof, the inconsistency revealed in Fig. 6 is avoided, and the argument can be successfully verified in modal logic S5 (indeed, as shown, only symmetry of the accessibility relation is actually needed)

Gödel 2

So we are done:

Theorem Th5: $\Box(\exists^E x. G x)$

Note that this proof works in modal logic **KB**, since only symmetry (Rsymm) of the accessibility between worlds was used (in the proof of theorem Th3).

```
1 theory GoedelVariantHOML2 imports HOMLinHOL ModalFilter
2 begin
3 consts PositiveProperty::"(e⇒σ)⇒σ" ("P")
4 axiomatization where Ax1: "[P φ ∧ P ψ ⊃ P (φ . ψ)]"
5 axiomatization where Ax2a: "[P φ ∨e P ~φ]"
6 definition God ("G") where "G x ≡ ∀φ. P φ ⊃ φ x"
7 abbreviation PropertyInclusion ("_⊃N_") where "φ ⊃N ψ ≡ □(∀Ey. φ y ⊃ ψ y)"
8 (**) definition Essence ("_Ess._") where "φ Ess. x ≡ φ x ∧ (∀ψ. ψ x ⊃ (φ ⊃N ψ))"
9 axiomatization where Ax2b: "[P φ ⊃ □ P φ]"
10 lemma Ax2b': "[¬P φ ⊃ □(¬P φ)]" using Ax2a Ax2b by blast
11 theorem Th1: "[G x ⊃ G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12 definition NecExist ("E") where "E x ≡ ∀φ. φ Ess. x ⊃ □(∃Ex. φ x)"
13 axiomatization where Ax3: "[P E]"
14 theorem Th2: "[G x ⊃ □(∃Ey. G y)]" using Ax3 Th1 God_def NecExist_def by smt
15 theorem Th3: "[◇(∃Ex. G x) ⊃ □(∃Ey. G y)]" sledgehammer(Th2 Rsymm) —<Proof found>
16 proof -
17   have 1: "[ (∃Ex. G x) ⊃ □(∃Ey. G y)]" using Th2 by blast
18   have 2: "[ ◇(∃Ex. G x) ⊃ ◇□(∃Ey. G y)]" using 1 by blast
19   have 3: "[ ◇(∃Ex. G x) ⊃ □(∃Ey. G y)]" using 2 Rsymm by blast
20   thus ?thesis by blast
21 qed
22 axiomatization where Ax4: "[P φ ∧ (φ ⊃N ψ) ⊃ P ψ]"
23 lemma True nitpick[satisfy,card=1,eval="[P (λx.⊥)]"] oops —<One model found of cardinality one>
24 abbreviation "PosProps Φ ≡ ∀φ. Φ φ ⊃ P φ"
25 abbreviation "ConjOfPropsFrom φ Φ ≡ □(∀Ez. φ z ↔ (∀ψ. Φ ψ ⊃ ψ z))"
26 axiomatization where Ax1Gen: "[ (PosProps Φ ∧ ConjOfPropsFrom φ Φ) ⊃ P φ]"
27 lemma L: "[P G]" using Ax1Gen God_def by smt
28 theorem Th4: "[ ◇(∃Ex. G x)]" using Ax2a Ax4 L by blast
29 theorem Th5: "[ □(∃Ex. G x)]" using Th3 Th4 by blast
```

Fig. 7 After an appropriate modification of the definition of essence in Gödel’s 1970 ontological proof, the inconsistency revealed in Fig. 6 is avoided, and the argument can be successfully verified in modal logic S5 (indeed, as shown, only symmetry of the accessibility relation is actually needed)

Gödel 2

Modal Collapse (what holds that holds necessarily) follows:

Modal Collapse MC: $\varphi \supset \Box\varphi$

This proof works again in modal logic **KB**.

```
1 theory GoedelVariantHOML2 imports HOMLinHOL ModalFilter
2 begin
3 consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4 axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5 axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6 definition God ("G") where "G x  $\equiv$   $\forall\varphi$ . P  $\varphi$   $\supset$   $\varphi$  x"
7 abbreviation PropertyInclusion ("_ $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\forall y. \varphi y \supset \psi y)$ "
8 (**) definition Essence ("_Ess._") where " $\varphi$  Ess. x  $\equiv$   $\varphi$  x  $\wedge$  ( $\forall\psi$ .  $\psi$  x  $\supset$  ( $\varphi \supset_N \psi$ ))"
9 axiomatization where Ax2b: "[P  $\varphi$   $\supset$   $\Box$  P  $\varphi$ ]"
10 lemma Ax2b': "[ $\neg$ P  $\varphi$   $\supset$   $\Box(\neg$ P  $\varphi$ )]" using Ax2a Ax2b by blast
11 theorem Th1: "[G x  $\supset$  G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12 definition NecExist ("E") where "E x  $\equiv$   $\forall\varphi$ .  $\varphi$  Ess. x  $\supset$   $\Box(\exists^E x. \varphi x)$ "
13 axiomatization where Ax3: "[P E]"
14 theorem Th2: "[G x  $\supset$   $\Box(\exists^E y. G y)]$ " using Ax3 Th1 God_def NecExist_def by smt
15 theorem Th3: "[ $\Diamond(\exists^E x. G x) \supset \Box(\exists^E y. G y)]$ " sledgehammer(Th2 Rsymm) —<Proof found>
16 proof -
17   have 1: "[ $(\exists^E x. G x) \supset \Box(\exists^E y. G y)]$ " using Th2 by blast
18   have 2: "[ $\Diamond(\exists^E x. G x) \supset \Diamond\Box(\exists^E y. G y)]$ " using 1 by blast
19   have 3: "[ $\Diamond(\exists^E x. G x) \supset \Box(\exists^E y. G y)]$ " using 2 Rsymm by blast
20   thus ?thesis by blast
21 qed
22 axiomatization where Ax4: "[P  $\varphi$   $\wedge$  ( $\varphi \supset_N \psi$ )  $\supset$  P  $\psi$ ]"
23 lemma True nitpick[satisfy,card=1,eval="[P ( $\lambda x. \bot$ )]"] oops —<One model found of cardinality one>
24 abbreviation "PosProps  $\Phi \equiv \forall\varphi$ .  $\Phi \varphi \supset P \varphi$ "
25 abbreviation "ConjOfPropsFrom  $\varphi \Phi \equiv \Box(\forall z. \varphi z \leftrightarrow (\forall\psi$ .  $\Phi \psi \supset \psi z))$ "
26 axiomatization where Ax1Gen: "[ (PosProps  $\Phi$   $\wedge$  ConjOfPropsFrom  $\varphi \Phi$ )  $\supset$  P  $\varphi$  ]"
27 lemma L: "[P G]" using Ax1Gen God_def by smt
28 theorem Th4: "[ $\Diamond(\exists^E x. G x)]$ " using Ax2a Ax4 L by blast
29 theorem Th5: "[ $\Box(\exists^E x. G x)]$ " using Th3 Th4 by blast
30 lemma MC: "[ $\varphi \supset \Box\varphi$ ]" sledgehammer(Ax2a Ax2b Th5 God_def Rsymm) —<Proof found>
31 proof - {fix w fix Q
32   have 1: " $\forall x. (G x w \longrightarrow (\forall z. Z x \supset \Box(\forall z. G z \supset Z z)) w)$ " using Ax2a Ax2b God_def by smt
33   have 2: " $(\exists x. G x w) \longrightarrow ((Q \supset \Box(\forall z. G z \supset Q)) w)$ " using 1 by force
34   have 3: " $(Q \supset \Box Q) w$ " using 2 Th5 Rsymm by blast}
35   thus ?thesis by auto
36 qed
```

Fig. 7 After an appropriate modification of the definition of essence in Gödel's 1970 ontological proof, the inconsistency revealed in Fig. 6 is avoided, and the argument can be successfully verified in modal logic S5 (indeed, as shown, only symmetry of the accessibility relation is actually needed)

Gödel 2

Examples of positive and negative properties ...

```
1 theory GoedelVariantHOML2 imports HOMLinHOL ModalFilter
2 begin
3 consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4 axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5 axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6 definition God ("G") where "G x  $\equiv$   $\forall\varphi$ . P  $\varphi$   $\supset$   $\varphi$  x"
7 abbreviation PropertyInclusion ("_ $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\forall y. \varphi y \supset \psi y)$ "
8 (**) definition Essence ("_Ess._") where " $\varphi$  Ess. x  $\equiv$   $\varphi$  x  $\wedge$  ( $\forall\psi. \psi x \supset (\varphi \supset_N \psi)$ )"
9 axiomatization where Ax2b: "[P  $\varphi$   $\supset$   $\Box$  P  $\varphi$ ]"
10 lemma Ax2b': "[ $\neg$ P  $\varphi$   $\supset$   $\Box(\neg$ P  $\varphi$ )]" using Ax2a Ax2b by blast
11 theorem Th1: "[G x  $\supset$  G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12 definition NecExist ("E") where "E x  $\equiv$   $\forall\varphi$ .  $\varphi$  Ess. x  $\supset$   $\Box(\exists^E x. \varphi x)$ "
13 axiomatization where Ax3: "[P E]"
14 theorem Th2: "[G x  $\supset$   $\Box(\exists^E y. G y)$ ]" using Ax3 Th1 God_def NecExist_def by smt
15 theorem Th3: "[ $\Diamond(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" sledgehammer(Th2 Rsymm) —<Proof found>
16 proof -
17   have 1: "[ $(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" using Th2 by blast
18   have 2: "[ $\Diamond(\exists^E x. G x) \supset \Diamond\Box(\exists^E y. G y)$ ]" using 1 by blast
19   have 3: "[ $\Diamond(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" using 2 Rsymm by blast
20   thus ?thesis by blast
21 qed
22 axiomatization where Ax4: "[P  $\varphi$   $\wedge$  ( $\varphi \supset_N \psi$ )  $\supset$  P  $\psi$ ]"
23 lemma True nitpick[satisfy,card=1,eval="[P ( $\lambda x. \bot$ )]" ] oops —<One model found of cardinality one>
24 abbreviation "PosProps  $\Phi \equiv \forall\varphi. \Phi \varphi \supset P \varphi$ "
25 abbreviation "ConjOfPropsFrom  $\varphi \Phi \equiv \Box(\forall^E z. \varphi z \leftrightarrow (\forall\psi. \Phi \psi \supset \psi z))$ "
26 axiomatization where Ax1Gen: "[ (PosProps  $\Phi$   $\wedge$  ConjOfPropsFrom  $\varphi \Phi$ )  $\supset$  P  $\varphi$  ]"
27 lemma L: "[P G]" using Ax1Gen God_def by smt
28 theorem Th4: "[ $\Diamond(\exists^E x. G x)$ ]" using Ax2a Ax4 L by blast
29 theorem Th5: "[ $\Box(\exists^E x. G x)$ ]" using Th3 Th4 by blast
30 lemma MC: "[ $\varphi \supset \Box\varphi$ ]" sledgehammer(Ax2a Ax2b Th5 God_def Rsymm) —<Proof found>
31 proof - {fix w fix Q
32   have 1: " $\forall x. (G x w \longrightarrow (\forall z. Z x \supset \Box(\forall^E z. G z \supset Z z)) w)$ " using Ax2a Ax2b God_def by smt
33   have 2: " $(\exists x. G x w) \longrightarrow ((Q \supset \Box(\forall^E z. G z \supset Q)) w)$ " using 1 by force
34   have 3: " $(Q \supset \Box Q) w$ " using 2 Th5 Rsymm by blast}
35   thus ?thesis by auto
36 qed
37 lemma PosProps: "[P ( $\lambda x. \top$ )  $\wedge$  P ( $\lambda x. x = x$ )]" using Ax2a Ax4 by blast
38 lemma NegProps: "[ $\neg$ P ( $\lambda x. \bot$ )  $\wedge$   $\neg$ P ( $\lambda x. x \neq x$ )]" using Ax2a Ax4 by blast
```

Fig. 7 After an appropriate modification of the definition of essence in Gödel's 1970 ontological proof, the inconsistency revealed in Fig. 6 is avoided, and the argument can be successfully verified in modal logic S5 (indeed, as shown, only symmetry of the accessibility relation is actually needed)

Gödel 2

Any two essences of x are necessarily equivalent,
but not necessarily identical.

Entities are uniquely characterised by their essences.

```
1 theory GoedelVariantHOML2 imports HOMLinHOL ModalFilter
2 begin
3 consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4 axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5 axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6 definition God ("G") where "G x  $\equiv$   $\forall\varphi$ . P  $\varphi$   $\supset$   $\varphi$  x"
7 abbreviation PropertyInclusion ("_ $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\forall y. \varphi y \supset \psi y)$ "
8 (**) definition Essence ("_Ess._") where " $\varphi$  Ess. x  $\equiv$   $\varphi$  x  $\wedge$  ( $\forall\psi$ .  $\psi$  x  $\supset$  ( $\varphi \supset_N \psi$ ))"
9 axiomatization where Ax2b: "[P  $\varphi$   $\supset$   $\Box$  P  $\varphi$ ]"
10 lemma Ax2b': "[ $\neg$ P  $\varphi$   $\supset$   $\Box(\neg$ P  $\varphi$ )]" using Ax2a Ax2b by blast
11 theorem Th1: "[G x  $\supset$  G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12 definition NecExist ("E") where "E x  $\equiv$   $\forall\varphi$ .  $\varphi$  Ess. x  $\supset$   $\Box(\exists^E x. \varphi x)$ "
13 axiomatization where Ax3: "[P E]"
14 theorem Th2: "[G x  $\supset$   $\Box(\exists^E y. G y)$ ]" using Ax3 Th1 God_def NecExist_def by smt
15 theorem Th3: "[ $\Diamond(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" sledgehammer(Th2 Rsymm) —<Proof found>
16 proof -
17   have 1: "[ $(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" using Th2 by blast
18   have 2: "[ $\Diamond(\exists^E x. G x) \supset \Diamond\Box(\exists^E y. G y)$ ]" using 1 by blast
19   have 3: "[ $\Diamond(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" using 2 Rsymm by blast
20   thus ?thesis by blast
21 qed
22 axiomatization where Ax4: "[P  $\varphi$   $\wedge$  ( $\varphi \supset_N \psi$ )  $\supset$  P  $\psi$ ]"
23 lemma True nitpick[satisfy,card=1,eval="[P ( $\lambda x. \bot$ )]" ] oops —<One model found of cardinality one>
24 abbreviation "PosProps  $\Phi \equiv \forall\varphi$ .  $\Phi \varphi \supset$  P  $\varphi$ "
25 abbreviation "ConjOfPropsFrom  $\varphi \Phi \equiv \Box(\forall^E z. \varphi z \leftrightarrow (\forall\psi$ .  $\Phi \psi \supset \psi z))$ "
26 axiomatization where Ax1Gen: "[ (PosProps  $\Phi$   $\wedge$  ConjOfPropsFrom  $\varphi \Phi$ )  $\supset$  P  $\varphi$  ]"
27 lemma L: "[P G]" using Ax1Gen God_def by smt
28 theorem Th4: "[ $\Diamond(\exists^E x. G x)$ ]" using Ax2a Ax4 L by blast
29 theorem Th5: "[ $\Box(\exists^E x. G x)$ ]" using Th3 Th4 by blast
30 lemma MC: "[ $\varphi \supset \Box\varphi$ ]" sledgehammer(Ax2a Ax2b Th5 God_def Rsymm) —<Proof found>
31 proof - {fix w fix Q
32   have 1: " $\forall x. (G x w \longrightarrow (\forall z. Z x \supset \Box(\forall^E z. G z \supset Z z)) w)$ " using Ax2a Ax2b God_def by smt
33   have 2: " $(\exists x. G x w) \longrightarrow ((Q \supset \Box(\forall^E z. G z \supset Q)) w)$ " using 1 by force
34   have 3: " $(Q \supset \Box Q) w$ " using 2 Th5 Rsymm by blast}
35   thus ?thesis by auto
36 qed
37 lemma PosProps: "[P ( $\lambda x. \top$ )  $\wedge$  P ( $\lambda x. x = x$ )]" using Ax2a Ax4 by blast
38 lemma NegProps: "[ $\neg$ P ( $\lambda x. \bot$ )  $\wedge$   $\neg$ P ( $\lambda x. x \neq x$ )]" using Ax2a Ax4 by blast
39 lemma UniqueEss1: "[ $\varphi$  Ess. x  $\wedge$   $\psi$  Ess. x  $\supset$   $\Box(\forall^E y. \varphi y \leftrightarrow \psi y)$ ]" using Essence_def by smt
40 lemma UniqueEss2: "[ $\varphi$  Ess. x  $\wedge$   $\psi$  Ess. x  $\supset$   $\Box(\varphi \equiv \psi)$ ]" nitpick[card i=1] oops —<Countermodel found>
41 lemma UniqueEss3: "[ $\varphi$  Ess. x  $\supset$   $\Box(\forall^E y. \varphi y \supset y \equiv x)$ ]" using Essence_def MC by auto
```

Fig. 7 After an appropriate modification of the definition of essence in Gödel's 1970 ontological proof, the inconsistency revealed in Fig. 6 is avoided, and the argument can be successfully verified in modal logic S5 (indeed, as shown, only symmetry of the accessibility relation is actually needed)

Gödel 2

We have monotheism (modulo Leibniz equality)

```
1 theory GoedelVariantHOML2 imports HOMLinHOL ModalFilter
2 begin
3 consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4 axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5 axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6 definition God ("G") where "G x  $\equiv$   $\forall\varphi$ . P  $\varphi$   $\supset$   $\varphi$  x"
7 abbreviation PropertyInclusion ("_ $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\forall y. \varphi y \supset \psi y)$ "
8 (**) definition Essence ("_Ess._") where " $\varphi$  Ess. x  $\equiv$   $\varphi$  x  $\wedge$  ( $\forall\psi$ .  $\psi$  x  $\supset$  ( $\varphi \supset_N \psi$ ))"
9 axiomatization where Ax2b: "[P  $\varphi$   $\supset$   $\Box$  P  $\varphi$ ]"
10 lemma Ax2b': "[ $\neg$ P  $\varphi$   $\supset$   $\Box(\neg$ P  $\varphi$ )]" using Ax2a Ax2b by blast
11 theorem Th1: "[G x  $\supset$  G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12 definition NecExist ("E") where "E x  $\equiv$   $\forall\varphi$ .  $\varphi$  Ess. x  $\supset$   $\Box(\exists^E x. \varphi x)$ "
13 axiomatization where Ax3: "[P E]"
14 theorem Th2: "[G x  $\supset$   $\Box(\exists^E y. G y)$ ]" using Ax3 Th1 God_def NecExist_def by smt
15 theorem Th3: "[ $\Diamond(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" sledgehammer(Th2 Rsymm) —<Proof found>
16 proof -
17   have 1: "[ $(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" using Th2 by blast
18   have 2: "[ $\Diamond(\exists^E x. G x) \supset \Diamond\Box(\exists^E y. G y)$ ]" using 1 by blast
19   have 3: "[ $\Diamond(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" using 2 Rsymm by blast
20   thus ?thesis by blast
21 qed
22 axiomatization where Ax4: "[P  $\varphi$   $\wedge$  ( $\varphi \supset_N \psi$ )  $\supset$  P  $\psi$ ]"
23 lemma True nitpick[satisfy,card=1,eval="[P ( $\lambda x. \bot$ )]" ] oops —<One model found of cardinality one>
24 abbreviation "PosProps  $\Phi \equiv \forall\varphi$ .  $\Phi \varphi \supset$  P  $\varphi$ "
25 abbreviation "ConjOfPropsFrom  $\varphi \Phi \equiv \Box(\forall z. \varphi z \leftrightarrow (\forall\psi$ .  $\Phi \psi \supset \psi z))$ "
26 axiomatization where Ax1Gen: "[ (PosProps  $\Phi$   $\wedge$  ConjOfPropsFrom  $\varphi \Phi$ )  $\supset$  P  $\varphi$  ]"
27 lemma L: "[P G]" using Ax1Gen God_def by smt
28 theorem Th4: "[ $\Diamond(\exists^E x. G x)$ ]" using Ax2a Ax4 L by blast
29 theorem Th5: "[ $\Box(\exists^E x. G x)$ ]" using Th3 Th4 by blast
30 lemma MC: "[ $\varphi \supset \Box\varphi$ ]" sledgehammer(Ax2a Ax2b Th5 God_def Rsymm) —<Proof found>
31 proof - {fix w fix Q
32   have 1: " $\forall x. (G x w \longrightarrow (\forall z. Z x \supset \Box(\forall z. G z \supset Z z)) w)$ " using Ax2a Ax2b God_def by smt
33   have 2: " $(\exists x. G x w) \longrightarrow ((Q \supset \Box(\forall z. G z \supset Q)) w)$ " using 1 by force
34   have 3: " $(Q \supset \Box Q) w$ " using 2 Th5 Rsymm by blast}
35   thus ?thesis by auto
36 qed
37 lemma PosProps: "[P ( $\lambda x. \top$ )  $\wedge$  P ( $\lambda x. x = x$ )]" using Ax2a Ax4 by blast
38 lemma NegProps: "[ $\neg$ P( $\lambda x. \bot$ )  $\wedge$   $\neg$ P( $\lambda x. x \neq x$ )]" using Ax2a Ax4 by blast
39 lemma UniqueEss1: "[ $\varphi$  Ess. x  $\wedge$   $\psi$  Ess. x  $\supset$   $\Box(\forall y. \varphi y \leftrightarrow \psi y)$ ]" using Essence_def by smt
40 lemma UniqueEss2: "[ $\varphi$  Ess. x  $\wedge$   $\psi$  Ess. x  $\supset$   $\Box(\varphi \equiv \psi)$ ]" nitpick[card i=1] oops —<Countermodel found>
41 lemma UniqueEss3: "[ $\varphi$  Ess. x  $\supset$   $\Box(\forall y. \varphi y \supset y \equiv x)$ ]" using Essence_def MC by auto
42 lemma Monotheism: "[G x  $\wedge$  G y  $\supset$  x  $\equiv$  y]" using Ax2a God_def by smt
```

Fig. 7 After an appropriate modification of the definition of essence in Gödel's 1970 ontological proof, the inconsistency revealed in Fig. 6 is avoided, and the argument can be successfully verified in modal logic S5 (indeed, as shown, only symmetry of the accessibility relation is actually needed)

Gödel 2

The set of positive properties forms an (Ultra)Filter

```
1 theory GoedelVariantHOML2 imports HOMLinHOL ModalFilter
2 begin
3 consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4 axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5 axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6 definition God ("G") where "G x  $\equiv$   $\forall\varphi$ . P  $\varphi$   $\supset$   $\varphi$  x"
7 abbreviation PropertyInclusion ("_ $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\forall^e y. \varphi y \supset \psi y)$ "
8 (**) definition Essence ("_Ess._") where " $\varphi$  Ess. x  $\equiv$   $\varphi$  x  $\wedge$  ( $\forall\psi$ .  $\psi$  x  $\supset$  ( $\varphi \supset_N \psi$ ))"
9 axiomatization where Ax2b: "[P  $\varphi$   $\supset$   $\Box$  P  $\varphi$ ]"
10 lemma Ax2b': "[ $\neg$ P  $\varphi$   $\supset$   $\Box(\neg$ P  $\varphi$ )]" using Ax2a Ax2b by blast
11 theorem Th1: "[G x  $\supset$  G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12 definition NecExist ("E") where "E x  $\equiv$   $\forall\varphi$ .  $\varphi$  Ess. x  $\supset$   $\Box(\exists^e x. \varphi x)$ "
13 axiomatization where Ax3: "[P E]"
14 theorem Th2: "[G x  $\supset$   $\Box(\exists^e y. G y)$ ]" using Ax3 Th1 God_def NecExist_def by smt
15 theorem Th3: "[ $\Diamond(\exists^e x. G x) \supset \Box(\exists^e y. G y)$ ]" sledgehammer(Th2 Rsymm) —<Proof found>
16 proof -
17   have 1: "[ $(\exists^e x. G x) \supset \Box(\exists^e y. G y)$ ]" using Th2 by blast
18   have 2: "[ $\Diamond(\exists^e x. G x) \supset \Diamond\Box(\exists^e y. G y)$ ]" using 1 by blast
19   have 3: "[ $\Diamond(\exists^e x. G x) \supset \Box(\exists^e y. G y)$ ]" using 2 Rsymm by blast
20   thus ?thesis by blast
21 qed
22 axiomatization where Ax4: "[P  $\varphi$   $\wedge$  ( $\varphi \supset_N \psi$ )  $\supset$  P  $\psi$ ]"
23 lemma True nitpick[satisfy,card=1,eval="[P ( $\lambda x. \bot$ )]"] oops —<One model found of cardinality one>
24 abbreviation "PosProps  $\Phi \equiv \forall\varphi$ .  $\Phi \varphi \supset$  P  $\varphi$ "
25 abbreviation "ConjOfPropsFrom  $\varphi \Phi \equiv \Box(\forall^e z. \varphi z \leftrightarrow (\forall\psi$ .  $\Phi \psi \supset \psi z))$ "
26 axiomatization where Ax1Gen: "[ (PosProps  $\Phi$   $\wedge$  ConjOfPropsFrom  $\varphi \Phi$ )  $\supset$  P  $\varphi$  ]"
27 lemma L: "[P G]" using Ax1Gen God_def by smt
28 theorem Th4: "[ $\Diamond(\exists^e x. G x)$ ]" using Ax2a Ax4 L by blast
29 theorem Th5: "[ $\Box(\exists^e x. G x)$ ]" using Th3 Th4 by blast
30 lemma MC: "[ $\varphi \supset \Box\varphi$ ]" sledgehammer(Ax2a Ax2b Th5 God_def Rsymm) —<Proof found>
31 proof - {fix w fix Q
32   have 1: " $\forall x. (G x w \longrightarrow (\forall z. z x \supset \Box(\forall^e z. G z \supset z z)) w)$ " using Ax2a Ax2b God_def by smt
33   have 2: " $(\exists x. G x w) \longrightarrow ((Q \supset \Box(\forall^e z. G z \supset Q)) w)$ " using 1 by force
34   have 3: " $(Q \supset \Box Q) w$ " using 2 Th5 Rsymm by blast}
35   thus ?thesis by auto
36 qed
37 lemma PosProps: "[P ( $\lambda x. \top$ )  $\wedge$  P ( $\lambda x. x = x$ )]" using Ax2a Ax4 by blast
38 lemma NegProps: "[ $\neg$ P ( $\lambda x. \bot$ )  $\wedge$   $\neg$ P ( $\lambda x. x \neq x$ )]" using Ax2a Ax4 by blast
39 lemma UniqueEss1: "[ $\varphi$  Ess. x  $\wedge$   $\psi$  Ess. x  $\supset$   $\Box(\forall^e y. \varphi y \leftrightarrow \psi y)$ ]" using Essence_def by smt
40 lemma UniqueEss2: "[ $\varphi$  Ess. x  $\wedge$   $\psi$  Ess. x  $\supset$   $\Box(\varphi \equiv \psi)$ ]" nitpick[card i=1] oops —<Countermodel found>
41 lemma UniqueEss3: "[ $\varphi$  Ess. x  $\supset$   $\Box(\forall^e y. \varphi y \supset y \equiv x)$ ]" using Essence_def MC by auto
42 lemma Monotheism: "[G x  $\wedge$  G y  $\supset$  x  $\equiv$  y]" using Ax2a God_def by smt
43 lemma Filter: "[Filter P]" using Ax1 Ax4 MC NegProps PosProps Rsymm by smt
44 lemma UltraFilter: "[UFilter P]" using Ax2a Filter by smt
45 lemma True nitpick[satisfy,card=1,eval="[P ( $\lambda x. \bot$ )]"] oops —<One model found of cardinality one>
46 end
```

Fig. 7 After an appropriate modification of the definition of essence in Gödel's 1970 ontological proof, the inconsistency revealed in Fig. 6 is avoided, and the argument can be successfully verified in modal logic S5 (indeed, as shown, only symmetry of the accessibility relation is actually needed)

Gödel 2

Consistency check is positive

```
1 theory GoedelVariantHOML2 imports HOMLinHOL ModalFilter
2 begin
3 consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4 axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5 axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6 definition God ("G") where "G x  $\equiv$   $\forall\varphi$ . P  $\varphi$   $\supset$   $\varphi$  x"
7 abbreviation PropertyInclusion ("_ $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\forall^e y. \varphi y \supset \psi y)$ "
8 (**) definition Essence ("_Ess._") where " $\varphi$  Ess. x  $\equiv$   $\varphi$  x  $\wedge$  ( $\forall\psi$ .  $\psi$  x  $\supset$  ( $\varphi \supset_N \psi$ ))"
9 axiomatization where Ax2b: "[P  $\varphi$   $\supset$   $\Box$  P  $\varphi$ ]"
10 lemma Ax2b': "[ $\neg$ P  $\varphi$   $\supset$   $\Box(\neg$ P  $\varphi$ )]" using Ax2a Ax2b by blast
11 theorem Th1: "[G x  $\supset$  G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12 definition NecExist ("E") where "E x  $\equiv$   $\forall\varphi$ .  $\varphi$  Ess. x  $\supset$   $\Box(\exists^e x. \varphi x)$ "
13 axiomatization where Ax3: "[P E]"
14 theorem Th2: "[G x  $\supset$   $\Box(\exists^e y. G y)$ ]" using Ax3 Th1 God_def NecExist_def by smt
15 theorem Th3: "[ $\Diamond(\exists^e x. G x) \supset \Box(\exists^e y. G y)$ ]" sledgehammer(Th2 Rsymm) —<Proof found>
16 proof -
17   have 1: "[ $(\exists^e x. G x) \supset \Box(\exists^e y. G y)$ ]" using Th2 by blast
18   have 2: "[ $\Diamond(\exists^e x. G x) \supset \Diamond\Box(\exists^e y. G y)$ ]" using 1 by blast
19   have 3: "[ $\Diamond(\exists^e x. G x) \supset \Box(\exists^e y. G y)$ ]" using 2 Rsymm by blast
20   thus ?thesis by blast
21 qed
22 axiomatization where Ax4: "[P  $\varphi$   $\wedge$  ( $\varphi \supset_N \psi$ )  $\supset$  P  $\psi$ ]"
23 lemma True nitpick[satisfy,card=1,eval="[P ( $\lambda x. \bot$ )]" ] oops —<One model found of cardinality one>
24 abbreviation "PosProps  $\Phi \equiv \forall\varphi$ .  $\Phi \varphi \supset$  P  $\varphi$ "
25 abbreviation "ConjOfPropsFrom  $\varphi \Phi \equiv \Box(\forall^e z. \varphi z \leftrightarrow (\forall\psi$ .  $\Phi \psi \supset \psi z))$ "
26 axiomatization where Ax1Gen: "[ (PosProps  $\Phi$   $\wedge$  ConjOfPropsFrom  $\varphi \Phi$ )  $\supset$  P  $\varphi$  ]"
27 lemma L: "[P G]" using Ax1Gen God_def by smt
28 theorem Th4: "[ $\Diamond(\exists^e x. G x)$ ]" using Ax2a Ax4 L by blast
29 theorem Th5: "[ $\Box(\exists^e x. G x)$ ]" using Th3 Th4 by blast
30 lemma MC: "[ $\varphi \supset \Box\varphi$ ]" sledgehammer(Ax2a Ax2b Th5 God_def Rsymm) —<Proof found>
31 proof - {fix w fix Q
32   have 1: " $\forall x. (G x w \longrightarrow (\forall z. z x \supset \Box(\forall^e z. G z \supset z z)) w)$ " using Ax2a Ax2b God_def by smt
33   have 2: " $(\exists x. G x w) \longrightarrow ((Q \supset \Box(\forall^e z. G z \supset Q)) w)$ " using 1 by force
34   have 3: " $(Q \supset \Box Q) w$ " using 2 Th5 Rsymm by blast}
35   thus ?thesis by auto
36 qed
37 lemma PosProps: "[P ( $\lambda x. \top$ )  $\wedge$  P ( $\lambda x. x = x$ )]" using Ax2a Ax4 by blast
38 lemma NegProps: "[ $\neg$ P( $\lambda x. \bot$ )  $\wedge$   $\neg$ P( $\lambda x. x \neq x$ )]" using Ax2a Ax4 by blast
39 lemma UniqueEss1: "[ $\varphi$  Ess. x  $\wedge$   $\psi$  Ess. x  $\supset$   $\Box(\forall^e y. \varphi y \leftrightarrow \psi y)$ ]" using Essence_def by smt
40 lemma UniqueEss2: "[ $\varphi$  Ess. x  $\wedge$   $\psi$  Ess. x  $\supset$   $\Box(\varphi \equiv \psi)$ ]" nitpick[card i=1] oops —<Countermodel found>
41 lemma UniqueEss3: "[ $\varphi$  Ess. x  $\supset$   $\Box(\forall^e y. \varphi y \supset y \equiv x)$ ]" using Essence_def MC by auto
42 lemma Monotheism: "[G x  $\wedge$  G y  $\supset$  x  $\equiv$  y]" using Ax2a God_def by smt
43 lemma Filter: "[Filter P]" using Ax1 Ax4 MC NegProps PosProps Rsymm by smt
44 lemma UltraFilter: "[UFilter P]" using Ax2a Filter by smt
45 lemma True nitpick[satisfy,card=1,eval="[P ( $\lambda x. \bot$ )]" ] oops —<One model found of cardinality one>
46 end
```

Fig. 7 After an appropriate modification of the definition of essence in Gödel's 1970 ontological proof, the inconsistency revealed in Fig. 6 is avoided, and the argument can be successfully verified in modal logic S5 (indeed, as shown, only symmetry of the accessibility relation is actually needed)

Gödel 2

Note that only logic **KB** (Rsymm) was needed here

How about Logic S4?

Neither a proof nor a counterexample has been found in logic S4 for the variant shown in Fig. 7, where the definition of essence has been adapted.

This question thus remains open.

```
1 theory GoedelVariantHOML2 imports HOMLinHOL ModalFilter
2 begin
3 consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4 axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5 axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6 definition God ("G") where "G x  $\equiv$   $\forall\varphi$ . P  $\varphi$   $\supset$   $\varphi$  x"
7 abbreviation PropertyInclusion ("_ $\supset_N$ ") where " $\varphi$   $\supset_N$   $\psi$   $\equiv$   $\Box(\forall y$ .  $\varphi$  y  $\supset$   $\psi$  y)"
8 (**) definition Essence ("_Ess._") where " $\varphi$  Ess. x  $\equiv$   $\varphi$  x  $\wedge$  ( $\forall\psi$ .  $\psi$  x  $\supset$  ( $\varphi$   $\supset_N$   $\psi$ ))"
9 axiomatization where Ax2b: "[P  $\varphi$   $\supset$   $\Box$  P  $\varphi$ ]"
10 lemma Ax2b': "[ $\neg$ P  $\varphi$   $\supset$   $\Box(\neg$ P  $\varphi$ )]" using Ax2a Ax2b by blast
11 theorem Th1: "[G x  $\supset$  G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12 definition NecExist ("E") where "E x  $\equiv$   $\forall\varphi$ .  $\varphi$  Ess. x  $\supset$   $\Box(\exists^E$  x.  $\varphi$  x)"
13 axiomatization where Ax3: "[P E]"
14 theorem Th2: "[G x  $\supset$   $\Box(\exists^E$  y. G y)]" using Ax3 Th1 God_def NecExist_def by smt
15 theorem Th3: "[ $\Diamond(\exists^E$  x. G x)  $\supset$   $\Box(\exists^E$  y. G y)]" sledgehammer(Th2 Rsymm) —<Proof found>
16 proof -
17   have 1: "[ $(\exists^E$  x. G x)  $\supset$   $\Box(\exists^E$  y. G y)]" using Th2 by blast
18   have 2: "[ $\Diamond(\exists^E$  x. G x)  $\supset$   $\Diamond\Box(\exists^E$  y. G y)]" using 1 by blast
19   have 3: "[ $\Diamond(\exists^E$  x. G x)  $\supset$   $\Box(\exists^E$  y. G y)]" using 2 Rsymm by blast
20   thus ?thesis by blast
21 qed
22 axiomatization where Ax4: "[P  $\varphi$   $\wedge$  ( $\varphi$   $\supset_N$   $\psi$ )  $\supset$  P  $\psi$ ]"
23 lemma True nitpick[satisfy,card=1,eval="[P ( $\lambda$ x.  $\bot$ )]"] oops —<One model found of cardinality one>
24 abbreviation "PosProps  $\Phi$   $\equiv$   $\forall\varphi$ .  $\Phi$   $\varphi$   $\supset$  P  $\varphi$ "
25 abbreviation "ConjOfPropsFrom  $\varphi$   $\Phi$   $\equiv$   $\Box(\forall z$ .  $\varphi$  z  $\leftrightarrow$  ( $\forall\psi$ .  $\Phi$   $\psi$   $\supset$   $\psi$  z))"
26 axiomatization where Ax1Gen: "[ (PosProps  $\Phi$   $\wedge$  ConjOfPropsFrom  $\varphi$   $\Phi$ )  $\supset$  P  $\varphi$  ]"
27 lemma L: "[P G]" using Ax1Gen God_def by smt
28 theorem Th4: "[ $\Diamond(\exists^E$  x. G x)]" using Ax2a Ax4 L by blast
29 theorem Th5: "[ $\Box(\exists^E$  x. G x)]" using Th3 Th4 by blast
30 lemma MC: "[ $\varphi$   $\supset$   $\Box\varphi$ ]" sledgehammer(Ax2a Ax2b Th5 God_def Rsymm) —<Proof found>
31 proof - {fix w fix Q
32   have 1: " $\forall x$ . (G x w  $\longrightarrow$  ( $\forall z$ . Z x  $\supset$   $\Box(\forall z$ . G z  $\supset$  Z z)) w)" using Ax2a Ax2b God_def by smt
33   have 2: " $(\exists x$ . G x w) $\longrightarrow$  ((Q  $\supset$   $\Box(\forall z$ . G z  $\supset$  Q)) w)" using 1 by force
34   have 3: "(Q  $\supset$   $\Box$ Q) w" using 2 Th5 Rsymm by blast
35   thus ?thesis by auto
36 qed
37 lemma PosProps: "[P ( $\lambda$ x.  $\top$ )  $\wedge$  P ( $\lambda$ x. x = x)]" using Ax2a Ax4 by blast
38 lemma NegProps: "[ $\neg$ P ( $\lambda$ x.  $\bot$ )  $\wedge$   $\neg$ P ( $\lambda$ x. x  $\neq$  x)]" using Ax2a Ax4 by blast
39 lemma UniqueEss1: "[ $\varphi$  Ess. x  $\wedge$   $\psi$  Ess. x  $\supset$   $\Box(\forall y$ .  $\varphi$  y  $\leftrightarrow$   $\psi$  y)]" using Essence_def by smt
40 lemma UniqueEss2: "[ $\varphi$  Ess. x  $\wedge$   $\psi$  Ess. x  $\supset$   $\Box(\varphi$   $\equiv$   $\psi$ )]" nitpick[card i=1] oops —<Countermodel found>
41 lemma UniqueEss3: "[ $\varphi$  Ess. x  $\supset$   $\Box(\forall y$ .  $\varphi$  y  $\supset$  y  $\equiv$  x)]" using Essence_def MC by auto
42 lemma Monotheism: "[G x  $\wedge$  G y  $\supset$  x  $\equiv$  y]" using Ax2a God_def by smt
43 lemma Filter: "[Filter P]" using Ax1 Ax4 MC NegProps PosProps Rsymm by smt
44 lemma UltraFilter: "[UFilter P]" using Ax2a Filter by smt
45 lemma True nitpick[satisfy,card=1,eval="[P ( $\lambda$ x.  $\bot$ )]"] oops —<One model found of cardinality one>
46 end
```

Fig. 7 After an appropriate modification of the definition of essence in Gödel's 1970 ontological proof, the inconsistency revealed in Fig. 6 is avoided, and the argument can be successfully verified in modal logic S5 (indeed, as shown, only symmetry of the accessibility relation is actually needed)

Gödel 2

Discussion:

With **Ax1** (which introduces only countably many positive properties) no proofs are found for L, Th4 and Th5.

With **Ax1Gen** (which generally introduces uncountably many positive properties) — proofs are found for L, Th4 and Th5.

If we assume (with Gödel) that e.g. the objects of mathematics (such as e.g. natural numbers) do exist, then we already start with at least countably many positive properties. With **Ax1Gen** we then get uncountably many new positive properties, etc. (we increase cardinalities).

Claim: An a priori reduction of Gödel's postulates to countable or even finite structures and means seems in conflict with Gödel's original intention.

This requires further attention.

```
1 theory GoedelVariantHOML2 imports HOMLinHOL ModalFilter
2 begin
3 consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4 axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5 axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6 definition God ("G") where "G x  $\equiv$   $\forall\varphi$ . P  $\varphi$   $\supset$   $\varphi$  x"
7 abbreviation PropertyInclusion ("_ $\supset_N$ ") where " $\varphi$   $\supset_N$   $\psi$   $\equiv$   $\Box(\forall y. \varphi y \supset \psi y)$ "
8 (**) definition Essence ("_Ess._") where " $\varphi$  Ess. x  $\equiv$   $\varphi$  x  $\wedge$  ( $\forall\psi. \psi x \supset$  ( $\varphi \supset_N \psi$ ))"
9 axiomatization where Ax2b: "[P  $\varphi$   $\supset$   $\Box$  P  $\varphi$ ]"
10 lemma Ax2b': "[ $\neg$ P  $\varphi$   $\supset$   $\Box(\neg$ P  $\varphi$ )]" using Ax2a Ax2b by blast
11 theorem Th1: "[G x  $\supset$  G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12 definition NecExist ("E") where "E x  $\equiv$   $\forall\varphi$ .  $\varphi$  Ess. x  $\supset$   $\Box(\exists^E x. \varphi x)$ "
13 axiomatization where Ax3: "[P E]"
14 theorem Th2: "[G x  $\supset$   $\Box(\exists^E y. G y)$ ]" using Ax3 Th1 God_def NecExist_def by smt
15 theorem Th3: "[ $\Diamond(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" sledgehammer(Th2 Rsymm) —<Proof found>
16 proof -
17   have 1: "[ $(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" using Th2 by blast
18   have 2: "[ $\Diamond(\exists^E x. G x) \supset \Diamond\Box(\exists^E y. G y)$ ]" using 1 by blast
19   have 3: "[ $\Diamond(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" using 2 Rsymm by blast
20   thus ?thesis by blast
21 qed
22 axiomatization where Ax4: "[P  $\varphi$   $\wedge$  ( $\varphi \supset_N \psi$ )  $\supset$  P  $\psi$ ]"
23 lemma True nitpick[satisfy,card=1,eval="[P ( $\lambda x. \bot$ )]" ] oops —<One model found of cardinality one>
24 abbreviation "PosProps  $\Phi \equiv \forall\varphi. \Phi \varphi \supset P \varphi$ "
25 abbreviation "ConjOfPropsFrom  $\varphi \Phi \equiv \Box(\forall^E z. \varphi z \leftrightarrow (\forall\psi. \Phi \psi \supset \psi z))$ "
26 axiomatization where Ax1Gen: "[ (PosProps  $\Phi$   $\wedge$  ConjOfPropsFrom  $\varphi \Phi$ )  $\supset$  P  $\varphi$  ]"
27 lemma L: "[P G]" using Ax1Gen God_def by smt
28 theorem Th4: "[ $\Diamond(\exists^E x. G x)$ ]" using Ax2a Ax4 L by blast
29 theorem Th5: "[ $\Box(\exists^E x. G x)$ ]" using Th3 Th4 by blast
30 lemma MC: "[ $\varphi \supset \Box\varphi$ ]" sledgehammer(Ax2a Ax2b Th5 God_def Rsymm) —<Proof found>
31 proof - {fix w fix Q
32   have 1: " $\forall x. (G x w \longrightarrow (\forall z. z x \supset \Box(\forall^E z. G z \supset z z)) w)$ " using Ax2a Ax2b God_def by smt
33   have 2: " $(\exists x. G x w) \longrightarrow ((Q \supset \Box(\forall^E z. G z \supset Q)) w)$ " using 1 by force
34   have 3: " $(Q \supset \Box Q) w$ " using 2 Th5 Rsymm by blast
35   thus ?thesis by auto
36 qed
37 lemma PosProps: "[P ( $\lambda x. \top$ )  $\wedge$  P ( $\lambda x. x = x$ )]" using Ax2a Ax4 by blast
38 lemma NegProps: "[ $\neg$ P( $\lambda x. \bot$ )  $\wedge$   $\neg$ P( $\lambda x. x \neq x$ )]" using Ax2a Ax4 by blast
39 lemma UniqueEss1: "[ $\varphi$  Ess. x  $\wedge$   $\psi$  Ess. x  $\supset$   $\Box(\forall^E y. \varphi y \leftrightarrow \psi y)$ ]" using Essence_def by smt
40 lemma UniqueEss2: "[ $\varphi$  Ess. x  $\wedge$   $\psi$  Ess. x  $\supset$   $\Box(\varphi \equiv \psi)$ ]" nitpick[card i=1] oops —<Countermodel found>
41 lemma UniqueEss3: "[ $\varphi$  Ess. x  $\supset$   $\Box(\forall^E y. \varphi y \supset y \equiv x)$ ]" using Essence_def MC by auto
42 lemma Monotheism: "[G x  $\wedge$  G y  $\supset$  x  $\equiv$  y]" using Ax2a God_def by smt
43 lemma Filter: "[Filter P]" using Ax1 Ax4 MC NegProps PosProps Rsymm by smt
44 lemma UltraFilter: "[UFilter P]" using Ax2a Filter by smt
45 lemma True nitpick[satisfy,card=1,eval="[P ( $\lambda x. \bot$ )]" ] oops —<One model found of cardinality one>
46 end
```

Fig. 7 After an appropriate modification of the definition of essence in Gödel's 1970 ontological proof, the inconsistency revealed in Fig. 6 is avoided, and the argument can be successfully verified in modal logic S5 (indeed, as shown, only symmetry of the accessibility relation is actually needed)

Formalisation of
Gödel's 1970 manuscript
of the modal ontological argument
(adapted property inclusion: Gödel3)
— proof in logic KB —

Gödel 3

Same as before in Gödel 1 & 2

```
1 theory GoedelVariantHOML3 imports HOMLinHOL ModalFilter
2 begin
3 consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4 axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi \supset$  P ( $\varphi . \psi$ )]"
5 axiomatization where Ax2a: "[P  $\varphi \vee^e$  P  $\sim\varphi$ ]"
6 definition God ("G") where "G x  $\equiv \forall\varphi. P \varphi \supset \varphi x$ "
```

Fig. 8 After an appropriate modification of the notion of necessary property inclusion in Gödel's 1970 ontological proof, the inconsistency revealed in Fig. 6 is avoided, and the argument can be successfully verified in modal logic S5 (indeed, as shown, only symmetry of the accessibility relation is actually needed)

Gödel 3

Now property inclusion is modified
(to exclude the empty property to imply any other property)

```
1 theory GoedelVariantHOML3 imports HOMLinHOL ModalFilter
2 begin
3   consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4   axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5   axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6   definition God ("G") where "G x  $\equiv$   $\forall\varphi$ . P  $\varphi$   $\supset$   $\varphi$  x"
7   (**) abbreviation PropertyInclusion ("_  $\supset_N$  _") where " $\varphi \supset_N \psi \equiv \Box(\varphi \neq (\lambda x. \bot) \wedge (\forall^E y. \varphi y \supset \psi y))$ "
8   definition Essence ("_Ess._") where " $\varphi$  Ess. x  $\equiv \forall\psi. \psi x \supset (\varphi \supset_N \psi)$ "
```

Fig. 8 After an appropriate modification of the notion of necessary property inclusion in Gödel’s 1970 ontological proof, the inconsistency revealed in Fig. 6 is avoided, and the argument can be successfully verified in modal logic S5 (indeed, as shown, only symmetry of the accessibility relation is actually needed)

Gödel 3

Same as before in Gödel 2

```
1 theory GoedelVariantHOML3 imports HOMLinHOL ModalFilter
2 begin
3 consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4 axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5 axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6 definition God ("G") where "G x  $\equiv$   $\forall\varphi$ . P  $\varphi$   $\supset$   $\varphi$  x"
7 (**) abbreviation PropertyInclusion (" $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\varphi \neq (\lambda x. \bot) \wedge (\forall^E y. \varphi y \supset \psi y))$ "
8 definition Essence ("_Ess._") where " $\varphi$  Ess. x  $\equiv$   $\forall\psi$ .  $\psi$  x  $\supset$  ( $\varphi \supset_N \psi$ )"
9 axiomatization where Ax2b: "[P  $\varphi$   $\supset$   $\Box$  P  $\varphi$ ]"
10 lemma Ax2b': "[ $\neg$ P  $\varphi$   $\supset$   $\Box(\neg$ P  $\varphi$ )]" using Ax2a Ax2b by blast
11 theorem Th1: "[G x  $\supset$  G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12 definition NecExist ("E") where "E x  $\equiv$   $\forall\varphi$ .  $\varphi$  Ess. x  $\supset$   $\Box(\exists^E x. \varphi x)$ "
13 axiomatization where Ax3: "[P E]"
14 theorem Th2: "[G x  $\supset$   $\Box(\exists^E y. G y)$ ]" using Ax3 Th1 God_def NecExist_def by smt
15 theorem Th3: "[ $\Diamond(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" sledgehammer(Th2 Rsymm) —<Proof found>
16 proof -
17   have 1: "[ $(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" using Th2 by blast
18   have 2: "[ $\Diamond(\exists^E x. G x) \supset \Diamond\Box(\exists^E y. G y)$ ]" using 1 by blast
19   have 3: "[ $\Diamond(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" using 2 Rsymm by blast
20   thus ?thesis by blast
21 qed
22 axiomatization where Ax4: "[P  $\varphi$   $\wedge$  ( $\varphi \supset_N \psi$ )  $\supset$  P  $\psi$ ]"
23 lemma True nitpick[satisfy,card=1,eval="[P ( $\lambda x. \bot$ )]"] oops —<Two models found of cardinality one>
24 abbreviation "PosProps  $\Phi \equiv \forall\varphi$ .  $\Phi \varphi \supset$  P  $\varphi$ "
25 abbreviation "ConjOfPropsFrom  $\varphi \Phi \equiv \Box(\forall^E z. \varphi z \leftrightarrow (\forall\psi$ .  $\Phi \psi \supset \psi z))$ "
26 axiomatization where Ax1Gen: "[ (PosProps  $\Phi$   $\wedge$  ConjOfPropsFrom  $\varphi \Phi$ )  $\supset$  P  $\varphi$  ]"
27 lemma L: "[P G]" using Ax1Gen God_def by (smt (verit))
```

Fig. 8 After an appropriate modification of the notion of necessary property inclusion in Gödel's 1970 ontological proof, the inconsistency revealed in Fig. 6 is avoided, and the argument can be successfully verified in modal logic S5 (indeed, as shown, only symmetry of the accessibility relation is actually needed)

Gödel 3

Same as before in Gödel 2

```
1 theory GoedelVariantHOML3 imports HOMLinHOL ModalFilter
2 begin
3 consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4 axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5 axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6 definition God ("G") where "G x  $\equiv$   $\forall\varphi$ . P  $\varphi$   $\supset$   $\varphi$  x"
7 (**) abbreviation PropertyInclusion (" $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\varphi \neq (\lambda x. \bot) \wedge (\forall^E y. \varphi y \supset \psi y))$ "
8 definition Essence ("_Ess._") where " $\varphi$  Ess. x  $\equiv \forall\psi$ .  $\psi$  x  $\supset$  ( $\varphi \supset_N \psi$ )"
9 axiomatization where Ax2b: "[P  $\varphi$   $\supset$   $\Box$  P  $\varphi$ ]"
10 lemma Ax2b': "[ $\neg$ P  $\varphi$   $\supset$   $\Box(\neg$ P  $\varphi$ )]" using Ax2a Ax2b by blast
11 theorem Th1: "[G x  $\supset$  G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12 definition NecExist ("E") where "E x  $\equiv \forall\varphi$ .  $\varphi$  Ess. x  $\supset$   $\Box(\exists^E x. \varphi x)$ "
13 axiomatization where Ax3: "[P E]"
14 theorem Th2: "[G x  $\supset$   $\Box(\exists^E y. G y)$ ]" using Ax3 Th1 God_def NecExist_def by smt
15 theorem Th3: "[ $\Diamond(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" sledgehammer(Th2 Rsymm) —<Proof found>
16 proof -
17   have 1: "[ $(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" using Th2 by blast
18   have 2: "[ $\Diamond(\exists^E x. G x) \supset \Diamond\Box(\exists^E y. G y)$ ]" using 1 by blast
19   have 3: "[ $\Diamond(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" using 2 Rsymm by blast
20   thus ?thesis by blast
21 qed
22 axiomatization where Ax4: "[P  $\varphi$   $\wedge$  ( $\varphi \supset_N \psi$ )  $\supset$  P  $\psi$ ]"
23 lemma True nitpick[satisfy,card=1,eval="[P ( $\lambda x. \bot$ )]"] oops —<Two models found of cardinality one>
24 abbreviation "PosProps  $\Phi \equiv \forall\varphi$ .  $\Phi \varphi \supset$  P  $\varphi$ "
25 abbreviation "ConjOfPropsFrom  $\varphi \Phi \equiv \Box(\forall^E z. \varphi z \leftrightarrow (\forall\psi$ .  $\Phi \psi \supset \psi z))$ "
26 axiomatization where Ax1Gen: "[ (PosProps  $\Phi$   $\wedge$  ConjOfPropsFrom  $\varphi \Phi$ )  $\supset$  P  $\varphi$  ]"
27 lemma L: "[P G]" using Ax1Gen God_def by (smt (verit))
```

Fig. 8 After an appropriate modification of the notion of necessary property inclusion in Gödel's 1970 ontological proof, the inconsistency revealed in Fig. 6 is avoided, and the argument can be successfully verified in modal logic S5 (indeed, as shown, only symmetry of the accessibility relation is actually needed)

Gödel 3

Here proofs get a bit more difficult to find (and reconstruct)

```
1 theory GoedelVariantHOML3 imports HOMLinHOL ModalFilter
2 begin
3 consts PositiveProperty::"(e⇒σ)⇒σ" ("P")
4 axiomatization where Ax1: "[P φ ∧ P ψ ⊃ P (φ . ψ)]"
5 axiomatization where Ax2a: "[P φ ∨e P ~φ]"
6 definition God ("G") where "G x ≡ ∀φ. P φ ⊃ φ x"
7 (**) abbreviation PropertyInclusion ("⊃N") where "φ ⊃N ψ ≡ □(φ ≠ (λx. ⊥) ∧ (∀Ey. φ y ⊃ ψ y))"
8 definition Essence ("_Ess._") where "φ Ess. x ≡ ∀ψ. ψ x ⊃ (φ ⊃N ψ)"
9 axiomatization where Ax2b: "[P φ ⊃ □ P φ]"
10 lemma Ax2b': "[¬P φ ⊃ □(¬P φ)]" using Ax2a Ax2b by blast
11 theorem Th1: "[G x ⊃ G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12 definition NecExist ("E") where "E x ≡ ∀φ. φ Ess. x ⊃ □(∃Ex. φ x)"
13 axiomatization where Ax3: "[P E]"
14 theorem Th2: "[G x ⊃ □(∃Ey. G y)]" using Ax3 Th1 God_def NecExist_def by smt
15 theorem Th3: "[◇(∃Ex. G x) ⊃ □(∃Ey. G y)]" sledgehammer(Th2 Rsymm) —<Proof found>
16 proof -
17   have 1: "[ (∃Ex. G x) ⊃ □(∃Ey. G y)]" using Th2 by blast
18   have 2: "[ ◇(∃Ex. G x) ⊃ ◇□(∃Ey. G y)]" using 1 by blast
19   have 3: "[ ◇(∃Ex. G x) ⊃ □(∃Ey. G y)]" using 2 Rsymm by blast
20   thus ?thesis by blast
21 qed
22 axiomatization where Ax4: "[P φ ∧ (φ ⊃N ψ) ⊃ P ψ]"
23 lemma True nitpick[satisfy,card=1,eval="[P (λx.⊥)]"] oops —<Two models found of cardinality one>
24 abbreviation "PosProps Φ ≡ ∀φ. Φ φ ⊃ P φ"
25 abbreviation "ConjOfPropsFrom φ Φ ≡ □(∀Ez. φ z ↔ (∀ψ. Φ ψ ⊃ ψ z))"
26 axiomatization where Ax1Gen: "[ (PosProps Φ ∧ ConjOfPropsFrom φ Φ) ⊃ P φ]"
27 lemma L: "[P G]" using Ax1Gen God_def by (smt (verit))
28 theorem Th4: "[◇(∃Ex. G x)]" sledgehammer[timeout=200](Ax2a L Ax1Gen) sorry —<Proof found>
```

Fig. 8 After an appropriate modification of the notion of necessary property inclusion in Gödel’s 1970 ontological proof, the inconsistency revealed in Fig. 6 is avoided, and the argument can be successfully verified in modal logic S5 (indeed, as shown, only symmetry of the accessibility relation is actually needed)

Gödel 3

Similar to before in Gödel 2 (proofs get more complicated)

```
1 theory GoedelVariantHOML3 imports HOMLinHOL ModalFilter
2 begin
3   consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4   axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5   axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6   definition God ("G") where "G x  $\equiv$   $\forall\varphi$ . P  $\varphi$   $\supset$   $\varphi$  x"
7   (**) abbreviation PropertyInclusion ("_ $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\varphi \neq (\lambda x. \bot) \wedge (\forall^E y. \varphi y \supset \psi y))$ "
8   definition Essence ("_Ess._") where " $\varphi$  Ess. x  $\equiv \forall\psi$ .  $\psi$  x  $\supset$  ( $\varphi \supset_N \psi$ )"
9   axiomatization where Ax2b: "[P  $\varphi$   $\supset$   $\Box$  P  $\varphi$ ]"
10  lemma Ax2b': "[ $\neg$ P  $\varphi$   $\supset$   $\Box(\neg$ P  $\varphi$ )]" using Ax2a Ax2b by blast
11  theorem Th1: "[G x  $\supset$  G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12  definition NecExist ("E") where "E x  $\equiv \forall\varphi$ .  $\varphi$  Ess. x  $\supset$   $\Box(\exists^E x. \varphi x)$ "
13  axiomatization where Ax3: "[P E]"
14  theorem Th2: "[G x  $\supset$   $\Box(\exists^E y. G y)$ ]" using Ax3 Th1 God_def NecExist_def by smt
15  theorem Th3: "[ $\Diamond(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" sledgehammer(Th2 Rsymm) —<Proof found>
16  proof -
17    have 1: "[ $(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" using Th2 by blast
18    have 2: "[ $\Diamond(\exists^E x. G x) \supset \Diamond\Box(\exists^E y. G y)$ ]" using 1 by blast
19    have 3: "[ $\Diamond(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" using 2 Rsymm by blast
20    thus ?thesis by blast
21  qed
22  axiomatization where Ax4: "[P  $\varphi$   $\wedge$  ( $\varphi \supset_N \psi$ )  $\supset$  P  $\psi$ ]"
23  lemma True nitpick[satisfy,card=1,eval="[P ( $\lambda x. \bot$ )]"] oops —<Two models found of cardinality one>
24  abbreviation "PosProps  $\Phi \equiv \forall\varphi$ .  $\Phi \varphi \supset$  P  $\varphi$ "
25  abbreviation "ConjOfPropsFrom  $\varphi \Phi \equiv \Box(\forall^E z. \varphi z \leftrightarrow (\forall\psi$ .  $\Phi \psi \supset \psi z))$ "
26  axiomatization where Ax1Gen: "[ (PosProps  $\Phi$   $\wedge$  ConjOfPropsFrom  $\varphi \Phi$ )  $\supset$  P  $\varphi$  ]"
27  lemma L: "[P G]" using Ax1Gen God_def by (smt (verit))
28  theorem Th4: "[ $\Diamond(\exists^E x. G x)$ ]" sledgehammer[timeout=200](Ax2a L Ax1Gen) sorry —<Proof found>
29  theorem Th5: "[ $\Box(\exists^E x. G x)$ ]" using Th3 Th4 by blast
30  lemma MC: "[ $\varphi \supset \Box\varphi$ ]" sledgehammer(Ax2a Ax2b Th5 God_def Rsymm) —<Proof found>
31  proof - {fix w fix Q
32    have 1: " $\forall x. (G x w \longrightarrow (\forall z. z x \supset \Box(\forall^E z. G z \supset z z)) w)$ " using Ax2a Ax2b God_def by smt
33    have 2: " $(\exists x. G x w) \longrightarrow ((Q \supset \Box(\forall^E z. G z \supset Q)) w)$ " using 1 by force
34    have 3: " $(Q \supset \Box Q) w$ " using 2 Th5 Rsymm by blast}
35  thus ?thesis by auto
36  qed
37  lemma PosProps: "[P ( $\lambda x. \top$ )  $\wedge$  P( $\lambda x. x = x$ )]" using Ax2a Ax4 L Th4 by smt
38  lemma NegProps: "[ $\neg$ P( $\lambda x. \bot$ )  $\wedge$   $\neg$ P( $\lambda x. x \neq x$ )]" using Ax2a Ax4 L Th4 by smt
```

Fig. 8 After an appropriate modification of the notion of necessary property inclusion in Gödel's 1970 ontological proof, the inconsistency revealed in Fig. 6 is avoided, and the argument can be successfully verified in modal logic S5 (indeed, as shown, only symmetry of the accessibility relation is actually needed)

Gödel 3

Open questions about uniqueness of essences

```

1 theory GoedelVariantHOML3 imports HOMLinHOL ModalFilter
2 begin
3   consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4   axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5   axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6   definition God ("G") where "G x  $\equiv$   $\forall\varphi$ . P  $\varphi$   $\supset$   $\varphi$  x"
7   (**) abbreviation PropertyInclusion ("_ $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\varphi \neq (\lambda x. \perp) \wedge (\forall^E y. \varphi y \supset \psi y))$ "
8   definition Essence ("_Ess._") where " $\varphi$  Ess. x  $\equiv$   $\forall\psi$ .  $\psi$  x  $\supset$  ( $\varphi \supset_N \psi$ )"
9   axiomatization where Ax2b: "[P  $\varphi$   $\supset$   $\Box$  P  $\varphi$ ]"
10  lemma Ax2b': "[ $\neg$ P  $\varphi$   $\supset$   $\Box(\neg$ P  $\varphi$ )]" using Ax2a Ax2b by blast
11  theorem Th1: "[G x  $\supset$  G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12  definition NecExist ("E") where "E x  $\equiv$   $\forall\varphi$ .  $\varphi$  Ess. x  $\supset$   $\Box(\exists^E x. \varphi x)$ "
13  axiomatization where Ax3: "[P E]"
14  theorem Th2: "[G x  $\supset$   $\Box(\exists^E y. G y)$ ]" using Ax3 Th1 God_def NecExist_def by smt
15  theorem Th3: "[ $\Diamond(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" sledgehammer(Th2 Rsymm) —<Proof found>
16  proof -
17    have 1: "[ $(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" using Th2 by blast
18    have 2: "[ $\Diamond(\exists^E x. G x) \supset \Diamond\Box(\exists^E y. G y)$ ]" using 1 by blast
19    have 3: "[ $\Diamond(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" using 2 Rsymm by blast
20    thus ?thesis by blast
21  qed
22  axiomatization where Ax4: "[P  $\varphi$   $\wedge$  ( $\varphi \supset_N \psi$ )  $\supset$  P  $\psi$ ]"
23  lemma True nitpick[satisfy,card=1,eval="[P ( $\lambda x. \perp$ )]"] oops —<Two models found of cardinality one>
24  abbreviation "PosProps  $\Phi \equiv \forall\varphi$ .  $\Phi \varphi \supset$  P  $\varphi$ "
25  abbreviation "ConjOfPropsFrom  $\varphi \Phi \equiv \Box(\forall^E z. \varphi z \leftrightarrow (\forall\psi$ .  $\Phi \psi \supset \psi z))$ "
26  axiomatization where Ax1Gen: "[ (PosProps  $\Phi$   $\wedge$  ConjOfPropsFrom  $\varphi \Phi$ )  $\supset$  P  $\varphi$  ]"
27  lemma L: "[P G]" using Ax1Gen God_def by (smt (verit))
28  theorem Th4: "[ $\Diamond(\exists^E x. G x)$ ]" sledgehammer[timeout=200](Ax2a L Ax1Gen) sorry —<Proof found>
29  theorem Th5: "[ $\Box(\exists^E x. G x)$ ]" using Th3 Th4 by blast
30  lemma MC: "[ $\varphi \supset \Box\varphi$ ]" sledgehammer(Ax2a Ax2b Th5 God_def Rsymm) —<Proof found>
31  proof - {fix w fix Q
32    have 1: " $\forall x. (G x w \longrightarrow (\forall Z. Z x \supset \Box(\forall^E z. G z \supset Z z)) w)$ " using Ax2a Ax2b God_def by smt
33    have 2: " $(\exists x. G x w) \longrightarrow ((Q \supset \Box(\forall^E z. G z \supset Q)) w)$ " using 1 by force
34    have 3: " $(Q \supset \Box Q) w$ " using 2 Th5 Rsymm by blast}
35  thus ?thesis by auto
36  qed
37  lemma PosProps: "[P ( $\lambda x. \top$ )  $\wedge$  P( $\lambda x. x = x$ )]" using Ax2a Ax4 L Th4 by smt
38  lemma NegProps: "[ $\neg$ P( $\lambda x. \perp$ )  $\wedge$   $\neg$ P( $\lambda x. x \neq x$ )]" using Ax2a Ax4 L Th4 by smt
39  lemma UniqueEss1: "[ $\varphi$  Ess. x  $\wedge$   $\psi$  Ess. x  $\supset$   $\Box(\forall^E y. \varphi y \leftrightarrow \psi y)$ ]" oops —<Unclear, open question>
40  lemma UniqueEss2: "[ $\varphi$  Ess. x  $\wedge$   $\psi$  Ess. x  $\supset$   $\Box(\varphi \equiv \psi)$ ]" oops —<Unclear, open question>

```

Fig. 8 After an appropriate modification of the notion of necessary property inclusion in Gödel's 1970 ontological proof, the inconsistency revealed in Fig. 6 is avoided, and the argument can be successfully verified in modal logic S5 (indeed, as shown, only symmetry of the accessibility relation is actually needed)

Gödel 3

Same as before in Gödel 2

```
1 theory GoedelVariantHOML3 imports HOMLinHOL ModalFilter
2 begin
3   consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4   axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5   axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6   definition God ("G") where "G x  $\equiv$   $\forall\varphi$ . P  $\varphi$   $\supset$   $\varphi$  x"
7   (**) abbreviation PropertyInclusion (" $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\varphi \neq (\lambda x. \bot) \wedge (\forall^E y. \varphi y \supset \psi y))$ "
8   definition Essence ("_Ess._") where " $\varphi$  Ess. x  $\equiv$   $\forall\psi$ .  $\psi$  x  $\supset$  ( $\varphi \supset_N \psi$ )"
9   axiomatization where Ax2b: "[P  $\varphi$   $\supset$   $\Box$  P  $\varphi$ ]"
10  lemma Ax2b': "[ $\neg$ P  $\varphi$   $\supset$   $\Box(\neg$ P  $\varphi$ )]" using Ax2a Ax2b by blast
11  theorem Th1: "[G x  $\supset$  G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12  definition NecExist ("E") where "E x  $\equiv$   $\forall\varphi$ .  $\varphi$  Ess. x  $\supset$   $\Box(\exists^E x. \varphi x)$ "
13  axiomatization where Ax3: "[P E]"
14  theorem Th2: "[G x  $\supset$   $\Box(\exists^E y. G y)$ ]" using Ax3 Th1 God_def NecExist_def by smt
15  theorem Th3: "[ $\Diamond(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" sledgehammer(Th2 Rsymm) —<Proof found>
16  proof -
17    have 1: "[ $(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" using Th2 by blast
18    have 2: "[ $\Diamond(\exists^E x. G x) \supset \Diamond\Box(\exists^E y. G y)$ ]" using 1 by blast
19    have 3: "[ $\Diamond(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" using 2 Rsymm by blast
20    thus ?thesis by blast
21  qed
22  axiomatization where Ax4: "[P  $\varphi$   $\wedge$  ( $\varphi \supset_N \psi$ )  $\supset$  P  $\psi$ ]"
23  lemma True nitpick[satisfy,card=1,eval="[P ( $\lambda x. \bot$ )]"] oops —<Two models found of cardinality one>
24  abbreviation "PosProps  $\Phi \equiv \forall\varphi$ .  $\Phi \varphi \supset$  P  $\varphi$ "
25  abbreviation "ConjOfPropsFrom  $\varphi \Phi \equiv \Box(\forall^E z. \varphi z \leftrightarrow (\forall\psi$ .  $\Phi \psi \supset \psi z))$ "
26  axiomatization where Ax1Gen: "[ (PosProps  $\Phi$   $\wedge$  ConjOfPropsFrom  $\varphi \Phi$ )  $\supset$  P  $\varphi$  ]"
27  lemma L: "[P G]" using Ax1Gen God_def by (smt (verit))
28  theorem Th4: "[ $\Diamond(\exists^E x. G x)$ ]" sledgehammer[timeout=200](Ax2a L Ax1Gen) sorry —<Proof found>
29  theorem Th5: "[ $\Box(\exists^E x. G x)$ ]" using Th3 Th4 by blast
30  lemma MC: "[ $\varphi \supset \Box\varphi$ ]" sledgehammer(Ax2a Ax2b Th5 God_def Rsymm) —<Proof found>
31  proof - {fix w fix Q
32    have 1: " $\forall x. (G x w \longrightarrow (\forall Z. Z x \supset \Box(\forall^E z. G z \supset Z z)) w)$ " using Ax2a Ax2b God_def by smt
33    have 2: " $(\exists x. G x w) \longrightarrow ((Q \supset \Box(\forall^E z. G z \supset Q)) w)$ " using 1 by force
34    have 3: " $(Q \supset \Box Q) w$ " using 2 Th5 Rsymm by blast}
35  thus ?thesis by auto
36  qed
37  lemma PosProps: "[P ( $\lambda x. \top$ )  $\wedge$  P( $\lambda x. x = x$ )]" using Ax2a Ax4 L Th4 by smt
38  lemma NegProps: "[ $\neg$ P( $\lambda x. \bot$ )  $\wedge$   $\neg$ P( $\lambda x. x \neq x$ )]" using Ax2a Ax4 L Th4 by smt
39  lemma UniqueEss1: "[ $\varphi$  Ess. x  $\wedge$   $\psi$  Ess. x  $\supset$   $\Box(\forall^E y. \varphi y \leftrightarrow \psi y)$ ]" oops —<Unclear, open question>
40  lemma UniqueEss2: "[ $\varphi$  Ess. x  $\wedge$   $\psi$  Ess. x  $\supset$   $\Box(\varphi \equiv \psi)$ ]" oops —<Unclear, open question>
41  lemma UniqueEss3: "[ $\varphi$  Ess. x  $\supset$   $\Box(\forall^E y. \varphi y \supset y \equiv x)$ ]" using Essence_def MC by auto
42  lemma Monotheism: "[G x  $\wedge$  G y  $\supset$  x  $\equiv$  y]" using Ax2a God_def by smt
43  lemma Filter: "[Filter P]" using Ax1 Ax4 MC NegProps PosProps Rsymm by smt
44  lemma UltraFilter: "[UFilter P]" using Ax2a Filter by smt
45  lemma True nitpick[satisfy,card=1,eval="[P ( $\lambda x. \top$ )]"] oops —<One model found of cardinality one>
46 end
```

Fig. 8 After an appropriate modification of the notion of necessary property inclusion in Gödel's 1970 ontological proof, the inconsistency revealed in Fig. 6 is avoided, and the argument can be successfully verified in modal logic S5 (indeed, as shown, only symmetry of the accessibility relation is actually needed)

```

1 theory GoedelVariantHOML2 imports HOMLinHOL ModalFilter
2 begin
3 consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4 axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5 axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6 definition God ("G") where "G x  $\equiv$   $\forall\varphi$ . P  $\varphi$   $\supset$   $\varphi$  x"
7 abbreviation PropertyInclusion (" $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\forall y. \varphi y \supset \psi y)$ "
8 (**) definition Essence ("_Ess_") where " $\varphi$  Ess. x  $\equiv$   $\varphi$  x  $\wedge$  ( $\forall\psi. \psi x \supset$  ( $\varphi \supset_N \psi$ ))"
9 axiomatization where Ax2b: "[P  $\varphi$   $\supset$   $\Box$  P  $\varphi$ ]"
10 lemma Ax2b': "[ $\neg$ P  $\varphi$   $\supset$   $\Box(\neg$ P  $\varphi$ )]" using Ax2a Ax2b by blast
11 theorem Th1: "[G x  $\supset$  G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12 definition NecExist ("E") where "E x  $\equiv$   $\forall\varphi$ .  $\varphi$  Ess. x  $\supset$   $\Box(\exists^E x. \varphi x)$ "
13 axiomatization where Ax3: "[P E]"
14 theorem Th2: "[G x  $\supset$   $\Box(\exists^E y. G y)$ ]" using Ax3 Th1 God_def NecExist_def by smt
15 theorem Th3: "[ $\Diamond(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" sledgehammer(Th2 Rsymm) —<Proof found>
16 proof -
17   have 1: "[ $(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" using Th2 by blast
18   have 2: "[ $\Diamond(\exists^E x. G x) \supset \Diamond\Box(\exists^E y. G y)$ ]" using 1 by blast
19   have 3: "[ $\Diamond(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" using 2 Rsymm by blast
20   thus ?thesis by blast
21 qed
22 axiomatization where Ax4: "[P  $\varphi$   $\wedge$  ( $\varphi \supset_N \psi$ )  $\supset$  P  $\psi$ ]"
23 lemma True nitpick[satisfy,card=1,eval="[P ( $\lambda x. \bot$ )]" ] oops —<One model found of cardinality one>
24 abbreviation "PosProps  $\Phi \equiv \forall\varphi. \Phi \varphi \supset P \varphi$ "
25 abbreviation "ConjOfPropsFrom  $\varphi \Phi \equiv \Box(\forall^E z. \varphi z \leftrightarrow (\forall\psi. \Phi \psi \supset \psi z))$ "
26 axiomatization where Ax1Gen: "[ (PosProps  $\Phi$   $\wedge$  ConjOfPropsFrom  $\varphi \Phi$ )  $\supset$  P  $\varphi$  ]"
27 lemma L: "[P G]" using Ax1Gen God_def by smt
28 theorem Th4: "[ $\Diamond(\exists^E x. G x)$ ]" using Ax2a Ax4 L by blast
29 theorem Th5: "[ $\Box(\exists^E x. G x)$ ]" using Th3 Th4 by blast
30 lemma MC: "[ $\varphi \supset \Box\varphi$ ]" sledgehammer(Ax2a Ax2b Th5 God_def Rsymm) —<Proof found>
31 proof - {fix w fix Q
32   have 1: " $\forall x. (G x w \longrightarrow (\forall Z. Z x \supset \Box(\forall^E z. G z \supset Z z))) w$ " using Ax2a Ax2b God_def by smt
33   have 2: " $(\exists x. G x w) \longrightarrow ((Q \supset \Box(\forall^E z. G z \supset Q)) w)$ " using 1 by force
34   have 3: " $(Q \supset \Box Q) w$ " using 2 Th5 Rsymm by blast}
35   thus ?thesis by auto
36 qed
37 lemma PosProps: "[P ( $\lambda x. T$ )  $\wedge$  P ( $\lambda x. x = x$ )]" using Ax2a Ax4 by blast
38 lemma NegProps: "[ $\neg$ P ( $\lambda x. \bot$ )  $\wedge$   $\neg$ P ( $\lambda x. x \neq x$ )]" using Ax2a Ax4 by blast
39 lemma UniqueEss1: "[ $\varphi$  Ess. x  $\wedge$   $\psi$  Ess. x  $\supset$   $\Box(\forall^E y. \varphi y \leftrightarrow \psi y)$ ]" using Essence_def by smt
40 lemma UniqueEss2: "[ $\varphi$  Ess. x  $\wedge$   $\psi$  Ess. x  $\supset$   $\Box(\varphi \equiv \psi)$ ]" nitpick[card i=1] oops —<Countermodel found>
41 lemma UniqueEss3: "[ $\varphi$  Ess. x  $\supset$   $\Box(\forall^E y. \varphi y \supset y \equiv x)$ ]" using Essence_def MC by auto
42 lemma Monotheism: "[G x  $\wedge$  G y  $\supset$  x  $\equiv$  y]" using Ax2a God_def by smt
43 lemma Filter: "[Filter P]" using Ax1 Ax4 MC NegProps PosProps Rsymm by smt
44 lemma UltraFilter: "[UFilter P]" using Ax2a Filter by smt
45 lemma True nitpick[satisfy,card=1,eval="[P ( $\lambda x. \bot$ )]" ] oops —<One model found of cardinality one>
46 end

```

Fig. 7 After an appropriate modification of the definition of essence in Gödel's 1970 ontological proof, the inconsistency revealed in Fig. 6 is avoided, and the argument can be successfully verified in modal logic S5 (indeed, as shown, only symmetry of the accessibility relation is actually needed)

```

1 theory GoedelVariantHOML3 imports HOMLinHOL ModalFilter
2 begin
3 consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4 axiomatization where Ax1: "[P  $\varphi$   $\wedge$  P  $\psi$   $\supset$  P ( $\varphi$  .  $\psi$ )]"
5 axiomatization where Ax2a: "[P  $\varphi$   $\vee^e$  P  $\sim\varphi$ ]"
6 definition God ("G") where "G x  $\equiv$   $\forall\varphi$ . P  $\varphi$   $\supset$   $\varphi$  x"
7 (**) abbreviation PropertyInclusion (" $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\varphi \neq (\lambda x. \bot) \wedge (\forall^E y. \varphi y \supset \psi y))$ "
8 definition Essence ("_Ess_") where " $\varphi$  Ess. x  $\equiv$   $\forall\psi. \psi x \supset$  ( $\varphi \supset_N \psi$ )"
9 axiomatization where Ax2b: "[P  $\varphi$   $\supset$   $\Box$  P  $\varphi$ ]"
10 lemma Ax2b': "[ $\neg$ P  $\varphi$   $\supset$   $\Box(\neg$ P  $\varphi$ )]" using Ax2a Ax2b by blast
11 theorem Th1: "[G x  $\supset$  G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12 definition NecExist ("E") where "E x  $\equiv$   $\forall\varphi$ .  $\varphi$  Ess. x  $\supset$   $\Box(\exists^E x. \varphi x)$ "
13 axiomatization where Ax3: "[P E]"
14 theorem Th2: "[G x  $\supset$   $\Box(\exists^E y. G y)$ ]" using Ax3 Th1 God_def NecExist_def by smt
15 theorem Th3: "[ $\Diamond(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" sledgehammer(Th2 Rsymm) —<Proof found>
16 proof -
17   have 1: "[ $(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" using Th2 by blast
18   have 2: "[ $\Diamond(\exists^E x. G x) \supset \Diamond\Box(\exists^E y. G y)$ ]" using 1 by blast
19   have 3: "[ $\Diamond(\exists^E x. G x) \supset \Box(\exists^E y. G y)$ ]" using 2 Rsymm by blast
20   thus ?thesis by blast
21 qed
22 axiomatization where Ax4: "[P  $\varphi$   $\wedge$  ( $\varphi \supset_N \psi$ )  $\supset$  P  $\psi$ ]"
23 lemma True nitpick[satisfy,card=1,eval="[P ( $\lambda x. \bot$ )]" ] oops —<Two models found of cardinality one>
24 abbreviation "PosProps  $\Phi \equiv \forall\varphi. \Phi \varphi \supset P \varphi$ "
25 abbreviation "ConjOfPropsFrom  $\varphi \Phi \equiv \Box(\forall^E z. \varphi z \leftrightarrow (\forall\psi. \Phi \psi \supset \psi z))$ "
26 axiomatization where Ax1Gen: "[ (PosProps  $\Phi$   $\wedge$  ConjOfPropsFrom  $\varphi \Phi$ )  $\supset$  P  $\varphi$  ]"
27 lemma L: "[P G]" using Ax1Gen God_def by (smt (verit))
28 theorem Th4: "[ $\Diamond(\exists^E x. G x)$ ]" sledgehammer[timeout=200](Ax2a L Ax1Gen) sorry —<Proof found>
29 theorem Th5: "[ $\Box(\exists^E x. G x)$ ]" using Th3 Th4 by blast
30 lemma MC: "[ $\varphi \supset \Box\varphi$ ]" sledgehammer(Ax2a Ax2b Th5 God_def Rsymm) —<Proof found>
31 proof - {fix w fix Q
32   have 1: " $\forall x. (G x w \longrightarrow (\forall Z. Z x \supset \Box(\forall^E z. G z \supset Z z))) w$ " using Ax2a Ax2b God_def by smt
33   have 2: " $(\exists x. G x w) \longrightarrow ((Q \supset \Box(\forall^E z. G z \supset Q)) w)$ " using 1 by force
34   have 3: " $(Q \supset \Box Q) w$ " using 2 Th5 Rsymm by blast}
35   thus ?thesis by auto
36 qed
37 lemma PosProps: "[P ( $\lambda x. T$ )  $\wedge$  P ( $\lambda x. x = x$ )]" using Ax2a Ax4 L Th4 by smt
38 lemma NegProps: "[ $\neg$ P ( $\lambda x. \bot$ )  $\wedge$   $\neg$ P ( $\lambda x. x \neq x$ )]" using Ax2a Ax4 L Th4 by smt
39 lemma UniqueEss1: "[ $\varphi$  Ess. x  $\wedge$   $\psi$  Ess. x  $\supset$   $\Box(\forall^E y. \varphi y \leftrightarrow \psi y)$ ]" oops —<Unclear, open question>
40 lemma UniqueEss2: "[ $\varphi$  Ess. x  $\wedge$   $\psi$  Ess. x  $\supset$   $\Box(\varphi \equiv \psi)$ ]" oops —<Unclear, open question>
41 lemma UniqueEss3: "[ $\varphi$  Ess. x  $\supset$   $\Box(\forall^E y. \varphi y \supset y \equiv x)$ ]" using Essence_def MC by auto
42 lemma Monotheism: "[G x  $\wedge$  G y  $\supset$  x  $\equiv$  y]" using Ax2a God_def by smt
43 lemma Filter: "[Filter P]" using Ax1 Ax4 MC NegProps PosProps Rsymm by smt
44 lemma UltraFilter: "[UFilter P]" using Ax2a Filter by smt
45 lemma True nitpick[satisfy,card=1,eval="[P ( $\lambda x. T$ )]" ] oops —<One model found of cardinality one>
46 end

```

Fig. 8 After an appropriate modification of the notion of necessary property inclusion in Gödel's 1970 ontological proof, the inconsistency revealed in Fig. 6 is avoided, and the argument can be successfully verified in modal logic S5 (indeed, as shown, only symmetry of the accessibility relation is actually needed)

```

1 theory GoedelVariantHOML2poss imports HOMLinHOL ModalFilter
2 begin
3 consts PositiveProperty::"(e⇒σ)⇒σ" ("P")
4 axiomatization where Ax1: "[P φ ∧ P ψ ⊃ P (φ . ψ)]"
5 axiomatization where Ax2a: "[P φ ∨e P ~φ]"
6 definition God ("G") where "G x ≡ ∀φ. P φ ⊃ φ x"
7 (**) abbreviation PropertyInclusion ("⊃N") where "φ ⊃N ψ ≡ □(∀y::e. φ y ⊃ ψ y)"
8 definition Essence ("_Ess._") where "φ Ess. x ≡ φ x ∧ (∀ψ. ψ x ⊃ (φ ⊃N ψ))"
9 axiomatization where Ax2b: "[P φ ⊃ □ P φ]"
10 lemma Ax2b': "[¬P φ ⊃ □(¬P φ)]" using Ax2a Ax2b by blast
11 theorem Th1: "[G x ⊃ G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12 (**) definition NecExist ("E") where "E x ≡ ∀φ. φ Ess. x ⊃ □(∃x. φ x)"
13 axiomatization where Ax3: "[P E]"
14 (**) theorem Th2: "[G x ⊃ □(∃y. G y)]" using Ax3 Th1 God_def NecExist_def by smt
15 (**) theorem Th3: "[◇(∃x. G x) ⊃ □(∃y. G y)]" sledgehammer(Th2 Rsymm) —<Proof found>
16 proof -
17 (**) have 1: "[ (∃x. G x) ⊃ □(∃y. G y)]" using Th2 by blast
18 (**) have 2: "[◇(∃x. G x) ⊃ ◇□(∃y. G y)]" using 1 by blast
19 (**) have 3: "[◇(∃x. G x) ⊃ □(∃y. G y)]" using 2 Rsymm by blast
20 thus ?thesis by blast
21 qed
22 axiomatization where Ax4: "[P φ ∧ (φ ⊃N ψ) ⊃ P ψ]"
23 lemma True nitpick[satisfy,card=1,eval="[P (λx.⊥)]"] oops —<One model found of cardinality one>
24 abbreviation "PosProps Φ ≡ ∀φ. Φ φ ⊃ P φ"
25 (**) abbreviation "ConjOfPropsFrom φ Φ ≡ □(∀z. φ z ↔ (∀ψ. Φ ψ ⊃ ψ z))"
26 axiomatization where Ax1Gen: "[ (PosProps Φ ∧ ConjOfPropsFrom φ Φ) ⊃ P φ]"
27 lemma L: "[P G]" using Ax1Gen God_def by smt
28 (**) theorem Th4: "[◇(∃x. G x)]" using Ax2a Ax4 L by blast
29 (**) theorem Th5: "[□(∃x. G x)]" using Th3 Th4 by blast
30 lemma MC: "[φ ⊃ □φ]" sledgehammer(Ax2a Ax2b Th5 God_def Rsymm) —<Proof found>
31 proof - {fix w fix Q
32 (**) have 1: "∀x.(G x w → (∀z. Z x ⊃ □(∀z. G z ⊃ Z z)) w)" using Ax2a Ax2b God_def by smt
33 (**) have 2: "(∃x. G x w) → ((Q ⊃ □(∀z. G z ⊃ Q)) w)" using 1 by force
34 have 3: "(Q ⊃ □Q) w" using 2 Th5 Rsymm by blast}
35 thus ?thesis by auto
36 qed
37 lemma PosProps: "[P (λx.T) ∧ P (λx. x = x)]" using Ax2a Ax4 by blast
38 lemma NegProps: "[¬P(λx.⊥) ∧ ¬P(λx. x ≠ x)]" using Ax2a Ax4 by blast
39 (**) lemma UniqueEss1: "[φ Ess. x ∧ ψ Ess. x ⊃ □(∀y. φ y ↔ ψ y)]" using Essence_def by smt
40 (**) lemma UniqueEss2: "[φ Ess. x ∧ ψ Ess. x ⊃ □(φ ≡ ψ)]" nitpick[card i=2] oops —<Countermodel found>
41 (**) lemma UniqueEss3: "[φ Ess. x ⊃ □(∀y. φ y ⊃ y ≡ x)]" using Essence_def MC by auto
42 lemma Monotheism: "[G x ∧ G y ⊃ x ≡ y]" using Ax2a God_def by smt
43 lemma Filter: "[FilterP P]" using Ax1 Ax4 MC NegProps PosProps Rsymm by smt
44 lemma UltraFilter: "[UFilterP P]" using Ax2a Filter by smt
45 lemma True nitpick[satisfy,card=1,eval="[P (λx.⊥)]"] oops —<One model found of cardinality one>
46 end

```

Fig. 17 After an appropriate modification of the definition of essence, the inconsistency revealed in Fig. 16 is avoided, and the argument can be successfully verified in modal logic S5 (indeed, as shown, only the modal schema *B* is actually needed). In contrast to Fig. 7 we here use only possibilist quantifiers to obtain these results. Modified lines compared to Fig. 7 are marked with “(**)” in red

```

1 theory GoedelVariantHOML3poss imports HOMLinHOL ModalFilter
2 begin
3 consts PositiveProperty::"(e⇒σ)⇒σ" ("P")
4 axiomatization where Ax1: "[P φ ∧ P ψ ⊃ P (φ . ψ)]"
5 axiomatization where Ax2a: "[P φ ∨e P ~φ]"
6 definition God ("G") where "G x ≡ ∀φ. P φ ⊃ φ x"
7 (**) abbreviation PropertyInclusion ("⊃N") where "φ ⊃N ψ ≡ □(φ ≠ (λx. ⊥) ∧ (∀y::e. φ y ⊃ ψ y))"
8 definition Essence ("_Ess._") where "φ Ess. x ≡ ∀ψ. ψ x ⊃ (φ ⊃N ψ)"
9 axiomatization where Ax2b: "[P φ ⊃ □ P φ]"
10 lemma Ax2b': "[¬P φ ⊃ □(¬P φ)]" using Ax2a Ax2b by blast
11 theorem Th1: "[G x ⊃ G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12 (**) definition NecExist ("E") where "E x ≡ ∀φ. (φ Ess. x) ⊃ □(∃x. φ x)"
13 axiomatization where Ax3: "[P E]"
14 (**) theorem Th2: "[G x ⊃ □(∃y. G y)]" using Ax3 Th1 God_def NecExist_def by smt
15 (**) theorem Th3: "[◇(∃x. G x) ⊃ □(∃y. G y)]" sledgehammer(Th2 Rsymm) —<Proof found>
16 proof -
17 (**) have 1: "[ (∃x. G x) ⊃ □(∃y. G y)]" using Th2 by blast
18 (**) have 2: "[◇(∃x. G x) ⊃ ◇□(∃y. G y)]" using 1 by blast
19 (**) have 3: "[◇(∃x. G x) ⊃ □(∃y. G y)]" using 2 Rsymm by blast
20 thus ?thesis by blast
21 qed
22 axiomatization where Ax4: "[P φ ∧ (φ ⊃N ψ) ⊃ P ψ]"
23 lemma True nitpick[satisfy,card=1,eval="[P (λx.⊥)]"] oops —<One model found of cardinality one>
24 abbreviation "PosProps Φ ≡ ∀φ. Φ φ ⊃ P φ"
25 (**) abbreviation "ConjOfPropsFrom φ Φ ≡ □(∀z. φ z ↔ (∀ψ. Φ ψ ⊃ ψ z))"
26 axiomatization where Ax1Gen: "[ (PosProps Φ ∧ ConjOfPropsFrom φ Φ) ⊃ P φ]"
27 lemma L: "[P G]" using Ax1Gen God_def by (smt (verit))
28 (**) theorem Th4: "[◇(∃x. G x)]" sledgehammer[timeout=200](Ax2a L Ax1Gen) sorry —<Proof found>
29 (**) theorem Th5: "[□(∃x. G x)]" using Th3 Th4 by blast
30 lemma MC: "[φ ⊃ □φ]" sledgehammer(Ax2a Ax2b Th5 God_def Rsymm) —<Proof found>
31 proof - {fix w fix Q
32 (**) have 1: "∀x.(G x w → (∀z. Z x ⊃ □(∀z. G z ⊃ Z z)) w)" using Ax2a Ax2b God_def by smt
33 (**) have 2: "(∃x. G x w) → ((Q ⊃ □(∀z. G z ⊃ Q)) w)" using 1 by force
34 have 3: "(Q ⊃ □Q) w" using 2 Th5 Rsymm by blast}
35 thus ?thesis by auto
36 qed
37 lemma PosProps: "[P (λx.T) ∧ P (λx. x = x)]" using Ax2a Ax4 L Th4 by smt
38 lemma NegProps: "[¬P(λx.⊥) ∧ ¬P(λx. x ≠ x)]" using Ax2a Ax4 L Th4 by smt
39 (**) lemma UniqueEss1: "[ (φ Ess. x) ∧ (ψ Ess. x) ⊃ □(∀y. φ y ↔ ψ y)]" oops —<Unclear, open question>
40 lemma UniqueEss2: "[ (φ Ess. x) ∧ (ψ Ess. x) ⊃ □(φ ≡ ψ)]" oops —<Unclear, open question>
41 (**) lemma UniqueEss3: "[φ Ess. x ⊃ □(∀y. φ y ⊃ y ≡ x)]" using Essence_def MC by auto
42 lemma Monotheism: "[G x ∧ G y ⊃ x ≡ y]" using Ax2a God_def by smt
43 lemma Filter: "[FilterP P]" using Ax1 Ax4 MC NegProps PosProps Rsymm by smt
44 lemma UltraFilter: "[UFilterP P]" using Ax2a Filter by smt
45 lemma True nitpick[satisfy,card=1,eval="[P (λx.T)]"] oops —<One model found of cardinality one>
46 end

```

Fig. 18 After an appropriate modification of the definition of necessary property implication, the inconsistency shown in Fig. 16 is avoided, and the argument can be successfully verified. As shown here, this still holds when using possibilist quantifiers only. Modified lines compared to Fig. 8 are marked with “(**)” in red

Gödel 2

Summary

- modifies notion of **essence**
- no existential import
- consistent
- derives poss. and nec. existence of G
- works already in logic KB
- requires generalized axiom A1Gen
- implies
 - modal collapse
 - monotheism
 - P is an ultrafilter
- works also for only possibilist quantifiers

Gödel 3

Summary

- modifies notion of **property inclusion**
- no existential import
- consistent
- derives poss. and nec. existence of G
- works already in logic KB
- requires generalized axiom A1Gen
- implies
 - modal collapse
 - monotheism
 - P is an ultrafilter
- works also for only possibilist quantifiers

How about symmetries?

```
1 theory ForPeterFritz3a imports HOMLinHOL ModalFilter
2 begin
3   consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4
5   axiomatization where UF: "[UFilter P]"
6   definition God ("G") where "G x  $\equiv$   $\forall \varphi. P \varphi \supset \varphi x$ "
7
8   lemma True nitpick[satisfy] oops —<Model found>
9
10  axiomatization where OtherAxiom: "[ $\Diamond(\neg(\exists^E x. G x))$ ]"
11  lemma True nitpick[satisfy] oops —<Model found>
12 end
```

```
1 theory ForPeterFritz3b imports HOMLinHOL ModalFilter
2 begin
3   consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4
5   axiomatization where UF: "[UFilter P]"
6   definition God ("G") where "G x  $\equiv$   $\forall \varphi. P \varphi \supset \varphi x$ "
7
8   lemma True nitpick[satisfy] oops —<Model found>
9
10  axiomatization where OtherAxiom: "[ $\Diamond(\exists^E x. G x)$ ]"
11  lemma True nitpick[satisfy] oops —<Model found>
12 end
```

Symmetry lost: A modal ontological argument for atheism?

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Abstract

The modal ontological argument for God's existence faces a symmetry problem: a seemingly equally plausible reverse modal ontological argument can be given for God's nonexistence. Here, we argue that there are significant asymmetries between the modal ontological argument and its reverse that render the latter more compelling than the former. Specifically, the latter requires a weaker logic than the former and, unlike the former, avoids the symmetry problem. We also explore to what extent these observations represent a new pathway to atheism.

1 | INTRODUCTION

- (P1) Necessarily, if God exists, then God necessarily exists.
- (P2*) God possibly does not exist.
- (C*) Therefore, God does not exist.

We will call this the *reverse modal ontological argument*. In propositional modal logic, we can formalize it as follows:

- (P1) $\Box(g \rightarrow \Box g)$
- (P2*) $\Diamond \neg g$
- (C*) $\neg g$

In S5, the argument from P1 and P2* to C* is also valid.

Poss. nonexistence of God is not compatible with Gödel's axioms that characterise the set of positive properties as an ultrafilter.

How about symmetries?

```
1 theory ForPeterFritz3a imports HOMLinHOL ModalFilter
2 begin
3   consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4
5   axiomatization where UF: "[UFILTER P]"
6   definition God ("G") where "G x  $\equiv$   $\forall \varphi. P \varphi \supset \varphi x$ "
7
8   lemma True nitpick[satisfy] oops —<Model found>
9
10  axiomatization where OtherAxiom: "[ $\Diamond(\neg(\exists^E x. G x))$ ]"
11  lemma True nitpick[satisfy] oops —<Model found>
12 end
```

```
1 theory ForPeterFritz3b imports HOMLinHOL ModalFilter
2 begin
3   consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4
5   axiomatization where UF: "[UFILTER P]"
6   definition God ("G") where "G x  $\equiv$   $\forall \varphi. P \varphi \supset \varphi x$ "
7
8   lemma True nitpick[satisfy] oops —<Model found>
9
10  axiomatization where OtherAxiom: "[ $\Diamond(\exists^E x. G x)$ ]"
11  lemma True nitpick[satisfy] oops —<Model found>
12 end
```

```
1 theory ForPeterFritz imports HOMLinHOL ModalFilter
2 begin
3   consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4   —<Some shorthand notation>
5   abbreviation "PosProps  $\Phi \equiv \forall \varphi. \Phi \varphi \supset P \varphi$ "
6   abbreviation "ConjOfPropsFrom  $\varphi \Phi \equiv \Box(\forall^E z. \varphi z \leftrightarrow (\forall \psi. \Phi \psi \supset \psi z))$ "
7   abbreviation PropertyInclusion (" $\supset_N$ ") where " $\varphi \supset_N \psi \equiv \Box(\forall^E y. \varphi y \supset \psi y)$ "
8   —<Axiom 1 (generalized version)>
9   axiomatization where Ax1Gen: "[ (PosProps  $\Phi \wedge$  ConjOfPropsFrom  $\varphi \Phi$ )  $\supset P \varphi$  ]"
10  —<Axiom 2a>
11  axiomatization where Ax2a: "[ $P \varphi \vee^E P \sim \varphi$ ]"
12  —<Axiom 2b>
13  axiomatization where Ax2b: "[ $P \varphi \supset \Box P \varphi$ ]"
14  —<Theorem 1 (generalized version)>
15  definition God ("G") where "G x  $\equiv \forall \varphi. P \varphi \supset \varphi x$ "
16  definition Essence ("_Ess._") where " $\varphi$  Ess. x  $\equiv \varphi x \wedge (\forall \psi. \psi x \supset (\varphi \supset_N \psi))$ "
17  theorem Th1: "[ $G x \supset G$  Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
18  —<Axiom 3>
19  definition NecExist ("E") where "E x  $\equiv \forall \varphi. \varphi$  Ess. x  $\supset \Box(\exists^E x. \varphi x)$ "
20  axiomatization where Ax3: "[ $P E$ ]"
21  —<Axiom 4>
22  axiomatization where Ax4: "[ $P \varphi \wedge (\varphi \supset_N \psi) \supset P \psi$ ]"
23  lemma True nitpick[satisfy] oops —<Model found>
24  —<PeterAxiom>
25  axiomatization where PeterAxiom: "[ $\Diamond(\neg(\exists^E x. G x))$ ]"
26  lemma True nitpick[satisfy] oops —<No model found>
27  lemma False by (metis (no_types, lifting) Ax1Gen Ax2a Ax3 Ax4 God_def NecExist_def PeterAxiom Th1)
28 end
```

After postulating the
possible nonexistence of God
we can derive falsity

Poss. nonexistence of God is not compatible with Gödel's axioms that characterise the set of positive properties as an ultrafilter.

How about Evil?

(thanks to Chad Brown)

```
1 theory ThereIsNoEvil1 imports GoedelVariantHOML2
2 begin
3 definition Evil ("Evil") where "Evil x  $\equiv \forall \varphi. \neg P \varphi \supset \varphi x$ "
4 theorem NecNoEvil: " $\Box(\neg(\exists^E x. \text{Evil } x))$ " sledgehammer(Ax2a Ax4 Evil_def) —<Proof found>
5 proof -
6   have " $\neg P(\lambda y. \bot)$ " using Ax2a Ax4 by blast
7   hence " $(\forall^E x. \text{Evil } x \supset (\lambda y. \bot) x)$ " using Evil_def by auto
8   hence " $(\forall^E x. \text{Evil } x \supset \bot)$ " by auto
9   hence " $(\exists^E x. \text{Evil } x) \supset \bot$ " by auto
10  hence " $\neg(\exists^E x. \text{Evil } x)$ " by blast
11  thus ?thesis by blast
12 qed
13 end
```

Fig. 10 Importing Gödel's modified axioms from Fig. 7 we can prove that necessarily there exists no entity that possesses all non-positive (=negative) properties

```
1 theory ThereIsNoEvil2 imports GoedelVariantHOML3
2 begin
3 definition Evil ("Evil") where "Evil x  $\equiv \forall \varphi. \neg P \varphi \supset \varphi x$ "
4 theorem NecNoEvil: " $\Box(\neg(\exists^E x. \text{Evil } x))$ " sledgehammer(Ax1Gen Ax2a Evil_def) oops —<Proof found>
5 end
```

Fig. 11 Importing Gödel's modified axioms from Fig. 8 we can prove that necessarily there exists no entity that possesses all non-positive (=negative) properties

How about Evil?

Appendix E: Necessary existence of an Evil-like entity derived from a modified set of axioms

By rejecting Gödel's assumptions and instead postulating corresponding negative versions of them, as shown in the Fig. 20, the necessary existence of Evil becomes derivable. Note that the empty property $\lambda x. \perp$ and self-difference $\lambda x. x \neq x$ now become positive properties, and the tautologous property $\lambda x. \top$ and self-identity $\lambda x. x = x$ are now non-positive. The non-positive properties of this "Evil-like" entity are thus apparently identical to the positive properties of Gödel's God-like entity.

```
1 theory EvilDerivable imports HOMLinHOL ModalFilter
2 begin
3   consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4   definition Evil ("Evil") where "Evil x  $\equiv$   $\forall \varphi. \neg P \varphi \supset \varphi x$ "
5   definition Essence ("_Ess._") where " $\varphi$  Ess. x  $\equiv$   $\varphi x \wedge (\forall \psi. \psi x \supset \Box(\forall^E y. \varphi y \supset \psi y))$ "
6   definition NecExist ("E") where "E x  $\equiv$   $\forall \varphi. \varphi$  Ess. x  $\supset \Box(\exists^E x. \varphi x)$ "
7   axiomatization where A1: " $\neg P \varphi \leftrightarrow P \sim \varphi$ "
8   axiomatization where A2: " $\neg P \varphi \wedge \Box(\forall^E y. \varphi y \supset \psi y) \supset \neg P \psi$ "
9   axiomatization where A4: " $\neg P$  Evil"
10  axiomatization where A3: " $\neg P \varphi \supset \Box(\neg P \varphi)$ "
11  axiomatization where A5: " $\neg P$  E"
12  lemma True nitpick[satisfy,card i=1,eval="P ( $\lambda x. \perp$ )",eval="P ( $\lambda x. \top$ )"] oops —<Model found>
13  theorem T1: " $\neg P \varphi \supset \Diamond(\exists^E x. \varphi x)$ " using A1 A2 by blast
14  theorem T2: " $\Diamond(\exists^E x. \text{Evil } x)$ " using A4 T1 by blast
15  theorem T3: " $\text{Evil } x \supset \text{Evil Ess. } x$ " using A1 A3 Essence_def Evil_def by (smt (verit, best))
16  theorem T4: " $\Diamond(\exists^E x. \text{Evil } x) \supset \Box(\exists^E y. \text{Evil } y)$ " using A5 Evil_def NecExist_def Rsymm T3 by smt
17  theorem T5: " $\Box(\exists^E x. \text{Evil } x)$ " using T2 T4 by presburger
18  lemma MC: " $\varphi \supset \Box \varphi$ " sledgehammer(A1 A3 T5 Evil_def Rsymm) oops —<proof found>
19  lemma PosProps: " $P (\lambda x. \perp) \wedge P (\lambda x. x \neq x)$ " using A1 A2 by blast
20  lemma NegProps: " $\neg P (\lambda x. \top) \wedge \neg P (\lambda x. x = x)$ " using A1 A2 by blast
21  lemma UniqueEss1: " $\varphi$  Ess. x  $\wedge \psi$  Ess. x  $\supset \Box(\forall^E y. \varphi y \leftrightarrow \psi y)$ " using Essence_def by smt
22  lemma UniqueEss2: " $\varphi$  Ess. x  $\wedge \psi$  Ess. x  $\supset \Box(\varphi = \psi)$ " nitpick[card i=2] oops —<Countermodel found>
23  lemma Monoevilism: " $\text{Evil } x \wedge \text{Evil } y \supset x \equiv y$ " using A1 Evil_def by smt
24  lemma Filter: " $\text{Filter } (\lambda \varphi. \neg P \varphi)$ " using A1 Evil_def Rsymm T1 T5 by (smt (verit, best))
25  lemma UltraFilter: " $\text{UFilter } (\lambda \varphi. \neg P \varphi)$ " using Filter A1 by blast
26 end
```

Fig. 20 The necessary existence of an Evil-like entity proved from (controversially) modified assumptions

The properties of this Evil-like entity are identical to the properties of Gödel's original God-like entity!

Formalisation of Scott's variant

— proof in logic KB —

Scott

Axiom A1: $\neg P \varphi \leftrightarrow P \sim \varphi$

Axiom A2: $P \varphi \wedge \Box(\forall^E y. \varphi y \supset \psi y) \supset P \psi$

```
1 theory ScottVariantHOML imports HOMLinHOL ModalFilter
2 begin
3 consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4 axiomatization where A1: "[ $\neg P \varphi \leftrightarrow P \sim \varphi$ ]"
5 axiomatization where A2: "[ $P \varphi \wedge \Box(\forall^E y. \varphi y \supset \psi y) \supset P \psi$ ]"
```

Fig. 12 Scott's variant of Gödel's ontological argument

Scott

Theorem T1: $P \varphi \supset \Diamond(\exists^E x. \varphi x)$

```
1 theory ScottVariantHOML imports HOMLinHOL ModalFilter
2 begin
3 consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4 axiomatization where A1: "[ $\neg P \varphi \leftrightarrow P \sim \varphi$ ]"
5 axiomatization where A2: "[ $P \varphi \wedge \Box(\forall^E y. \varphi y \supset \psi y) \supset P \psi$ ]"
6 theorem T1: "[ $P \varphi \supset \Diamond(\exists^E x. \varphi x)$ ]" using A1 A2 by blast
```

Fig. 12 Scott's variant of Gödel's ontological argument

Scott

Definition of God-like: $G x \equiv \forall \varphi. P \varphi \supset \varphi x$

Axiom A3: $P G$

Theorem Coro: $\Diamond(\exists^E x. G x)$

```
1 theory ScottVariantHOML imports HOMLinHOL ModalFilter
2 begin
3 consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4 axiomatization where A1: "[ $\neg P \varphi \leftrightarrow P \sim \varphi$ ]"
5 axiomatization where A2: "[ $P \varphi \wedge \Box(\forall^E y. \varphi y \supset \psi y) \supset P \psi$ ]"
6 theorem T1: "[ $P \varphi \supset \Diamond(\exists^E x. \varphi x)$ ]" using A1 A2 by blast
7 definition God ("G") where "[ $G x \equiv \forall \varphi. P \varphi \supset \varphi x$ ]"
8 axiomatization where A3: "[ $P G$ ]"
9 theorem Coro: "[ $\Diamond(\exists^E x. G x)$ ]" using A3 T1 by blast
```

Fig. 12 Scott's variant of Gödel's ontological argument

Scott

Axiom A4: $P \varphi \supset \Box P \varphi$

Definition of essence: $\varphi \text{ Ess. } x \equiv \varphi x \wedge \forall \psi. \psi x \supset \Box (\forall^E y. \varphi y \supset \psi y)$

Theorem T2: $G x \supset G \text{ Ess. } x$

```
1 theory ScottVariantHOML imports HOMLinHOL ModalFilter
2 begin
3   consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4   axiomatization where A1: "[ $\neg P \varphi \leftrightarrow P \sim \varphi$ ]"
5   axiomatization where A2: "[ $P \varphi \wedge \Box (\forall^E y. \varphi y \supset \psi y) \supset P \psi$ ]"
6   theorem T1: "[ $P \varphi \supset \Diamond (\exists^E x. \varphi x)$ ]" using A1 A2 by blast
7   definition God ("G") where "G x  $\equiv \forall \varphi. P \varphi \supset \varphi x$ "
8   axiomatization where A3: "[P G]"
9   theorem Coro: "[ $\Diamond (\exists^E x. G x)$ ]" using A3 T1 by blast
10  axiomatization where A4: "[P  $\varphi \supset \Box P \varphi$ ]"
11  definition Ess ("_Ess._") where " $\varphi \text{ Ess. } x \equiv \varphi x \wedge (\forall \psi. \psi x \supset \Box (\forall^E y. \varphi y \supset \psi y))$ "
12  theorem T2: "[G x  $\supset G \text{ Ess. } x$ ]" using A1 A4 Ess_def God_def by fastforce
```

Fig. 12 Scott's variant of Gödel's ontological argument

Scott

Definition of Necessary Existence: $NE\ x \equiv \forall\varphi. \varphi\ Ess. x \supset \Box(\exists^E x. \varphi\ x)$

Axiom A5: $P\ NE$

Consistency of the assumptions

```
1 theory ScottVariantHOML imports HOMLinHOL ModalFilter
2 begin
3 consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4 axiomatization where A1: "[ $\neg P\ \varphi \leftrightarrow P\ \sim\varphi$ ]"
5 axiomatization where A2: "[ $P\ \varphi \wedge \Box(\forall^E y. \varphi\ y \supset \psi\ y) \supset P\ \psi$ ]"
6 theorem T1: "[ $P\ \varphi \supset \Diamond(\exists^E x. \varphi\ x)$ ]" using A1 A2 by blast
7 definition God ("G") where "G x  $\equiv \forall\varphi. P\ \varphi \supset \varphi\ x$ "
8 axiomatization where A3: "[P G]"
9 theorem Coro: "[ $\Diamond(\exists^E x. G\ x)$ ]" using A3 T1 by blast
10 axiomatization where A4: "[ $P\ \varphi \supset \Box P\ \varphi$ ]"
11 definition Ess ("_Ess._") where " $\varphi\ Ess. x \equiv \varphi\ x \wedge (\forall\psi. \psi\ x \supset \Box(\forall^E y. \varphi\ y \supset \psi\ y))$ "
12 theorem T2: "[G x  $\supset G\ Ess. x$ ]" using A1 A4 Ess_def God_def by fastforce
13 definition NecExist ("NE") where "NE x  $\equiv \forall\varphi. \varphi\ Ess. x \supset \Box(\exists^E x. \varphi\ x)$ "
14 axiomatization where A5: "[P NE]"
15 lemma True nitpick[satisfy,card=1,eval="[P ( $\lambda x. T$ )]"] oops —<One model found of cardinality one>
```

Fig. 12 Scott's variant of Gödel's ontological argument

Scott

Theorem T3: $\Box(\exists^E x. G x)$

Logik KB (Rsymm) needed here

```
1 theory ScottVariantHOML imports HOMLinHOL ModalFilter
2 begin
3   consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4   axiomatization where A1: "[ $\neg P \varphi \leftrightarrow P \sim \varphi$ ]"
5   axiomatization where A2: "[ $P \varphi \wedge \Box(\forall^E y. \varphi y \supset \psi y) \supset P \psi$ ]"
6   theorem T1: "[ $P \varphi \supset \Diamond(\exists^E x. \varphi x)$ ]" using A1 A2 by blast
7   definition God ("G") where "G x  $\equiv \forall \varphi. P \varphi \supset \varphi x$ "
8   axiomatization where A3: "[P G]"
9   theorem Coro: "[ $\Diamond(\exists^E x. G x)$ ]" using A3 T1 by blast
10  axiomatization where A4: "[P  $\varphi \supset \Box P \varphi$ ]"
11  definition Ess ("_Ess._") where " $\varphi$  Ess. x  $\equiv \varphi x \wedge (\forall \psi. \psi x \supset \Box(\forall^E y. \varphi y \supset \psi y))$ "
12  theorem T2: "[G x  $\supset G$  Ess. x]" using A1 A4 Ess_def God_def by fastforce
13  definition NecExist ("NE") where "NE x  $\equiv \forall \varphi. \varphi$  Ess. x  $\supset \Box(\exists^E x. \varphi x)$ "
14  axiomatization where A5: "[P NE]"
15  lemma True nitpick[satisfy,card=1,eval="[P ( $\lambda x. T$ )]"] oops —<One model found of cardinality one>
16  theorem T3: "[ $\Box(\exists^E x. G x)$ ]" sledgehammer(A5 Coro God_def NecExist_def Rsymm T2) —<Proof found>
17  proof -
18    have 1: "[ $(G x \supset NE x) \wedge (G$  Ess. x  $\supset \Box(\exists^E x. G x))$ ]" using A5 Ess_def God_def NecExist_def by smt
19    hence 2: "[ $(\exists^E x. G x) \supset \Box(\exists^E x. G x)$ ]" using A5 God_def NecExist_def T2 by smt
20    hence 3: "[ $\Diamond(\exists^E x. G x) \supset (\Diamond(\Box(\exists^E x. G x)) \supset \Box(\exists^E x. G x))$ ]" using Rsymm by blast
21    thus ?thesis using 2 Coro by blast
22  qed
```

Fig. 12 Scott's variant of Gödel's ontological argument

Scott

Further results as in Gödel 2

```
1 theory ScottVariantHOML imports HOMLinHOL ModalFilter
2 begin
3   consts PositiveProperty::"(e $\Rightarrow$  $\sigma$ ) $\Rightarrow$  $\sigma$ " ("P")
4   axiomatization where A1: "[ $\neg P \varphi \leftrightarrow P \sim \varphi$ ]"
5   axiomatization where A2: "[ $P \varphi \wedge \Box(\forall^E y. \varphi y \supset \psi y) \supset P \psi$ ]"
6   theorem T1: "[ $P \varphi \supset \Diamond(\exists^E x. \varphi x)$ ]" using A1 A2 by blast
7   definition God ("G") where
8     axiomatization where A3: "[ $P G$ ]"
9   theorem Coro: "[ $\Diamond(\exists^E x. G x)$ ]" using A3 T1 by blast
10  axiomatization where A4: "[ $P \varphi \supset \Box P \varphi$ ]"
11  definition Ess ("_Ess._") where
12    theorem T2: "[ $G x \supset G \text{Ess. } x$ ]" using A1 A4 Ess_def God_def by fastforce
13  definition NecExist ("NE") where
14    axiomatization where A5: "[ $P NE$ ]"
15  lemma True nitpick[satisfy,card=1,eval="[ $P (\lambda x. T)$ ]" ] oops —<One model found of cardinality one>
16  theorem T3: "[ $\Box(\exists^E x. G x)$ ]" sledgehammer(A5 Coro God_def NecExist_def Rsymm T2) —<Proof found>
17  proof -
18    have 1: "[ $(G x \supset NE x) \wedge (G \text{Ess. } x \supset \Box(\exists^E x. G x))$ ]" using A5 Ess_def God_def NecExist_def by smt
19    hence 2: "[ $(\exists^E x. G x) \supset \Box(\exists^E x. G x)$ ]" using A5 God_def NecExist_def T2 by smt
20    hence 3: "[ $\Diamond(\exists^E x. G x) \supset (\Diamond(\Box(\exists^E x. G x)) \supset \Box(\exists^E x. G x))$ ]" using Rsymm by blast
21    thus ?thesis using 2 Coro by blast
22  qed
23  lemma MC: "[ $\varphi \supset \Box \varphi$ ]" sledgehammer(A1 A4 God_def Rsymm T3) —<Proof found>
24  proof - {fix w fix Q
25    have 1: "[ $\forall x. (G x w \longrightarrow (\forall Z. Z x \supset \Box(\forall^E z. ((G z) \supset (Z z)))) w$ ]" using A1 A4 God_def by smt
26    have 2: "[ $(\exists x. G x w) \longrightarrow ((Q \supset \Box(\forall^E z. ((G z) \supset Q))) w)$ ]" using 1 by force
27    have 3: "[ $(Q \supset \Box Q) w$ ]" using 2 T3 Rsymm by blast
28    thus ?thesis by auto
29  qed
30  lemma PosProps: "[ $P (\lambda x. T) \wedge P (\lambda x. x = x)$ ]" using A1 A2 by blast
31  lemma NegProps: "[ $\neg P (\lambda x. \bot) \wedge \neg P (\lambda x. x \neq x)$ ]" using A1 A2 by blast
32  lemma UniqueEss1: "[ $\varphi \text{Ess. } x \wedge \psi \text{Ess. } x \supset \Box(\forall^E y. \varphi y \leftrightarrow \psi y)$ ]" using Ess_def by smt
33  lemma UniqueEss2: "[ $\varphi \text{Ess. } x \wedge \psi \text{Ess. } x \supset \Box(\varphi \equiv \psi)$ ]" nitpick[card i=1] oops —<Countermodel found>
34  lemma UniqueEss3: "[ $\varphi \text{Ess. } x \supset \Box(\forall^E y. \varphi y \supset y \equiv x)$ ]" using Ess_def MC by auto
35  lemma Monotheism: "[ $G x \wedge G y \supset x \equiv y$ ]" using A1 God_def by smt
36  lemma Filter: "[Filter P]" using A1 God_def Rsymm T1 T3 by (smt (verit, best))
37  lemma UltraFilter: "[UltraFilter P]" using Filter A1 by blast
38  lemma True nitpick[satisfy,card=1,eval="[ $P (\lambda x. \bot)$ ]" ] oops —<One model found of cardinality one>
39 end
```

Fig. 12 Scott's variant of Gödel's ontological argument

```

1 theory GoedelVariantHOML2 imports HOMLinHOL ModalFilter
2 begin
3   consts PositiveProperty::"(e⇒σ)⇒σ" ("P")
4   axiomatization where Ax1: "[P φ ∧ P ψ ⊃ P (φ . ψ)]"
5   axiomatization where Ax2a: "[P φ ∨e P ~φ]"
6   definition God ("G") where "G x ≡ ∀φ. P φ ⊃ φ x"
7   abbreviation PropertyInclusion ("⊃N") where "φ ⊃N ψ ≡ □(∀ey. φ y ⊃ ψ y)"
8   (**) definition Essence ("_Ess._") where "φ Ess. x ≡ φ x ∧ (∀ψ. ψ x ⊃ (φ ⊃N ψ))"
9   axiomatization where Ax2b: "[P φ ⊃ □ P φ]"
10  lemma Ax2b': "[¬P φ ⊃ □(¬P φ)]" using Ax2a Ax2b by blast
11  theorem Th1: "[G x ⊃ G Ess. x]" using Ax2a Ax2b Essence_def God_def by (smt (verit))
12  definition NecExist ("E") where "E x ≡ ∀φ. φ Ess. x ⊃ □(∃Ex. φ x)"
13  axiomatization where Ax3: "[P E]"
14  theorem Th2: "[G x ⊃ □(∃Ey. G y)]" using Ax3 Th1 God_def NecExist_def by smt
15  theorem Th3: "[◇(∃Ex. G x) ⊃ □(∃Ey. G y)]" sledgehammer(Th2 Rsymm) —<Proof found>
16  proof -
17    have 1: "[◇(∃Ex. G x) ⊃ □(∃Ey. G y)]" using Th2 by blast
18    have 2: "[◇(∃Ex. G x) ⊃ ◇□(∃Ey. G y)]" using 1 by blast
19    have 3: "[◇(∃Ex. G x) ⊃ □(∃Ey. G y)]" using 2 Rsymm by blast
20    thus ?thesis by blast
21  qed
22  axiomatization where Ax4: "[P φ ∧ (φ ⊃N ψ) ⊃ P ψ]"
23  lemma True nitpick[satisfy,card=1,eval="[P (λx.⊥)"]]" oops —<One model found of cardinality one>
24  abbreviation "PosProps Φ ≡ ∀φ. Φ φ ⊃ P φ"
25  abbreviation "ConjOfPropsFrom φ Φ ≡ □(∀Ez. φ z ↔ (∀ψ. Φ ψ ⊃ ψ z))"
26  axiomatization where Ax1Gen: "[ (PosProps Φ ∧ ConjOfPropsFrom φ Φ) ⊃ P φ]"
27  lemma L: "[P G]" using Ax1Gen God_def by smt
28  theorem Th4: "[◇(∃Ex. G x)]" using Ax2a Ax4 L by blast
29  theorem Th5: "[□(∃Ex. G x)]" using Th3 Th4 by blast
30  lemma MC: "[φ ⊃ □φ]" sledgehammer(Ax2a Ax2b Th5 God_def Rsymm) —<Proof found>
31  proof - {fix w fix Q
32    have 1: "∀x.(G x w → (∀Z. Z x ⊃ □(∀Ez. G z ⊃ Z z))) w" using Ax2a Ax2b God_def by smt
33    have 2: "(∃x. G x w) → ((Q ⊃ □(∀Ez. G z ⊃ Q)) w)" using 1 by force
34    have 3: "(Q ⊃ □Q) w" using 2 Th5 Rsymm by blast
35    thus ?thesis by auto
36  qed
37  lemma PosProps: "[P (λx.T) ∧ P (λx.x = x)]" using Ax2a Ax4 by blast
38  lemma NegProps: "[¬P(λx.⊥) ∧ ¬P(λx.x ≠ x)]" using Ax2a Ax4 by blast
39  lemma UniqueEss1: "[φ Ess. x ∧ ψ Ess. x ⊃ □(∀ey. φ y ↔ ψ y)]" using Essence_def by smt
40  lemma UniqueEss2: "[φ Ess. x ∧ ψ Ess. x ⊃ □(φ ≡ ψ)]" nitpick[card i=1] oops —<Countermodel found>
41  lemma UniqueEss3: "[φ Ess. x ⊃ □(∀ey. φ y ⊃ y ≡ x)]" using Essence_def MC by auto
42  lemma Monotheism: "[G x ∧ G y ⊃ x ≡ y]" using Ax2a God_def by smt
43  lemma Filter: "[Filter P]" using Ax1 Ax4 MC NegProps PosProps Rsymm by smt
44  lemma UltraFilter: "[UFilter P]" using Ax2a Filter by smt
45  lemma True nitpick[satisfy,card=1,eval="[P (λx.⊥)"]]" oops —<One model found of cardinality one>
46 end

```

Fig. 7 After an appropriate modification of the definition of essence in Gödel's 1970 ontological proof, the inconsistency revealed in Fig. 6 is avoided, and the argument can be successfully verified in modal logic S5 (indeed, as shown, only symmetry of the accessibility relation is actually needed)

```

1 theory ScottVariantHOML imports HOMLinHOL ModalFilter
2 begin
3   consts PositiveProperty::"(e⇒σ)⇒σ" ("P")
4   axiomatization where A1: "[¬P φ ↔ P ~φ]"
5   axiomatization where A2: "[P φ ∧ □(∀ey. φ y ⊃ ψ y) ⊃ P ψ]"
6   theorem T1: "[P φ ⊃ ◇(∃Ex. φ x)]" using A1 A2 by blast
7   definition God ("G") where "G x ≡ ∀φ. P φ ⊃ φ x"
8   axiomatization where A3: "[P G]"
9   theorem Coro: "[◇(∃Ex. G x)]" using A3 T1 by blast
10  axiomatization where A4: "[P φ ⊃ □ P φ]"
11  definition Ess ("_Ess._") where "φ Ess. x ≡ φ x ∧ (∀ψ. ψ x ⊃ □(∀ey. φ y ⊃ ψ y))"
12  theorem T2: "[G x ⊃ G Ess. x]" using A1 A4 Ess_def God_def by fastforce
13  definition NecExist ("NE") where "NE x ≡ ∀φ. φ Ess. x ⊃ □(∃Ex. φ x)"
14  axiomatization where A5: "[P NE]"
15  lemma True nitpick[satisfy,card=1,eval="[P (λx.T)"]]" oops —<One model found of cardinality one>
16  theorem T3: "[□(∃Ex. G x)]" sledgehammer(A5 Coro God_def NecExist_def Rsymm T2) —<Proof found>
17  proof -
18    have 1: "[ (G x ⊃ NE x) ∧ (G Ess. x ⊃ □(∃Ex. G x))]" using A5 Ess_def God_def NecExist_def by smt
19    hence 2: "[◇(∃Ex. G x) ⊃ □(∃Ex. G x)]" using A5 God_def NecExist_def T2 by smt
20    hence 3: "[◇(∃Ex. G x) ⊃ (◇(□(∃Ex. G x)) ⊃ □(∃Ex. G x))]" using Rsymm by blast
21    thus ?thesis using 2 Coro by blast
22  qed
23  lemma MC: "[φ ⊃ □φ]" sledgehammer(A1 A4 God_def Rsymm T3) —<Proof found>
24  proof - {fix w fix Q
25    have 1: "∀x.(G x w → (∀Z. Z x ⊃ □(∀Ez. ((G z) ⊃ (Z z)))) w)" using A1 A4 God_def by smt
26    have 2: "(∃x. G x w) → ((Q ⊃ □(∀Ez. ((G z) ⊃ Q))) w)" using 1 by force
27    have 3: "(Q ⊃ □Q) w" using 2 T3 Rsymm by blast
28    thus ?thesis by auto
29  qed
30  lemma PosProps: "[P (λx.T) ∧ P (λx.x = x)]" using A1 A2 by blast
31  lemma NegProps: "[¬P(λx.⊥) ∧ ¬P(λx.x ≠ x)]" using A1 A2 by blast
32  lemma UniqueEss1: "[φ Ess. x ∧ ψ Ess. x ⊃ □(∀ey. φ y ↔ ψ y)]" using Ess_def by smt
33  lemma UniqueEss2: "[φ Ess. x ∧ ψ Ess. x ⊃ □(φ ≡ ψ)]" nitpick[card i=1] oops —<Countermodel found>
34  lemma UniqueEss3: "[φ Ess. x ⊃ □(∀ey. φ y ⊃ y ≡ x)]" using Ess_def MC by auto
35  lemma Monotheism: "[G x ∧ G y ⊃ x ≡ y]" using A1 God_def by smt
36  lemma Filter: "[Filter P]" using A1 God_def Rsymm T1 T3 by (smt (verit, best))
37  lemma UltraFilter: "[UFilter P]" using Filter A1 by blast
38  lemma True nitpick[satisfy,card=1,eval="[P (λx.⊥)"]]" oops —<One model found of cardinality one>
39 end

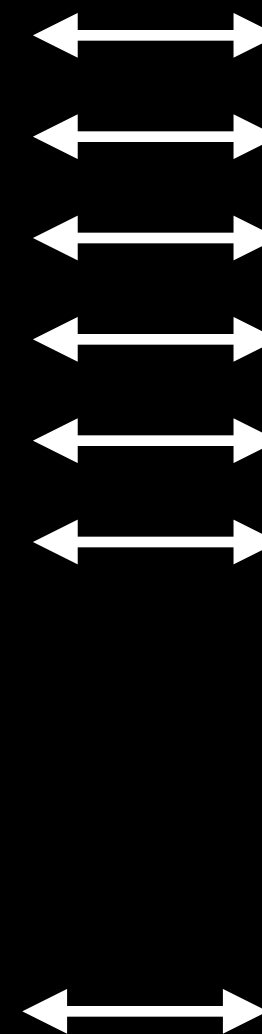
```

Fig. 12 Scott's variant of Gödel's ontological argument

Summary

- modifies notion of **essence / property incl.**
- no existential import
- consistent
- derives poss. and nec. existence of G
- works already in logic KB
- requires generalized axiom A1Gen
- implies
 - modal collapse
 - monotheism
 - P is an ultrafilter
- works also for only possibilist quantifiers

Ax1Gen generally implies uncountably infinite structures ...



Summary

- **various modifications and reorderings**
- no existential import
- consistent
- derives poss. and nec. existence of G
- works already in logic KB
- avoids generalized axiom A1Gen
- implies
 - modal collapse
 - monotheism
 - P is an ultrafilter
- works also for only possibilist quantifiers

Avoids Ax1Gen ...

Conclusion

- Demonstration of virtues of **Computational Metaphysics**
- First-time formalization of Gödel's 1970 manuscript
- New variant based on modified property inclusion
- Pointed the new insights gained
- Discussed the problem of oversimplification of Gödel's work

Future work

- Various smaller open questions
- Proof in logic S4?
- Study of the implications of Ax1Gen, full comprehension and impredicativity